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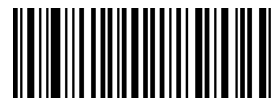
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Research Methodology

AMITY



Research Methodology

Research Methodology

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Module - I: Introduction

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Learning Objectives

At the end of this module, you will be able to:

- Assess Research
- Infer Research Process
- Mention Types of Research Design
- Explain Research Modelling

Introduction

Authors and management experts have defined research in various ways. Typically, research is said to commence with a question or a problem. It is described as the creation of new concepts, methodologies, and understandings through the development of new knowledge or the innovative application of existing knowledge. This process often involves synthesizing and analysing previous studies to produce new and groundbreaking results.

1.1 Research

Through research, we can identify solutions to problems by employing systematic and scientific methods. This may involve experimentation or innovation, with “risk” highlighting the inherent dangers of discovery.

1.1.1 Definition and Objectives of Research

What is research?

Investigation might lead to analysis, and tests could be conducted to verify hypotheses. This work is distinct and unprecedented, offering new insights as a result. It could introduce a novel method or new data source, leading to significant breakthroughs or incremental improvements in current practices. Research is essential for progress, as it is fundamentally tied to “research and development.”

Slesinger & Stephenson, Encyclopaedia of Social Sciences:

“the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in the construction of theory or in the practice of an art”.

- Redman & Mory: “Systematized effort to gain new knowledge”. It is an academic activity and should be used in a technical sense.
- Clifford Woody: Research comprises “defining and redefining problems, formulating hypotheses or suggested solutions; collecting, organizing and evaluating data; making deductions and reaching conclusions; and finally, carefully testing the conclusions to determine whether they fit the formulated hypotheses”.

Research Objectives

General Objectives

General objectives, also known as secondary objectives, provide a detailed view

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of a study's goal. In other words, by the end of your studies, you will have a general idea of what you want to accomplish. For instance, your overarching objective can be to look at the organisation's usage of renewable energy sources and sustainable practices in order to determine how much the organisation contributes to environmental sustainability.

Specific Objectives

Specific objectives outline the primary aim of a study. Typically, general objectives provide the basis for establishing specific goals. Essentially, specific objectives are general objectives that have been divided into smaller, logically interconnected goals. They assist you in defining the who, what, why, when, and how of your project. It's much easier to develop and carry out a research plan once you've identified the main goal.

Take, for example, a study of an organisation's contribution to environmental sustainability. The specific goals will be as follows:

- Determine how the organisation has changed its practises and adopted new solutions throughout its history.
- To determine the impact of new practices, technology, and strategies on overall effectiveness.

1.1.2: Characteristics of Research

- The research should concentrate on the most pressing issues.
- The investigation should be methodical. It emphasises the importance of following a structured procedure when conducting research.
- The research should follow a logical pattern. The scientific researcher cannot make much progress in any investigation without manipulating ideas logically.
- The study should be condensed. This means that a researcher's findings should be made available to other researchers so that they don't have to repeat the same research.
- The findings should be repeatable. This states that results from earlier studies have to be verifiable in new locations, environments, with fresh people, or at a different time.
- The research ought to be successful. The fact that a single research topic may spark the creation of several others is one of its most significant features.
- Action-oriented research is required. In other words, it should aim to find a solution that will allow its findings to be implemented.
- The research should take an integrated multidisciplinary approach, which means it will require research approaches from multiple disciplines.
- At all stages of the study, all parties involved (from policymakers to community members) should be invited to participate.
- The research has to have an easy-to-understand design and be conducted within a specific timeframe.
- As little money as feasible should be spent on the research.
- Formats that are most helpful to administrators, decision-makers, company managers, or community members should be used to deliver the study findings.

1.1.3 Differences b/w Research Methods and Research Methodology

Although the names sound similar, both Research methods and Research methodology are different, as explained below:

Research Methods

The various procedures, schemes, steps, and algorithms used in research are known as research methods. The term “research methods” refers to all of the methods used by a researcher during a research study. They’re primarily planned, scientific, and value-agnostic. Observations, theoretical procedures, experimental studies, numerical schemes, statistical approaches, and so on are all examples of these. We can use research methods to collect samples, data, and come up with a solution to a problem. Business and scientific research methods, in particular, demand explanations based on collected facts, measurements, and observations, rather than solely on reasoning. They only accept explanations that can be verified through experiments.

Research Methodology

A systematic approach to solving a problem is known as research methodology. It is a science that studies how research should be conducted. Research methodology is essentially the procedures by which researchers go about their work of describing, explaining, and predicting phenomena. It can also be defined as the study of methods for gaining knowledge. Its goal is to provide a research work plan.

1.2 Research Process

In the early decades human inquiry was primarily based on the examination of one’s own conscious thoughts and feelings that means the observation of any one and understanding through logical discussion to seek the truth. This process served as a well-established conceptual framework for comprehending the world and was accepted for a century. An essential component of the investigation process was the knowledge seeker. With time, this part was changed. The Scientific method introduced several major components in research procedures like Objectivity.

1.2.1 Research Criteria and Research Process

Research Criteria:

- The goal of the study should be made very clear, and common principles should be applied.
- There should be enough description of the research process. in such a way as to preserve the integrity of the work previously completed while enabling another researcher to carry out the research for future development.
- To get unbiased outcomes, the procedural design of the research should be carefully prepared.
- Any shortcomings in the technique design should be fully disclosed by the researcher, along with an assessment of how they could affect the results.
- The analytic techniques employed should be suitable, and the data should be sufficiently analysed to demonstrate its relevance. Verification of the data’s legitimacy and dependability is necessary.

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- Only conclusions that are bolstered by the study findings and for which the facts offer a sufficient basis should be drawn.
- If the researcher is experienced, has a good research reputation, and is a person of integrity, greater trust in the research is warranted.

Research Process:

Planning must be done methodically at every step of the marketing research process. The methods at each step are systematic, meticulously recorded, and, to the greatest degree possible, time-scheduled. In order to examine theories or preconceived beliefs, marketing research employs the scientific method, which involves gathering and analysing data.

For reliable information that accurately reflects the scenario, marketing research should be carried out impartially. The study should be impartial and devoid of the personal or political prejudices of the researcher or management, even if the researcher's research philosophy will always have an impact on the findings.

For instance, we could discover that the project's scope is insufficient, that our topic is too wide and has to be limited, that there aren't enough information resources accessible, or that the material we gather contradicts our thesis.

There are nine main steps of the research process that are followed at the time of designing a research project. They are as follows.

- Step 1: Problem Definition
- Step 2: Development of an Approach to the Problem
- Step 3: Research Design Formulation
- Step 4: Field Work or Data Collection
- Step 5: Data Preparation and Analysis
- Step 6: Report Preparation and Presentation

1.2.2: Approaches of Research: Deductive, Inductive, Quantitative and Qualitative

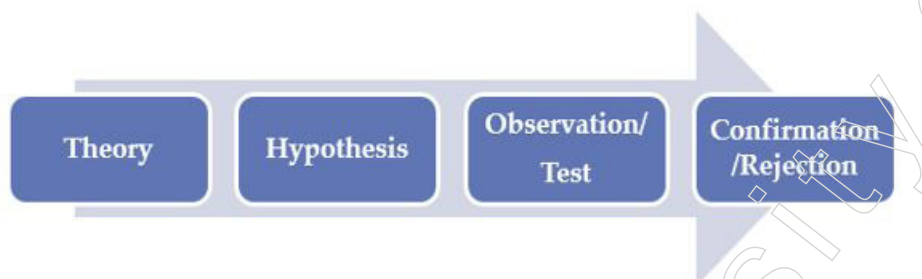
Deductive: Deductive approach starts with developing a hypothesis based on existing theory, and then prepares a research strategy for testing the hypothesis. To reason deductively is to go from the specific to the general. In many circumstances, a theory or case study that suggests a causal relationship or correlation may be accurate. The term "deductive approach" refers to a method of formulating hypotheses or drawing conclusions from theoretical ideas. The deductive method is concerned with drawing conclusions from statements or premises. By "testing against observations," deductive reasoning begins with a pattern, whereas induction methodology "begins with observations and seeks to find a pattern within them."

Benefits of Deductive:

- Probability to clarify the relationships between variables and concepts.
- Probability to measure concepts in a quantitative manner.
- Probability to generalize research findings to a certain extent.

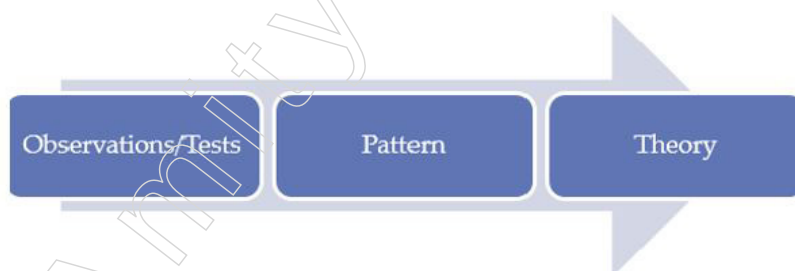
This approach mainly follows the given stages, like

1. Deducing hypothesis from theory
2. Formulating hypothesis in operational terms and proposing relationships between two specific variables
3. Testing hypothesis with the help of relevant methods like some quantitative methods such as regression and correlation analysis, mean, mode and median etc.
4. Examining the result and decide whether it is confirming or rejecting and it is important to compare research findings with the literature review findings
5. Modifying theory in instances when hypothesis is not confirmed.



Inductive: Inductive approach or inductive reasoning, starts with observations and theories are proposed which are related to the conclusion of the investigation due to the findings. It entails looking for patterns in observations and creating theories—or explanations—for those patterns by formulating a number of conjectures. In inductive studies, at the beginning of the research neither theories nor hypotheses would apply and the researcher is free to alter the direction for the study after the research process had commenced.

Using an inductive method to formulate research questions and objectives does not entail ignoring theories. By finding patterns and linkages in the data set, this method aids in the generation of interpretations from which a theory may be developed. This approach is mainly based on learning from experience. Previous patterns, resemblances and regularities are observed in order to reach conclusions or to generate theory.

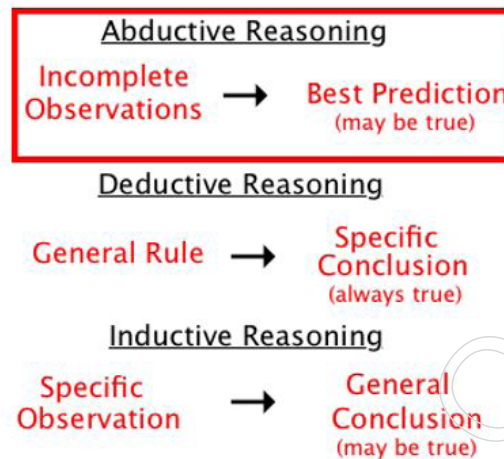


Abductive Reasoning or Abductive Approach

Abductive reasoning is set to address weaknesses associated with deductive and inductive approaches. Specifically for deductive reasoning 'how to select theory to be tested via formulating hypotheses' and for Inductive reasoning "no amount of empirical data will necessarily enable theory-building". It acts as a third alternative, overcoming these weaknesses via adopting a pragmatist perspective.

The figure below illustrates the main differences between abductive, deductive and inductive reasoning:

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It is clear that abductive reasoning is similar to deductive and inductive approaches in a way that it is applied to make logical inferences and construct theories.

Abductive approaches begin with “surprising facts” or “puzzles,” and the goal of the research process is to figure out how to explain them. A researcher may come into “surprising facts” or “puzzles” when they come across an empirical phenomenon that the current theories are unable to explain. In that approach, researchers search for the ‘best explanation among many alternatives to choose. Researchers can combine both, numerical and cognitive reasoning for explaining ‘surprising facts’ or ‘puzzles’.

Qualitative research: It is a non-statistical process of inquiry. It helps with in-depth understanding of problems or issues in their natural settings. It is highly based on the researchers’ background and the sample-probing questions they employed. Usually, there are restrictions on the sample size in between a minimum of 6 and a maximum of 10 people. Open-ended questions are asked (to gather as much information as possible from the sample) in a manner that encourages answers that lead to another question or a group of questions. The below methods are used for qualitative research:

1. One-to-one interview
2. Focus groups
3. Ethnographic research
4. Content/Text analysis
5. Case study research

Quantitative research: A systematic approach to gathering data and interpreting it to make inferences is known as quantitative research. This approach gathers and analyses data using a statistical and computational procedure. The main focus of quantitative data is numbers. A greater population is involved because more individuals will contribute more data which helps to obtain more accurate results. This research uses close-ended questions because the researchers are typically looking to gather statistical data. It involves use of data collection tools like online surveys, questionnaires, and polls. There are several ways to distribute questionnaires or surveys. Surveyors can reach a larger number of respondents or smaller focus groups with the use of online surveys for a variety of research purposes.

Advocacy/participatory Approach to Research

Researchers occasionally use an advocacy or participation strategy that ignores the needs or circumstances of members of vulnerable or marginalised groups. It is

frequently referred to as emancipatory when a researcher seeks to improve the lives of the research subjects. It's hardly an impartial position. Researchers want their work to either directly or indirectly lead to reform, and in order to prevent further marginalisation of the group they are studying, they engage them in the research process from the beginning to the end.

The researchers might not take the impartial stance that is often necessary in scientific study. This might entail living with the research participants (the co-researchers) or engaging in casual interactions. The results of the research can be communicated in a more intimate manner, frequently in the exact words of the study subjects.

1.2.3 Research Proposal and Aspects

A research proposal is designed to persuade others that we have an important research project and that we are capable of finishing it given our work schedule and level of expertise. Essentially, a research proposal needs to encompass all the essential elements of the research procedure and provide adequate details for the readers to evaluate the suggested investigation.

We need to answer the following questions about our study field and methodology: what we want to do, why we want to do it, and how we are going to do it.

Why is good preparation needed?

Good preparation of a research proposal is necessary such as:

This is important for grant of application in a competitive environment. Funding is very competitive.

It assists the researcher to formulate the project, to plan, to perform and to monitor the same.

A research proposal is comprised of:

- ❖ Title
- ❖ Introduction
- ❖ Literature review
- ❖ Methodology
- ❖ Plan - time frame and schedule of activities (Gantt chart)
- ❖ Budget
- ❖ Details of research team (signed CV)

Title

A study proposal's name has to be clear and descriptive. Make it as entertaining as it is educational. A strong title not only draws the reader in, but also makes the proposal seem appealing to them.

Introduction

Through the introduction researchers have to produce the required skeletal structure for the issues in research. One of the major challenges in drafting proposals is figuring out how to structure the research topic.

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A broad overview of the issue area, with an emphasis on a particular research question, usually opens an introduction. The rationale or reason for the suggested study comes next. The following topics are often covered in the introduction:

- Describe the research challenge, sometimes known as the study's goal.
- Describe the study's aims. Both broad and particular objectives should be specified.
- Give a detailed explanation of the proposed study's justification and why it is worthwhile.
- Give a brief explanation of the main concerns and minor difficulties that the project will identify.
- Determine the study's main independent and dependent variables.
- Describe the study's hypothesis.
- To ensure that the research has a clear focus, take attention while defining the boundaries or restrictions of the planned study.

Goals of the study

- ❖ It ought to be made explicit
- ❖ Be succinct and straightforward.
- ❖ It ought to be quantifiable and doable.

Literature Review

The literature review's goal is to give readers enough background knowledge.

The review should be succinct, highlight pertinent related research that has been done or is being done (citations should be given), and demonstrate that the review committee is aware of the numerous ongoing studies.

The literature review fulfils a number of crucial purposes:

- ❖ Acknowledges the contributions of those who have prepared the way for the planned research.
- ❖ Give evidence that you understand the research problem.
- ❖ Exhibits familiarity with the research question's theoretical and methodological underpinnings.
- ❖ Proves that you can analyse and assess pertinent literary works.
- ❖ Shows that you can pull information from several sources and put it all together.
- ❖ Offers novel theoretical understanding or creates a brand-new model to serve as the basis for our research.
- ❖ Ensures that our reader understands how our planned research would cover a big gap in the literature or solve a key theoretical difficulty.

Methodology

Our strategy for solving the research question. It will outline the steps we need to take to finish the project and serve as our work plan.

The sections describing the methods are provided below.

- How was the study conducted? Was it a survey or an experiment in a controlled environment? Do we go with a descriptive, cross-sectional, or case-control design?

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- Choosing the site for the study
- The people who will take part in our research are called subjects or participants. What type of process or methodology do we employ for sampling? We need to settle on the criteria for what counts as inclusion and what does not.
- We need to determine the sort of study we are undertaking in order to calculate the sample size. Calculating the sample size may be done using one of numerous formulae.
- Research tools—What kinds of surveys and evaluation tools do you employ? What makes them your top pick? How credible and legitimate are they?
- How will you go about gathering data for your study? Just what are the steps? When will it be finished?
- Plans for data processing and coding, software to be used (e.g., Statistical Package for the Social Sciences / SPSS, EPI-INFO, etc.), statistical techniques to be chosen, confidence levels, significance levels, etc. are all part of data analysis and interpretation. It is also helpful to include hypothetical tables describing the data that needs to be examined.
- For ethics reasons, we need to get approval from the Ethical Committee at both our workplace and the location where we intend to do the research. It all depends on the kind of research we want to undertake and the location where we are employed.

Plan

The proposed research should be planned out in detail, including the time range and activity schedule.

Make sure you factor in time for:

- ❖ acquiring the necessary supplies and equipment for the study
- ❖ carrying out the study
- ❖ analysing the data
- ❖ composing the project report

In order to effectively monitor the project, the activity schedule is crucial. Major actions and their associated deadlines should be detailed. Using a Gantt chart is the most efficient method for creating the schedule of activities.

Budget

It is critical to ask for a sufficient funding for the research you intend to carry out:

- Provide a total and yearly breakdown of the budget needed
- Give appropriate estimates of costs depending on the different areas, e.g.: travel and transportation, consumables, salaries, services, rentals, equipment, utilities, repairs, etc.
- Provide adequate justification, especially for costly items

Research Team

- Determine which areas of knowledge are necessary for your research.
- Make sure to include the resumes of all the important researchers.
- Collect written consent from all team members before proceeding.

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- Define roles for all researchers and make sure they have sufficient knowledge.

The common errors in proposal writing are as:

- ❖ Goals are overly general or excessively lofty.
- ❖ The goals do not correspond with the study's title or problem description.
- ❖ Missing references and literature review
- ❖ Lack of sufficient details on approach
- ❖ Too ambitious in terms of both time and schedule
- ❖ There is no rationale for Excessive or insufficient budget requests

1.2.4: Literature Survey and Review

Introduction

We humans are the most intelligent creatures in the universe. Did you know that? Our exceptional intelligence allows us to make use of information that has been passed down through the generations. There are three equally important stages of human understanding -

- namely preservation,
- transmission,
- and advancement.

Research contributes to the expansion of human understanding, which in turn allows for the creation and transmission of an up-to-date knowledge reservoir. Continuously striving to add to the enormous reservoir of knowledge in every conceivable sector, humans construct upon the documented and amassed information of the past, allowing the human race to progress. As researchers, it is our responsibility to check the literature to make sure that similar questions have been answered. We must have a thorough understanding of all prior work, theory, and research that pertains to the research topic. In order to guarantee such familiarity, we must do an exhaustive evaluation of Research and theoretical literature.

Meaning of Review of Literature:

The term "review" means "to organise the knowledge of the specific research area to create a knowledge pool so that our study adds on to and enriches the field of research." The term "literature" stands for "the knowledge of a specific area of investigation, related to a given discipline, which includes theoretical, research-oriented, and practical studies." In this way, reviewing the literature is like building a new knowledge pool in a specific area or expanding an existing one; it adds to and improves research domains.

Importance of Review of Literature

We have to learn why and how a successful review of literature would help us about our Research.

- Review of Literature provides researchers with theories, ideas, explanations or hypotheses that may prove useful in the formulation of a new problem
- Review of Literature guides researchers on the availability of adequate evidence that solves the problem sufficiently without the need of further investigation - this initiative avoids the replication of Research

- Review of Literature serves as prominent sources for hypothesis - researchers can formulate research hypotheses based on available studies
- Review of Literature suggests data sources, methodology, and statistical techniques apt for the solution of the research problem
- Review of Literature helps researchers locate comparative data and findings useful in the correct interpretation of results .

Sources of Literature

What documents would you consider the sources of literature?

Sources	Examples
Lists and indices of books, textbooks, and handbooks	Cumulative Book Index
	• Book Review Index
	• Books Review Digest
	• Subject Guide to Books
Periodicals (publication issued in successive volumes, at regular intervals)	National Union Catalogue
	Yearbook
	• Documents
	• Almanacs
	• Cumulative Book Index
	• International abstracts
	• Journals
	• Newspapers
Encyclopaedias (reservoirs of concise and convenient sources of information on various subjects; these include biographies and illustrations)	• Magazines
	• International Index to Periodicals
	The Encyclopaedia Britannica
	• World Book Encyclopaedia
	• The Encyclopaedia of Science Fiction
	• The Atlas of the Prehistoric World
	• Encyclopaedia of Cryptography and Security
	• Encyclopaedia of Public Health
	• Encyclopaedia of General Science

Apart from these, newspapers, dissertations, abstracts, theses, online libraries, etc. serve as good sources of review of literature.

Recommended Practices in Review of Literature:

Next, we are going to study about a few recommended practices in literature review.

- Get an overall view by referring to a generic source like a good textbook. This will help us grasp the theoretical concepts like the meaning, nature, and importance of the concept. We will also get a clear idea about the variables that we would use to formulate the hypothesis of our research problem.
- On developing a clear insight about the research problem, we have to review the empirical research related to our research problem. During this phase, our chief concern would be to gather specific details of the field of investigation by using good handbooks, recommended encyclopaedias, and good abstracts for this purpose.

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- b) Perform methodical and comprehensive Conducting research using study resources obtained from both online and offline libraries. Commence gathering citations from the educational index. Employ a meticulous level of accuracy while duplicating or inputting such references.
- d) Maintain a methodical approach by consistently and thoroughly documenting information on cards with precision. Document citations with comprehensive bibliographic details. Document citations on the front side of these cards and make annotations on the back sides.

1.2.5: Errors in Research

Various types of errors in Research are described below.

1. Population Specification

Population specification errors occur when the researcher does not understand whom they should survey. Because there are more than one people who may consume the product, but only one who purchases it, or they may miss a segment looking to purchase in the future.

How to avoid this: In the beginning analysis of customers buying the items have to be done. This is necessary as it will allow us to identify the appropriate customers and ways to satisfy them.

2. Sampling and Sample Frame Errors

When the wrong subpopulation is used to select a sample then survey sampling and sample frame errors occur, or because of variation in the number or representativeness (is a subset of a population that seeks to accurately reflect the characteristics of the larger group) of the sample that responds, but the resulting sample is not representative of the population concern.

Basically some element of sampling error is unavoidable, but it can be predictable. For instance, in the 1936 presidential election between Roosevelt and Landon, the sample frame was from car registrations and telephone directories. The researchers failed to realize that the majority of people that owned cars and telephones were Republicans, and wrongly predicted a Republican victory.

Example: Assume that we obtained a random sample of 5000 individuals from the whole adult population of India in order to determine their preferences about entertainment. Upon examination, it was determined that the composition of the group consisted of 70% females. The entertainment tastes of females would have more significance, making it difficult to accurately generalise to the overall adult population in India. The homogeneity of a sample has an impact on the sampling error (the quality of consisting of parts or people that are similar to each other) of the population being studied and sampled from and by the size of the sample.

How to avoid this: Since it cannot be avoided completely, then we should have multiple people reviewing our sample to account for an accurate representation of our target population. We can also increase the size of our sample so we get more participants to survey.

3. Selection

When a sample is selected using a non-probability method, selection error can occur. This error arises when respondents choose to self-participate in a study and only

those interested respond, potentially introducing inherent bias. Selection error can also occur when respondents irrelevant to the study participate or when there's a bias in the way participants are grouped.

Example: Interviewers conducting a small intercept study tend to select respondents who are most accessible and agreeable, often choosing friends and associates who share characteristics with the target population.

To avoid this, extra efforts should be made to ensure broad participation. A typical survey process includes pre-survey contact to request cooperation, actual surveying, and post-survey follow-up. If there is no response, a second survey request is sent, and interviews may be conducted using alternative modes such as telephone or in-person interviews.

4. Non-responsive

Non-response errors occur when there is a discrepancy between the original sample and the selected sample. This discrepancy arises when potential respondents are either not contacted or refuse to respond. The primary issue here is the absence of data rather than inaccurate data. Non-response errors can be categorized into two types:

- (a) Non-contact errors: These occur when the respondent cannot be reached, possibly because they are not at home.
- (b) Refusal errors: These occur when the respondent chooses not to respond to specific items or multiple items in the questionnaire. Topics such as monthly household income, religion, sex, and politics are more likely to result in item refusal.

Example: In the event of conducting telephone surveys, certain respondents may be deemed inaccessible due to their absence from home during the first call or subsequent call-backs. The remaining individuals have either relocated or are now absent from their residences during the survey period. Respondents who are not at home are usually younger and do not have young children.

Additionally, a larger percentage of these families have women who are employed compared to households where someone is at home. Individuals who have relocated or are now absent throughout the survey period exhibit a greater degree of geographical mobility compared to the general population. The majority of surveys may predict inaccuracies resulting from the lack of touch with respondents. Online surveys can prevent this problem by distributing them over email, therefore excluding respondents who are not at home.

How can one prevent or steer clear of this situation? When gathering replies, it is important to ensure that our initial respondents are actively engaged. If they do not respond initially, we should employ follow-up surveys and alternative methods of reaching out to them. Additionally, we may employ several channels, such as face-to-face interactions, web surveys, or SMS, to effectively communicate with our target audience.

5. Measurement

The measurement method can have its issues as well. These issues include the difference between the data produced and the data required by the analyst. There are multiple minute factors which can contribute to such issues.

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To circumvent this, we should double-check all measures to make sure they are accurate and make sure that our observers and measurement takers are knowledgeable with the experiment's parameters.

Although it may not be possible to totally prevent all of these mistakes, being aware of them will make things easier. Make sure you're doing all possible to avoid these typical pitfalls the next time we begin a study endeavour by using this blog as a checklist.

There are so many errors like

Surrogate Information Error:

That kind of error arises from either the inability or the unwillingness of the respondent to provide the needed information, in other words, information is obtained from substitutes rather than original sample. It is also described as the fact that the data the researcher is trying to collect does not match up with what is needed to address the issue.

Experimental Error: It can be defined as the non-correspondence of the "true impact of" and the "impact at-tribute to" the independent variable

For example

A research looking into cancer causes might indicate that bread is the culprit as it was consumed by all the patients studied. What followed was: By manipulating one or more independent variables and keeping an exogenous variable under control, researchers hope to quantify the effect of these variables on a dependent variable in an experiment.

1.3: Types of Research Design

Research design is a critical component of research methodology, serving as the blueprint for the collection, measurement, and analysis of data. It outlines the procedures and techniques that researchers follow to ensure their studies are reliable, valid, and capable of addressing their research questions or hypotheses. Understanding the different types of research designs is essential for choosing the most appropriate approach for a given study. Following are various types of Research Design.

1.3.1 Pure and Applied

Applied Research

Applied research is a type of study that aims to solve a specific problem or offer novel solutions to issues that affect a person, a group, or a society. Because it involves the practical application of scientific methods to everyday problems, it is often referred to as a scientific method of inquiry or contractual research.

When conducting applied research, the researcher pays special attention to identifying a problem, developing a research hypothesis, and then conducting an experiment to test these hypotheses. In many cases, empirical methods are used in this research approach to solve practical problems.

Because of its direct approach to finding a solution to a problem, applied research is sometimes considered a non-systematic inquiry. It's a type of follow-up research that

digs deeper into the findings of pure or basic research in order to validate them and use them to develop innovative solutions.

Applied Research Example in Business

- Applied research to improve the hiring process in a company.
- Applied research aimed at improving workplace efficiency and policies.
- Workplace skill gaps are being addressed through applied research.

Applied Research Examples in Education

- An investigation into how to improve teacher-student engagement in the classroom.
- An investigation into how to improve a school's student readiness.
- A research project aimed at piquing students' interest in mathematics

Pure/Fundamental Research or Basic Research

Basic research is a method of investigation that seeks to clarify fundamental aspects of a topic, phenomena, or natural law. The goal of this type of research is to advance knowledge rather than to solve a specific problem.

Pure research or fundamental research are other terms for basic research. Between the late 19th and early 20th centuries, the concept of basic research arose as a means of bridging the gaps in science's societal utility.

Basic research can be exploratory, descriptive, or explanatory in nature; however, in many cases, it is explanatory. The primary goal of this research method is to collect data in order to improve one's understanding, which can then be used to propose solutions to a problem.

Basic Research Example in Education

- How do humans retain memory?
How do different teaching methods affect students' concentration in class?

1.3.2: Causal and Correlational

Explanatory research, also known as causal research, is used to determine the scope and nature of cause-and-effect relationships. Causal research can be used to assess the effects of specific changes on existing norms, processes, and so on.

Causal studies examine a situation or a problem in order to explain the patterns of relationships between variables. In studies using a causal research design, the most common primary data collection method is experiments.

Examples of Causal Research (Explanatory Research)

- To determine the effects of foreign direct investment on Taiwanese economic growth.
- To investigate the impact of rebranding initiatives on customer loyalty.
- To determine the nature of the impact of work process re-engineering on employee motivation levels.

Did you know? Conceptual Research is related to certain abstract ideas or theories that are often applied by philosophers to develop new concepts or to rework on the existing ones.

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Without manipulating either variable, researchers use a correlational study design to find out how two variables relate to one another. Its stated goal is to determine if either: A favourable relationship. When one variable changes, the other follows suit. Naturalistic observation, survey research, and archive research are the three main approaches to correlational studies. Types vary in function and provide different advantages and disadvantages.

1.3.3: Crosssectional and Longitudinal

Cross-sectional study

The longitudinal and cross-sectional studies are both observational studies. This means that researchers record data about their subjects without tampering with the research environment. We would simply measure the cholesterol levels of daily walkers and non-walkers, as well as any other.

Characteristics of interest to us, in our study. We would not persuade non-walkers to start walking or advise daily walkers to change their habits. In a nutshell, we'd try not to get in the way.

A cross-sectional study is distinguished by the ability to compare different population groups at a single point in time. Consider it like taking a photograph. The findings are based on whatever fits into the frame.

To return to our previous example, we could compare cholesterol levels in daily walkers in two age groups, over 40 and under 40, to cholesterol levels in non-walkers in the same age groups. We might even create gender subgroups. We would not, however, consider past or future cholesterol levels because they would be outside the scope. We'd only examine cholesterol levels at a single point in time.

A cross-sectional study design has the advantage of allowing researchers to compare multiple variables at once. For instance, with minimal extrapolation, we may examine the correlation between walking and cholesterol levels and demographic variables such as age, gender, income, and education level.

Cross-sectional studies, on the other hand, may not provide conclusive evidence of cause-and-effect relationships. This is because such studies provide a snapshot of a single moment in time and do not consider what occurs before or after the snapshot. As a result, we can't say for sure whether our daily walkers had low cholesterol levels before starting their exercise routines or if the daily walking behaviour helped to lower cholesterol levels that were previously high.

Longitudinal Study

A longitudinal study is observational, just like a cross-sectional one. As a result, researchers do not interfere with their subjects once more. A longitudinal study, on the other hand, involves researchers making multiple observations of the same subjects over a long period of time, sometimes many years.

A longitudinal study has the advantage of allowing researchers to detect changes in the characteristics of the target population at both the group and individual level. The important thing to remember is that longitudinal studies go beyond a single point in time. As a result, they can create event sequences.

To return to our example, we could examine the change in cholesterol levels among women over 40 who have walked every day for the past 20 years. The

longitudinal study design would take into account cholesterol levels at the start of a walking programme and as the programme progressed. As a result of its scope, a longitudinal study is more likely than a cross-sectional study to suggest cause-and-effect relationships.

In general, the design should be driven by the research. However, the progression of the research can sometimes aid in determining which design is best. Longitudinal studies take longer to complete than cross-sectional studies.

1.3.4: Experimental, Semi-experimental and Non-Experimental

Experimental

Experimental research is a type of study that employs a scientific approach to manipulate one or more control variables of the research subject(s) and then measure the impact of the manipulation on the subject. It is well-known for allowing the manipulation of control variables.

Even though it can be difficult to execute, this research method is widely used in a variety of physical and social science fields. They are far more common in information systems research than in library and information management research within the information field.

When the goal of the research is to trace cause-and-effect relationships between defined variables, experimental research is usually used. The type of experimental research chosen, on the other hand, has a significant impact on the experiment's outcomes.

Semi Experimental: The prefix quasi means "similar to." As a result, quasi-experimental research is research that resembles experimental research but isn't actually experimental. Participants are not randomly assigned to conditions or orders of conditions, despite the fact that the independent variable is manipulated (Cook & Campbell, 1979). 1st The directionality problem is eliminated in quasi-experimental research because the independent variable is manipulated before the dependent variable is measured. However, because participants are not assigned at random, there is a chance that other differences exist between conditions. Thus, quasi-experimental research does not eliminate the problem of confounding variables.

Example: Non-equivalent groups design just like we hypothesized a new after-school program which assured the students of higher grades. We split the class into two equal halves; one half gets the new programme and the other half doesn't.

In field settings where random assignment is difficult or impossible, quasi-experiments are most likely to be conducted. They're frequently used to assess the efficacy of a treatment, such as psychotherapy or an educational intervention. There are numerous types of quasi-experiments, but we will focus on a few of the most common here.

Non-Experimental:

Non-experimental research is defined as research in which no control or independent variable is manipulated. Researchers in non-experimental research measure variables as they occur naturally, without any further manipulation.

When the researcher doesn't have a specific research question about a causal relationship between two variables and manipulating the independent variable is impossible, this type of research is used. They're also useful for:

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- It is impossible to assign subjects to conditions at random.
- The subject of the study is a causal relationship, but the independent variable cannot be changed.
- The study is broad and exploratory in nature.
- The study focuses on a variable-to-variable non-causal relationship.
- Only a limited amount of information about the research topic is available.

1.3.5 Descriptive and Exploratory

Descriptive Research

Providing a detailed account of the researched population or phenomena is the goal of descriptive research. This approach to study places greater emphasis on the “what” than the “why” of a topic.

Without asking “why” an event happens, descriptive research mainly aims to describe the characteristics of a demographic group. To rephrase, it only “describes” the research topic without addressing the question of “why” it occurs.

For example, surveys related to frequency of shopping, food habits, product preference, etc. are examples of descriptive research. Example: AA group of market researchers aims at identifying the impulse buying trends among various households pan India. The researchers would focus on collecting data related to “what is the impulse buying pattern of Indian consumers” and the scope of their research would be limited to that. The research does not explain the underlying reasons behind such impulse buying practices or “why” such buying patterns exist. Here, the scope of the research is just to report the existence of such buying trends and not why do people resort to impulse buying. This is, hence, an ideal example of descriptive research.

Exploratory Research

The goal of exploratory research is to learn more about a topic that hasn’t been covered extensively in the past. Although it seldom produces a definitive outcome, exploratory research is typically conducted to enhance comprehension of the current issue.

For the purpose of developing a more targeted issue statement, researchers often resort to exploratory studies in order to better understand pre-existing phenomena. Using the study findings to explore connected issues to the research topic begins with a broad notion.

The process of exploratory research varies depending on the discovery of new data or insight. The results of this research, also known as interpretative research or grounded theory approach, provide answers to questions like what, how, and why.

Exploratory Research Example on Product Research

When developing a new product or service, companies conduct two types of research. The first is carried out prior to the development of the product, while the second is carried out after it has been developed.

The exploratory research conducted after product development will be the focus of our attention. It’s known as the beta testing stage of product development for tech products.

For example, if a new feature is added to an existing app, product researchers will want to see how well the feature is received by users. The research is not exploratory if the feature added to the app is something that already exists.

If Telegram adds a status feature to its app, for example, the app's beta research stage is not exploratory. This is because this feature is already available, and they can easily obtain sufficient information from WhatsApp.

When it comes to a new feature, such as Snapchat filters when they first launched, the research is instructive. A focus group of beta testers is used to conduct exploratory research in this case.

1.4: Research Modelling

Uses of Research Model

The following are the uses of Research Model:

- Research models help in identifying basic concepts.
- Research model helps in defining meanings with great precision.
- Research models are a great way to approach and simulate reality.

Importance of Research Model:

The importance of using a research model is highlighted below:

- Model building is an integral part of the research design because models guide both theory development and research design.
- Computers, biotech, and automation have given scientists a new level of respect thanks to models, which have also elevated them in the public, private, and military sectors.
- Additionally, social scientists rely heavily on models as they offer a structure for examining crucial issues.

1.4.1: Types of Models

Research Models are classified broadly into two types as mentioned below:

- Qualitative Research Model
- Quantitative Research Model

Qualitative Research Model

It involves non-numerical data collection and analysis in order to understand concepts, opinions and experiences. This helps to gather in-depth insights into a problem or develop new ideas for research. Anthropology, sociology, education, health sciences, history, and other fields in the social and humanities tend to make heavy use of qualitative research methods. Qualitative research helps to visualize how people can experience the world. While there are many approaches to qualitative research, they are less desirable as they are flexible and focus on retaining rich meaning when interpreting data.

Quantitative Research Model

It is about collecting and analysing numerical data. Used for locating and defining patterns and averages, this research model can make predictions, test causal

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relationships, and help to generate results to wider populations. Quantitative research finds a wide use in the natural and social sciences: biology, chemistry, psychology, economics, sociology, marketing, etc.

1.4.2: Model Building and Stages

Model building is a crucial aspect of research methodology, especially in fields such as data science, engineering, and social sciences. It involves creating abstract representations of real-world systems or phenomena to understand, analyze, and predict their behavior. The process of model building in research follows several essential stages to develop a reliable and valid model.

Firstly, the problem must be defined clearly by identifying the objectives and understanding the scope, objectives, and constraints of the study. This includes gathering relevant data and information to understand the context and requirements of the model.

Next, conceptualization involves developing a theoretical framework based on existing literature and theories related to the problem. This helps in formulating hypotheses and identifying key variables. A preliminary model is then created to outline the relationships between different variables and components, often represented diagrammatically.

The data collection stage involves identifying and gathering data from primary and secondary sources such as surveys, experiments, observations, and databases. Ensuring data quality is crucial, as the data must be accurate, reliable, and relevant to the research problem, often requiring data cleaning and preprocessing.

In the model specification stage, the conceptual model is translated into a mathematical or statistical form. This involves selecting appropriate modeling techniques and specifying equations or algorithms. Any assumptions made during this stage must be clearly stated, as they can impact the model's validity.

Model estimation involves using statistical or computational methods to estimate the parameters of the model, employing techniques such as regression analysis or machine learning algorithms. The model parameters are then calibrated to improve accuracy and fit to the observed data.

Validation of the model is crucial to ensure it is correctly implemented and free from logical or computational errors. The model's predictions are compared with real-world data to assess accuracy and reliability, which may involve cross-validation, sensitivity analysis, and robustness checks.

Once validated, the model is implemented to analyze the research problem, make predictions, or provide insights. This stage involves using the model for decision-making or further research. Optimization of the model is also necessary based on feedback and results, potentially iterating through previous stages to enhance performance.

Finally, the results are interpreted in the context of the research problem, understanding the implications and limitations. The entire modeling process, including the methodology, assumptions, data sources, and findings, is documented to ensure transparency and reproducibility. The results are then reported clearly and concisely through research papers, reports, or presentations, including visualizations and explanations of the model's implications.

This systematic process ensures the development of robust, reliable, and applicable models to real-world scenarios, providing valuable insights and aiding informed decision-making based on research findings.

1.4.3: Data Consideration and Testing

In today's digitalized world, we generate a massive amount of data every minute. The sheer volume of data produced makes it challenging to store, manage, utilize, and analyse effectively. Even large business enterprises struggle to find ways to make this vast amount of data useful. The rate at which data is being produced by large business enterprises is growing rapidly, increasing by 40 to 60% per year. Simply storing this massive amount of data is not beneficial unless it is used for specific research purposes. This is why organisations are exploring options like data lakes and big data analysis tools to help manage and leverage big data more effectively.

There are a few steps in considering data:

Proper understanding of Data: Understanding is the key to any solution. Data when comes in huge format is not only difficult to understand but store and manage. Thus data consideration is a huge task in itself. A close look needs to be given to the research that is being done and hence its answers will be easy to find out on detailed understanding.

- Nature, scope and object of enquiry: This constitutes the most important factor affecting the choice of a particular consideration of data. Data is to be considered on its nature, scope and object of enquiry, when the data is available for research, checking its sources and utilisation in the research it will be judged and considered.
- Time Factor: Time is important as there can be huge data which is available but there will certainly be a deadline for the research hence, firstly the data collection needs to be done timely and post the collection there has to be enough time for the consideration and testing.
- Precision Required: it is another factor that the collection of data is ongoing and post the collection. While considering the data the right precision is required so that it follows the same wave length of the research program.

1.4.4: Heuristic and Simulation Modelling

Heuristic Research Model

There are a wide range of qualitative research models available and one of the lesser-known models is the Heuristic research model. This research model was developed by Clark Moustakas (an American psychologist and researcher). The name Heuristic was derived from the Greek work 'Heuriskein' (which means discover, find). The research model has six phases

- ❖ Initial engagement
- ❖ Immersion
- ❖ Incubation
- ❖ Illumination
- ❖ Explication
- ❖ Creative synthesis

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Shelly Chaiken developed the heuristic-systematic model of information processing (HSM), which attempts to explain how people receive and process persuasive messages. 1st Individuals can process messages in one of two ways, according to the model: heuristically or systematically. Heuristic processing, on the other hand, entails the use of simplifying decision rules or “heuristics” to quickly assess the message content, whereas systematic processing entails the careful and deliberate processing of a message. This model’s guiding belief is that people are more likely to use heuristics instead of cognitive resources, which affects message intake and processing. The elaboration likelihood model, or ELM, is very similar to the HSM. Many of the same ideas and concepts are shared by both models, which were mostly created in the early to mid-1980s.

Simulation Research Modelling

By using statistical descriptions of the activities involved, stimulation models attempt to replicate the workings and logic of a real system. For example, a line might produce 1000 units per hour on average. If we assume this is always the case, we lose sight of what happens when there is a breakdown or a stoppage for routine maintenance, for example. When we consider the effect on downstream units, the effect of such a delay may be amplified (or absorbed).

‘Entities’ (e.g. machines, materials, people, etc.) and ‘activities’ are two types of entities in a simulation model (e.g. processing, transporting, etc.). It also includes an explanation of the logic that governs each activity. A processing activity, for example, can only begin when a certain quantity of working material, a person to operate the machine, and an empty conveyor to transport the product are all available. Once an activity has begun, the time it will take to complete it is calculated, which is frequently done using a sample from a statistical distribution.

Summary

- Authors and management experts have defined research in various ways.
- Typically, research is said to commence with a question or a problem.
- It is described as the creation of new concepts, methodologies, and understandings through the development of new knowledge or the innovative application of existing knowledge.
- This process often involves synthesizing and analysing previous studies to produce new and groundbreaking results.
- Through research, we can identify solutions to problems by employing systematic and scientific methods. This may involve experimentation or innovation, with “risk” highlighting the inherent dangers of discovery.
- Investigation might lead to analysis, and tests could be conducted to verify hypotheses. This work is distinct and unprecedented, offering new insights as a result.
- It could introduce a novel method or new data source, leading to significant breakthroughs or incremental improvements in current practices. Research is essential for progress, as it is fundamentally tied to “research and development.”

Glossary

- General Objectives: General objectives, also known as secondary objectives, provide a detailed view of a study’s goal.

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- **Specific Objectives:** Specific objectives outline the primary aim of a study. Typically, general objectives provide the basis for establishing specific goals.
- **Research Methods:** The various procedures, schemes, steps, and algorithms used in research are known as research methods.
- **Research Methodology:** A systematic approach to solving a problem is known as research methodology.
- **Deductive:** Deductive approach starts with developing a hypothesis based on existing theory, and then prepares a research strategy for testing the hypothesis.
- **Inductive:** Inductive approach or inductive reasoning, starts with observations and theories are proposed which are related to the conclusion of the investigation due to the findings.
- **Abductive Reasoning or Abductive Approach:** Abductive reasoning is set to address weaknesses associated with deductive and inductive approaches.

Check Your Understanding

1. What is the typical starting point of research?
 - a) A solution to a problem
 - b) A question or a problem
 - c) A hypothesis to be tested
 - d) A new methodology to be developed
2. What is a key characteristic of research that leads to breakthroughs or improvements in current practices?
 - a) It involves only analysing existing data
 - b) It introduces novel methods or new data sources
 - c) It relies solely on existing knowledge
 - d) It avoids taking risks in the discovery process
3. What is the primary purpose of specific objectives in a research study?
 - a) To outline the primary aim of the study
 - b) To provide a detailed view of the study's goal
 - c) To develop a research plan
 - d) To collect samples and data
4. What is a key characteristic of research methods?
 - a) They are primarily based on reasoning
 - b) They are not used to collect samples and data
 - c) They are only used in business research
 - d) They are planned, scientific, and value-agnostic
5. What is the relationship between general objectives and specific objectives?
 - a) General objectives are divided into specific objectives
 - b) Specific objectives are broader than general objectives
 - c) General objectives provide the basis for establishing specific objectives
 - d) General objectives are unrelated to specific objectives

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Exercise

1. What is research typically said to commence with, and what is its ultimate goal?
2. What is the difference between general objectives and specific objectives in research, and how do they relate to each other?
3. What are the key attributes that contribute to the credibility of a research study, and how do these ensure the reliability and validity of the findings?
4. How has the research process evolved from early human inquiry to modern marketing research, and what fundamental principles are integral to conducting impartial and reliable research today?
5. What are the key stages involved in the deductive approach to research, and how do these stages contribute to the development of hypotheses, testing, and refinement of theories?

Learning Activity

1. Suggest some ways to generalise research findings while maintaining the validity of the deductive approach?
2. Are there any specific techniques or methods that can aid in the collection and analysis of data for this research project?

Check Your Understanding- Answers

1. b)
2. b)
3. a)
4. d)
5. c)

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Module - II: Sampling

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Learning Objectives

At the end of this module, you will be able to:

- Analyse Data Collection
- Explain Questionnaire Designing
- Mention Sampling

Introduction

Sampling is a fundamental concept in research methodology that involves selecting a subset of individuals, items, or data points from a larger population to make inferences about that population. This process allows researchers to draw conclusions without having to study the entire population, which is often impractical or impossible.

Purpose of Sampling

The main purpose of sampling is to gather data in a way that is manageable and cost-effective while still allowing for accurate and reliable conclusions. By studying a representative sample, researchers can make generalizations about the broader population.

Types of Sampling Methods

1. Probability Sampling: Every member of the population has a known, non-zero chance of being selected. This method includes techniques such as:
 - ❖ Simple Random Sampling: Every member has an equal chance of being selected.
 - ❖ Stratified Sampling: The population is divided into subgroups (strata) based on specific characteristics, and samples are drawn from each stratum.
 - ❖ Systematic Sampling: Selecting every n th member from a list of the population.
 - ❖ Cluster Sampling: Dividing the population into clusters and randomly selecting entire clusters.
2. Non-Probability Sampling: Not every member has a known chance of being selected. This method includes techniques such as:
 - ❖ Convenience Sampling: Selecting individuals who are easily accessible.
 - ❖ Judgmental or Purposive Sampling: Selecting individuals based on the researcher's judgment about who would be most beneficial to the study.
 - ❖ Quota Sampling: Ensuring certain characteristics are represented in the sample to the same proportion as in the population.
 - ❖ Snowball Sampling: Existing study subjects recruit future subjects from among their acquaintances.

Importance of Sample Size

The size of the sample is critical to the accuracy and reliability of the research findings. A larger sample size generally leads to more accurate results, reducing sampling

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error and increasing the study's statistical power. However, practical considerations such as time, cost, and resource availability also influence the sample size.

Steps in the Sampling Process

1. Define the Population: Clearly identify the group from which the sample will be drawn.
2. Choose the Sampling Frame: Create a list or database of the population members.
3. Select the Sampling Method: Decide whether to use probability or non-probability sampling.
4. Determine the Sample Size: Use statistical formulas and consider practical constraints to decide how many individuals or items to include.
5. Collect the Sample: Implement the chosen method to gather the sample.
6. Analyze the Sample: Conduct the research and analyze the data from the sample to draw conclusions about the population.

Challenges in Sampling

- Sampling Bias: Occurs when certain members of the population are systematically excluded or overrepresented.
- Non-response: When selected participants do not respond, potentially skewing the results.
- Sampling Error: The difference between the sample results and the actual population characteristics due to the sample not being perfectly representative.

T2.1: Data Collection

Information gathered through original or first-hand research is referred to as primary data.

2.1.1 Introduction to Data Collection

Surveys and focus group discussions, for example. Secondary data, on the other hand, is information that has already been gathered by someone else. For instance, internet research, newspaper articles, and company reports.

Any study's goal determines whether primary or secondary data will be collected. For example, if a company wants to enter the women's apparel market in India and wants to know the size of the market, it can use secondary data such as industry reports and newspaper articles, whereas if it wants to learn about consumer preferences for a new type of fabric or style, it must conduct primary research.

Primary data collection is usually more expensive and time-consuming than secondary data, but it serves a specific purpose and helps to eliminate biases.

2.1.2 Data Collection Methods

Primary & Secondary Data

There are two types of data:

1. Primary Data and
2. Secondary Data

Primary Data: Source data is what this word refers to. Referred to as “primary data,” this information is culled from unprocessed, first-hand sources such as surveys, experiments, and experimental results. Primary data refers to first-hand information that has been meticulously gathered with a specific goal in mind. Meaningful data collection occurred at the source level. More trustworthy, genuine, and impartial primary data has not been released just yet.

Secondary data: This means that the data has previously been gathered and evaluated by someone other than the user; in other words, the data is not new. Public and private databases, books, periodicals, newspapers, trade journals, and other written works are common places to find secondary data.

Collection of Primary Data

Experiments, surveys, observations, and interviews with participants are all examples of primary data collection methods used in descriptive and experimental research. The following are a few ways to get primary data:

1. Method of Observation Research in the field of behavioural science frequently makes use of it. When conducted methodically with a predetermined goal in mind and given to appropriate checks and controls, observation takes on the role of a scientific tool and the researcher’s primary means of gathering data according to this approach.
 - a. Structured (Descriptive) and Unstructured (Exploratory) Observation: When an observation is characterized by a careful definition of the units to be observed, the style of the observer, the conditions of the observation, and the selection of pertinent data, it is called structured observation. In contrast, when these characteristics are not predetermined or absent, it is referred to as unstructured observation.
 - b. Participant, Non-participant, and Disguised Observation: In participant observation, the observer becomes more or less a member of the group being observed. In non-participant observation, the observer remains detached from the group under observation. Disguised observation occurs when the observer’s presence is unknown to the people being observed.
 - c. Controlled (Laboratory) and Uncontrolled (Exploratory) Observation: Uncontrolled observation takes place in a natural setting, while controlled observation occurs according to pre-arranged plans, often involving experimental procedures.

Benefits of Primary Data

- ❖ Subjective bias is eliminated.
- ❖ Data is not affected by past behaviour or future intentions.
- ❖ Natural behaviour of the group

Drawbacks

- ❖ Expensive methodology.
- ❖ Information provided is limited.
- ❖ Unforeseen factors may interfere with the observational task

Interview Methods:

This method of data collection involves the presentation of oral verbal stimuli and

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the recording of responses in terms of oral-verbal answers. This can be accomplished through various types of interviews, such as:

- **Personal Interview:** This requires an interviewer to ask questions in a face-to-face interaction with the respondent. It can be further categorized into:
 - **Direct Personal Investigation:** The interviewer collects information directly from the subjects involved.
 - **Indirect Oral Examination:** The interviewer cross-examines individuals who are presumed to have knowledge about the issue.
- **Structured Interviews:** These interviews use predetermined questions and standardized recording techniques.
- **Unstructured Interviews:** These interviews do not follow a predetermined set of questions and are characterized by a flexible approach to questioning.
- **Focused Interview:** This type of interview aims to concentrate on the respondent's specific experiences and their impacts. The interviewer may ask questions in any order or manner to explore the respondent's reasons and motives.
- **Clinical Interviews:** These interviews delve into the underlying feelings, motives, and life experiences of individuals, using these aspects to gather information as directed by the interviewer.
- **Non-Directive Interview:** In this type, the interviewer's role is to encourage the respondent to discuss a given topic with minimal direction.

Advantages

- ❖ More information and in depth can be obtained.
- ❖ Samples can be controlled.
- ❖ There is greater flexibility under this method
- ❖ Personal information can as well be obtained.

An unstructured interview might help prevent misunderstandings.

Limitations

- ❖ It is an expensive method.
- ❖ More time consuming.
- ❖ Possibility of imaginary info and less frank responses.
- ❖ High skilled interviewer

Telephonic Interview:

In this method, the interviewer calls potential interviewees and asks them questions or solicits their thoughts over the phone.

2. Questionnaire

In this approach, respondents are given a questionnaire to read, comprehend, and respond to independently before sending it back. A form or series of forms with multiple-choice questions printed or written in a certain sequence make up the test. An expert-led "pilot study" can help identify potential problems with the survey's questions and methods before the actual survey takes place.

Fundamental to a good questionnaire-

- It needs to be brief and uncomplicated.
- It is important to digest questions in a logical order.
- You should stay away from jargon and technical words.
- It is essential to include control questions in order to assess the respondent's dependability.
- Please ensure that there is sufficient room for responses.
- The questionnaire has to be filled out according to the following brief instructions.
- In order to grab the respondent's attention, the physical appearances—the paper quality, colour, etc.—must be good.

Benefits

- ❖ Independent of the interviewer's prejudice.
- ❖ There is sufficient time for respondents to provide responses.
- ❖ Respondents are very accessible and easy to get in touch with.
- ❖ More trustworthy results can be obtained with more samples.

Drawbacks

- ❖ Return rate of properly completed questionnaires is low.
- ❖ After a query is sent, the ability to edit it is lost.
- ❖ It cannot be changed after it has been sent.
- ❖ The possibility of unclear responses not being provided.
- ❖ Protracted and laborious procedure

3. Schedules

The enumerations designated for the purpose fill out the schedules, which is a distinction from the questionnaire technique of data collecting. Enumerations clarify the goals of the study, which can assist respondents remember their responses and eliminate confusion. A good enumerator is knowledgeable about the tasks at hand, has the necessary skills, is trustworthy, and is patient. This data is useful for in-depth investigations, but it's pricey.

Collection of secondary data

There are a number of places where a researcher can find secondary data. Publicly available data and unpublished data are both forms of secondary data.

Publicly available data can be found in:

- ❖ Publications of the government.
- ❖ Technical and trade journals.
- ❖ Reports of various businesses, banks etc.
- ❖ Public records.
- ❖ Statistical or historical

Memoirs, journals, unpublished biographies, and other works may include information that has not been published. It is important to verify that secondary data meets certain criteria before utilising it:

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1. Data reliability—Who took the data? Where did you hear this? Which approach? Is it now? Could there be prejudice? What about precision?
2. Data appropriateness — After studying the original inquiry's scope and character, thoroughly examine the data to ensure it is suitable.
3. Adequacy- If the data are not accurate or pertain to a scope that is too limited or too broad for the current investigation, we will say that they are insufficient.

Types of Data Collection Methods

There are various methods to collect the two sources of data (Primary and Secondary) as mentioned and explained below:

Primary data is gathered from first-hand experience and has never been used before. The data gathered through primary data collection methods is highly accurate and specific to the research's purpose.

Quantitative and qualitative data collection methods are the two types of primary data collection methods.

Quantitative Methods:

Time Series Analysis

A time series is a sequential order of values of a variable at equal time intervals, also known as a trend. An organisation can forecast demand for its products and services for the future using patterns.

Smoothing Techniques

Smoothing techniques can be used when the time series lacks significant trends. They get rid of the random variation in historical demand. It aids in the identification of patterns and demand levels in order to forecast future demand. The simple moving average method and the weighted moving average method are the two most common methods for smoothing demand forecasting techniques.

Barometric Method

Researchers use this method, also known as the leading indicators approach, to predict future trends based on current events. When past events are used to forecast future events, they are referred to as leading indicators.

Qualitative Methods:

Surveys

Surveys are used to gather information about the target audience's preferences, opinions, choices, and feedback on their products and services. Most survey software allows you to choose from a variety of question types.

You can also save time and effort by using a pre-made survey template. By changing the theme, logo, and other elements, online surveys can be tailored to fit the brand of the company. They can be distributed via a variety of channels, including email, website, offline app, QR code, social media, and so on. You can choose the channel based on the type and source of your audience. Survey software can generate various reports and run analytics algorithms to uncover hidden insights once the data has been collected. A survey dashboard can show you statistics such as response rate, completion rate, demographic filters, export and sharing options, and so on. Integrating

survey builder with third-party apps can help you get the most out of your online data collection efforts.

Polls

One single or multiple choice question is asked in a poll. You can use polls when you need to get a quick pulse on the audience's feelings. It is easier to get responses from people because they are short in length.

Online polls, like surveys, can be integrated into a variety of platforms. After the respondents have responded to the question, they can see how their responses compare to those of others.

Interviews

The interviewer asks the respondents questions either face-to-face or over the phone in this method. In face-to-face interviews, the interviewer asks the interviewee a series of questions in person and takes notes on the answers. If meeting the person is not possible, the interviewer can conduct a telephonic interview. When there are only a few respondents, this method of data collection is appropriate. If there are many participants, repeating the same process is too time-consuming and tedious.

Delphi Technique

Market experts are given the estimates and assumptions of forecasts made by other industry experts in this method. Based on the information provided by other experts, experts may reconsider and revise their estimates and assumptions. The final demand forecast is based on the consensus of all experts on demand forecasts.

Focus Groups

A focus group is a small group of people (around 8-10 members) who meet to discuss the problem's common areas. Each person expresses his or her viewpoint on the subject at hand. The discussion among the group members is moderated by a moderator. The group comes to an agreement at the end of the discussion.

Questionnaire

Questionnaires can be either open-ended or closed-ended and are often printed out. Respondents are expected to provide responses that are grounded on their understanding and expertise of the subject matter. The survey includes the questionnaire, but the latter's ultimate purpose might not always be a survey.

Sources of Secondary data:

Generally speaking, there are two types of secondary data sources:

- Published Sources
- Unpublished Sources

Published Sources:

Statistical statistics, including the following, are published by international, governmental, and local agencies:

International Publications:

We have international institutions and bodies like I.M.F, I.B.R.D, I.C.A.F.E and U.N.O who occasionally publish occasional reports on statistical and economic matters.

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Official Publications of Central and State Governments:

Various branches of the federal and state governments make periodical reports available to the public. They gather all the supplementary data. The Reserve Bank of India Bulletin, Indian Trade Journal, Statistical Abstracts of States, Census of India, and Agricultural Statistics of India are some of the important publications in this category.

Semi-Official Publications:

Example: Municipal Corporations, District Boards, Panchayats, etc. that will publish reports relating to different matters of public concern.

Publications of Research Institutions:

Indian Statistical Institute (I.S.I), Indian Council of Agricultural Research (I.C.A.R), Indian Agricultural Statistics Research Institute (I.A.S.R.I), etc., publish the findings of their research programs.

Publications of various Commercial and Financial Institutions:

Reports of various Committees and Commissions appointed by the Government:

Such as the Raj Committee's Report on Agricultural Taxation, Wanchoo Committee's Report on Taxation and Black Money, etc. are also important sources of secondary data.

Journals and Newspapers:

Journals and Newspapers are the powerful sources from which data is obtained. Current and important materials on statistics and socio-economic problems are provided by journals and newspapers like Economic Times, Commerce, Capital, Indian Finance, Monthly Statistics of trade etc.

Unpublished Sources:

Records kept by public and commercial agencies, the work of many academics at colleges and universities, and so on are all examples of such sources of information.,

Precautions to be taken in the use of Secondary Data:

Since secondary data has already been collected, it is wise to review it to make sure it is accurate. When using this data, the investigator must use greater caution. "Secondary data should not be accepted at their face value," as Prof. Bowley correctly says. A number of factors, including the information collectors' inherent biases and prejudices, an insufficient sample size, faulty definitions, inaccurate calculations, and problems with replacement, can lead to inaccurate results. Even without error, such data still can be unsuitable for enquiry purposes. E According to Prof. Simon Kuznets (which is of importance), "the degree of reliability of secondary source is to be assessed from the source, the compiler and his capacity to produce correct statistics and the users also, for the most part, tend to accept a series particularly one issued by a government agency at its face value without enquiring its reliability".

Thus we need to follow some of the given factors:

The Suitability of Data:

This is possible by judging the scope and nature of the present enquiry with the original one. For example, if we are looking for trends in retail prices while the data provided is meant for wholesale prices, then it is of no use.

Adequacy of Data:

Once it is ensured that the data is suitable for investigation, it should be checked for the purpose of present analysis. Geographical area for the original enquiry can be studied in this respect along with the time for which we are getting the data. Using the previous example, the data collected would be useless if our goal was to analyse the retail pricing trend in India as a whole, rather than just the state of UP.

Reliability of Data:

This issue concerns whether research findings can be applied to a larger group than those who participated in the study. To put it another way, would similar results have been obtained if a different group of respondents or a different set of data points had been used? Is the information obtained from these 40 people sufficient to conclude how the entire sales force feels about company policies, for example, if 40 salespeople out of a 2,000-person corporate sales force participate in a research study focusing on company policy? Would the results be the same if the study was repeated with 40 different salespeople?

The main goal of reliability is to ensure that the data collection method produces consistent results. This can be measured in some types of research by having different researchers use the same methods to see if the results can be replicated. If the results are similar, the data collection method is most likely reliable. The scientific research method includes ensuring that research can be replicated and produces similar results.

The following should be kept in mind when altering primary data:

- ❖ It must be comprehensive in every way
- ❖ It must be precise
- ❖ It must be consistent
- ❖ It must be constant throughout.

The following forms of editing are necessary for data to have the aforementioned qualities:

2.1.3: Tabulating and Validating Data**Meaning of Tabulation**

To make comparisons and statistical analyses easier, numerical data is often tabulated in rows and columns. It aids in further statistical analysis and interpretation and makes comparison easier by bringing relevant data together.

Put simply, tabulation refers to the process of arranging structured data into a tabular format. Its complexity, duality, or simplicity are all dependent on the kind of classification.

Tabulation of Data Collected

The objectives of tabulation of collected data are as follows:

- Orderly arrangement of data in columns and rows
- To bring out essential features of the data collected
- To simplify the data collected
- Conserves space

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- Ease of comparison
- Summation of items
- Enables easy detection of errors and omissions
- Facilities Statistical Computation

Definition of Data Validation:

As defined by the United Nations Economic Commission of Europe (UNECE 2013), data validation is an activity aimed at verifying whether the value of a data item comes from the given (finite or infinite) set of acceptable values.

Data validation means checking the accuracy and quality of source data before using, importing or otherwise processing data. Different types of validation can be performed depending on destination constraints or objectives.

Data Validation is a form of Data Cleansing

To illustrate the point, an email query may be set up to automatically verify the validity of the provided email address. You may use the country code as the basis for a phone number query to verify if the number is valid.

Reasons for performing Data Validation:

The following are the reasons for validating data:

- It is cost-effective because the collection of datasets saves the right amount of time and money.
- Because it removes duplicates from the entire dataset, it is simple to use and compatible with existing processes.
- With improved information collection, data validation can directly aid in business improvement.
- It's made up of a data-efficient structure that provides information from a standard database as well as a cleaned dataset.

Types of Validity

There are several types of validity, each addressing different aspects of the research process. Here are the main types of validity:

1. Construct Validity

Construct validity assesses whether a test or instrument truly measures the theoretical construct it is intended to measure. It encompasses:

- ❖ Convergent Validity: The degree to which two measures of constructs that theoretically should be related are in fact related.
- ❖ Discriminant Validity: The extent to which a measure does not associate too strongly with measures from which it should differ.

2. Internal Validity

Internal validity refers to the extent to which a study can establish a causal relationship between variables. High internal validity indicates that the study design effectively rules out alternative explanations for the observed relationships.

- ❖ Threats to Internal Validity: Include confounding variables, selection bias, and measurement errors.

3. External Validity

External validity is the extent to which the findings of a study can be generalized to other settings, populations, or times.

- ❖ Population Validity: The degree to which the study results can be generalized to a larger population.
- ❖ Ecological Validity: The extent to which the findings can be applied to real-world settings.

4. Criterion Validity

Criterion validity evaluates how well one measure predicts an outcome based on another measure (the criterion).

- ❖ Concurrent Validity: The degree to which a test correlates with a measure that has been previously validated.
- ❖ Predictive Validity: The extent to which a test predicts future performance on a related measure.

5. Content Validity

Content validity examines whether a test represents all aspects of the given construct. This type of validity is often assessed qualitatively by experts.

- ❖ Face Validity: A subset of content validity, referring to the degree to which a test appears to measure what it claims to measure, based on subjective judgment.

6. Face Validity

Face validity is a superficial measure of whether a test appears to measure what it is supposed to measure. It is often considered the weakest form of validity, but it can be important for ensuring that stakeholders accept the measure.

7. Statistical Conclusion Validity

Statistical conclusion validity pertains to the extent to which conclusions about the relationship among variables based on the data are correct and reliable. It deals with issues like whether the statistical tests used are appropriate and whether the assumptions of these tests are met.

2.2 Questionnaire Designing

A questionnaire is a research tool designed to collect data from respondents through a series of questions or prompts.

Key Features of a Questionnaire:

1. Variety of Question Types:

- ❖ Open-Ended Questions: These questions allow respondents to provide detailed, qualitative responses.
- ❖ Closed-Ended Questions: These questions offer predefined answer choices, enabling quantitative data collection.

2. Purpose:

- ❖ Quantitative Data: Collected through closed-ended questions, this data is numerical and can be statistically analysed to draw conclusions.

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- ❖ Qualitative Data: Collected through open-ended questions, this data provides in-depth, specific explanations and insights.
3. Common Misconception:
- ❖ A questionnaire is often mistaken for a survey. While both are used to collect data, a survey refers to the entire process of data collection, which may include a questionnaire as one of its tools. Many people use the terms interchangeably, but they are not the same.

2.2.1 Steps in Constructing a Questionnaire

Designing a questionnaire involves multiple steps, each requiring meticulous attention to detail to ensure the collection of usable data. Researchers aim to create a survey that is easy to understand and complete, minimising the risk of respondents dropping out or providing inaccurate answers due to confusion.

Steps Involved in Questionnaire Design

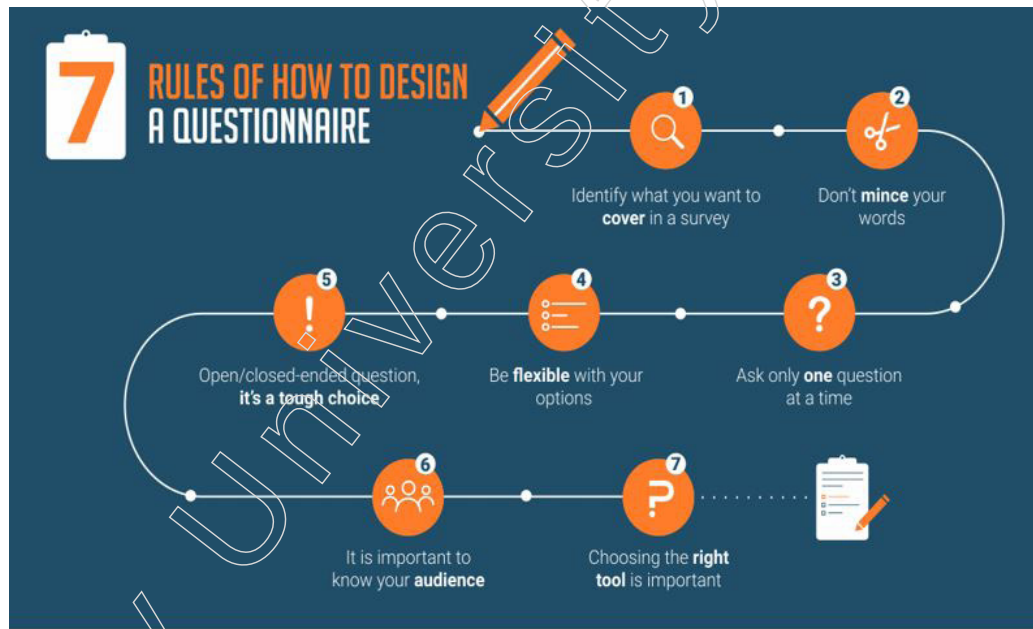


Image Source: <https://medium.com/@markquepro/7-platinum-rules-to-design-a-questionnaire-3aad839f604>

1. **Identify the Scope of Your Research:** Clearly define the purpose and topics to be covered in your questionnaire. Understanding the scope is crucial as it guides the entire design process.
2. **Keep It Simple:** Use clear, straightforward language. Avoid complex words or phrases to ensure respondents understand the questions accurately, preventing skewed data.
3. **Ask Only One Question at a Time:** Avoid double-barrelled questions that address multiple issues at once. Each question should focus on a single topic to ensure clarity and accurate responses.
4. **Be Flexible with Your Options:** Provide a variety of answer choices, including an "other" option, to accommodate respondents who may not find a suitable answer among the predefined options.
5. **Choose Between Open-Ended and Closed-Ended Questions:** Decide on the type of

questions based on the data you need. Open-ended questions are ideal for detailed responses, while closed-ended questions are better for specific, easily analysable data.

6. **Know Your Audience:** Tailor the questionnaire to your target audience's language and comprehension level. Avoid technical jargon and ensure the questions are relevant to their experiences and background.
7. **Choose the Right Tool:** Utilise a reliable survey platform, such as Questioner, which offers advanced features and a variety of templates. This can streamline the questionnaire creation process and enhance the quality of data collected.
8. **Save Personal Questions for Last:** Place sensitive or demographic questions at the end of the questionnaire. This approach increases the likelihood of completion, as respondents are more comfortable by the end of the survey.

2.2.2 Types of Questions

Utilising various question types in a questionnaire can enhance respondent engagement and yield more comprehensive data. Below are some commonly used types of questions:

1. **Open-Ended Questions:** Allow respondents to answer in their own words, providing qualitative data without restrictions. Useful for collecting detailed, nuanced responses and exploring respondents' thoughts and feelings.
2. **Dichotomous Questions:** A close-ended question type that typically offers two choices, such as "yes" or "no." Effective for obtaining straightforward, binary responses and validating specific information.
3. **Multiple-Choice Questions:** Present respondents with a list of options to select from, either as single-select or multi-select. Consists of a question stem, correct answer(s), incorrect answers, close alternatives, and distractors. Ideal for gauging opinions, preferences, and gathering categorical data.
4. **Scaling Questions:** These questions leverage measurement scales (nominal, ordinal, interval, ratio) to quantify responses. Useful for measuring intensity, frequency, and other scalable attributes.

Types:

- a. **Rank Order Questions:** Ask respondents to rank items in a specific order.
 - b. **Likert Scale Questions:** Measure attitudes or opinions on a scale (e.g., 1 to 5).
 - c. **Semantic Differential Scale Questions:** Use bipolar adjectives to measure respondents' attitudes.
 - d. **Stapel Scale Questions:** Rate items based on a single adjective.
5. **Pictorial Questions:** Similar to multiple-choice questions, but the answer choices are images instead of text. Enhances engagement by allowing respondents to quickly select answers without overthinking, leading to more accurate data.

2.2.3 Format of Questionnaire

Size

- **Compactness:** The questionnaire should be smaller than a schedule in size.
- **Appropriate Dimensions:** Ensure the length and breadth are suitable.

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- **Page Limit:** Ideally, it should not exceed two or three pages, depending on the nature of the research.

Appearance

- **Quality:** Use high-quality paper and printing for a professional look.
- **Layout:** Ensure an attractive and organised layout to enhance respondent engagement.

Clarity

- **Simplicity:** Questions should be short, clear, and easy to understand.
- **Expression:** Use concise and precise language to avoid ambiguity.

Sequence

- **Logical Order:** Arrange questions based on their importance and logical flow.
- **Preference:** Prioritise questions to maintain a smooth progression.

Communicability

- **Engagement:** Design questions to keep respondents interested and motivated to complete the questionnaire.

Span

- **Conciseness:** Keep questions as brief as possible without sacrificing clarity.
- **Length:** Ensure the overall questionnaire is not too long to maintain respondent interest and minimise dropout rates.

2.3: Sampling

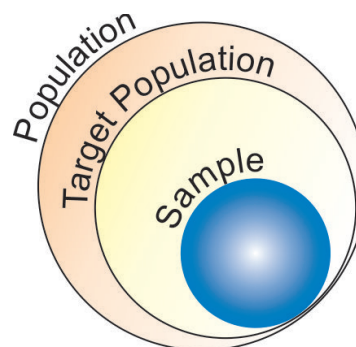
Introduction to Sampling

"Segment of population that is selected for investigation."

Bryman & Bell–

"Some of the elements of population"

Cooper & Schindler–



A sample, in research terms, is a group of people, objects, or items selected for measurement from a larger population. To ensure that the findings from the research sample can be applied to the entire population, the sample should be representative of the population.

What is the purpose of sampling?

By using inferential statistics to a small subset (or “sample”) of a larger population, we may learn about its features and extrapolate those findings to the whole.

We obtain a sample of the population for a variety of reasons, including the fact that it is rarely practical and almost never cost-effective.

Many Populations are Quite Large.

- Inaccessibility of some populations – Access to some populations is so difficult that only a sample can be used. Prisoners, people with severe mental illness, and disaster survivors are just a few examples. And so on. The inaccessibility could be due to a lack of funds, time, or simply access.
- Destructiveness of observation- Sometimes just observing a product's desired characteristic destroys it for its intended use. Quality control is a good example of this. For example, a fuse must be destroyed to determine its quality and whether it is defective. As a result, if you tested all of the fuses, they'd all blow up.
- Accuracy and sampling - A sample of the study population may be more accurate than the entire population. A population that has been incorrectly identified can provide less reliable data than a sample that has been carefully selected.

2.3.1: Sampling Plan

Definition of Sampling Plan

The phrase “sampling plan” describes the framework that research projects rely on to carry out their investigations. Included are instructions for selecting a representative sample from the population, the appropriate sample size, and the category to question.

The research begins with the sampling plan, which encompasses the following important decisions:

i. Choose the population

The primary and initial step in developing a sample strategy is deciding which subset of the target population will be polled.

ii. Determine the Sample Size

Decide how many items will be surveyed as part of the sample (the second option in the sampling strategy). Since “the larger the sample size, the more is the reliability” in general, researchers endeavour to cover a wide number of samples.

iii. Decide the Sampling Procedure

Choosing the sampling technique, or the approach that may be employed to ensure that every object in the population has an equal chance of being picked, is the last option that completes the sampling strategy. In most cases, researchers will employ probability sampling to pick items that are good representations of the sample.

2.3.2: Sampling Frame

Definition of Sampling Frame

A sampling frame is a list or database from which a sample can be used. In market research terms, a sampling frame is a database of potential respondents that can be drawn from, to invite to take part in a given research project.

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Steps involved in Sampling Process

The following are the series of steps that are involved in the sampling process:

- Classify the people
- Ascertain the sample interval
- Make your choice of sampling strategy
- Pick a number for the sample
- Complete the sampling procedure.

2.3.3: Sample Selection Methods

The sample selection methods can be broadly classified into:

- Probability Sampling
- Non-Probability Sampling

Methods of sampling that use the use of probability to choose individuals or items from a population.

There are several probability sampling methods available for examination and consideration. These are some of the most renowned choices.

1. Simple random sampling

Simple random sampling ensures that each element in the population has an equitable probability of being chosen as a component of the sample. It is like randomly selecting a name from a hat. Anonymizing the population by assigning a unique number to each object or person and then randomly selecting numbers is a method for conducting simple random sampling.

Conducting simple random sampling is a straightforward and cost-effective method that completely eliminates any potential bias in the sample process. Nevertheless, it lacks researcher control and may result in the selection of unrepresentative categories by chance.

2. Systematic sampling

Systematic sampling, often referred to as systematic clustering, involves randomly selecting only the first item, while the subsequent items are picked in a predetermined pattern. Subsequently, a rule is implemented to choose every n th item or person thereafter.

Despite the presence of randomness, the researcher has the ability to set the interval at which items are chosen, ensuring that the selections are not unintentionally grouped together.

3. Stratified sampling

Stratified sampling is a method that entails the random selection of individuals from pre-established groups. Researchers find it advantageous to possess knowledge about the target population, enabling them to strategically divide it (stratify) in a manner that is logical and relevant to the research.

For instance, when studying travel patterns within a population, it would be beneficial to distinguish between those who possess or have access to a car and those who rely on public transportation.

Stratified sampling offers advantages, but it also raises the issue of how to divide a population into strata, increasing the potential for bias.

4. Cluster sampling

Cluster sampling involves randomly selecting groupings, rather than individual units, from the target population. These groupings might potentially consist of pre-established entities, such as individuals residing in specific postal codes or students affiliated with a particular academic year.

Cluster sampling can be conducted by either picking the entire cluster or, in the case of two-stage cluster sampling, randomly selecting the cluster itself and then randomly selecting inside the cluster.

Non-probability Sampling Methods

Non-probability sampling methods do not have the same advantages in eliminating bias as probability sampling methods. However, there are situations where non-probability sampling methods are used for their convenience or simplicity. Below are some examples of non-probability sampling methods and their respective mechanisms.

1. Convenience sampling refers to a non-probability sampling technique where individuals are selected based on their easy accessibility and convenience.

Individuals or components within a sample are chosen based on their accessibility. If you are doing a research survey at a university, a convenience sample may include students or co-workers who are present on campus and have spare time, and are willing to participate in your questionnaire.

This type of sample can be valuable, particularly when used as an initial or preliminary measure, but it will inevitably introduce considerable bias.

2. Quota sampling is a method of selecting a sample that ensures representation of different subgroups in proportion to their presence in the population.

Similar to the probability-based stratified sampling method, this strategy seeks to ensure representation throughout the target population by selecting survey participants based on certain groups or criteria. For instance, your quota may encompass a specific count of males and females, or individuals belonging to particular age ranges or ethnicities.

During the selection process, bias might arise, such as volunteer bias, which may lead to a sample that is skewed towards those having enough free time and a specific interest in participating. It is possible that bias is inherent in the process of selecting categories for quotas by researchers.

3. Purposive sampling is a method of selecting individuals or elements for a study based on specific criteria or purpose.

Researchers deliberately select participants for the sample based on their expertise and comprehension of the study subject or their objectives. Also referred to as judgement sampling, this method is unlikely to yield a sample that accurately represents the entire population. However, it is a rapid and relatively simple approach to obtain a variety of outcomes or replies.

4. Snowball or referral sampling refers to a method of selecting participants for a study where existing participants are asked to recommend or refer to other potential participants.

Notes

This strategy involves recruiting individuals to participate in a sample and requesting them to invite their acquaintances to join. These acquaintances are then encouraged to invite their own friends and family members, creating a chain of invitations. The involvement spreads throughout a network of interconnected individuals, resembling the momentum of a snowball moving downhill.

This approach might be advantageous when the researcher possesses limited knowledge about the target group and lacks convenient means to contact or reach them. Nevertheless, it will result in prejudice, such as the exclusion of isolated individuals within a community or a tendency to favour specific age or interest groups that engage in self-recruitment.

2.3.4: Probability Sampling Techniques

The various Probability sampling methods are given below:

- Simple random sampling
- Systematic Sampling
- Stratified Sampling
- Cluster Sampling
- Multistage Sampling

Simple Random Sampling

There is a known and equal probability of selection for each constituent of the population in this probability sample.

E.g. Population = Total students in AMITY (591)

Sample size = 25 students

$$\text{Probability of selection} = \frac{\text{Sample size}}{\text{Population size}}$$

$$= 25/591$$

$$= 0.042 \text{ or } 4.2\%$$



Systematic Sampling

A systematic sample obtained by using a predetermined skip interval on a sample frame.

Formula to calculate skip interval-

Formula to calculate skip interval-

$$\text{Skip interval (k)} = \frac{\text{Population sample frame (N)}}{\text{Sample Size (n)}}$$

E.g. Population = Total students in AMITY (591)

Sample size = 25 students

$$\text{Skip interval (k)} = \frac{\text{Population sample frame (N)}}{\text{Sample Size (n)}}$$

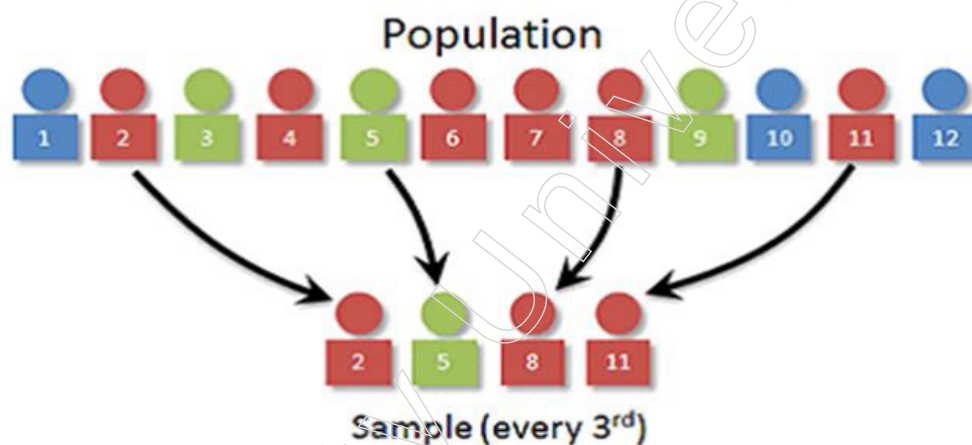
$$= 591 / 25$$

$$= 23.64 = 23$$

Select any number randomly between 1 – 23, and then select the remaining 24 numbers with a gap of 23 numbers.

$$k = 23$$

Select any number randomly between 1 – 23, and then select the remaining 24 numbers with a gap of 23 numbers.



Stratified Sampling

Stratified sampling is a method of probability sampling where the population is split into distinct subgroups or strata, and random samples are taken from each of these subgroups.

E.g. Population = Total students in JSPM (591)

Sample size = 25 students

Sub groups-

MBA = 272 MMM = 42

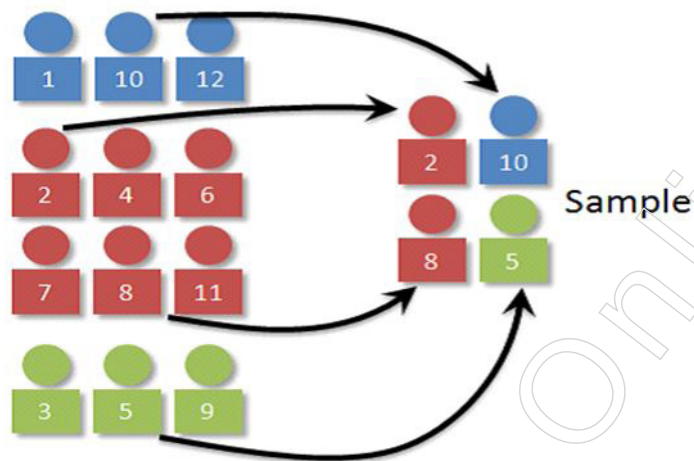
MPM = 59 DBM = 92

DIEM = 37 MCA = 59

MCM = 30

Notes

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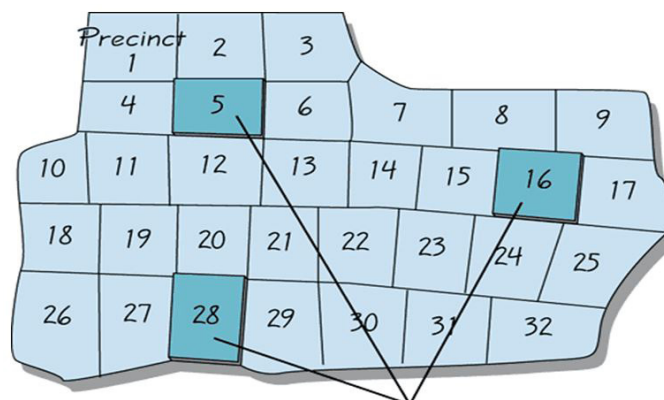
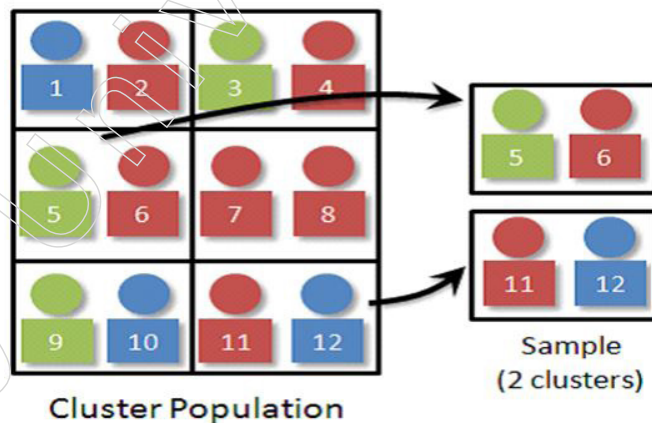


Cluster Sampling

A probability sampling technique in which the population is divided into several small sub groups and some groups selected randomly for study.

e.g. MBA = 272 MMM = 42
 MPM = 59 DBM = 92
 DIEM = 37 MCA = 59
 MCM = 30

MCM = 30 and MMM = 42 randomly selected from above groups.



Interview all voters in shaded precincts.

2.3.5: Non - Probability Sampling Techniques

Also known as 'deliberate,' 'purposeful,' or 'judgement' sampling. The items have been chosen with care.

The researcher's choice of elements to include in the sample. There is no guarantee that each element will be given an equal chance to be chosen as a sample.

Convenience Sampling

A non-probability sampling technique where researcher use any readily available individuals as participants

Most cheapest & easiest to conduct

Least reliable

Researcher is free to select anybody as sample

Purposive or Judgmental Sampling

A non-probability sample that conforms to certain criteria. The units or elements are purposely selected.

It is a way of selecting items for a study based on the specific objectives of the research. Purposive sampling entails the examination of either the complete population of a restricted group or a subset of a population. Similar to other non-probability sampling techniques, purposive sampling does not yield a sample that accurately represents a wider population. However, it can be very suitable for studying specific and well-defined groups such as organisations, communities, or other constrained populations.

Quota Sampling

A purposive sampling where relevant characteristics are used to stratify the samples.

This is useful to increase the representativeness of samples

MBA = 272	46%	12
MMM = 42	7%	2
MPM = 59	10%	2
DBM = 92	16%	4
DIEM = 37	6%	2
MCA = 59	10%	2
MCM = 30	5%	1
Total = 591	100%	25

Snowball Sampling

A non-probability sampling in which subsequent participants are referred to by current sample elements.

Initial individuals are discovered/identified

These initial individuals refer to others who are similar to them and so on. Like a snowball gathers subjects as it rolls along.

Notes

2.3.6: Sampling and Non-Sampling Errors

Definition of Sampling Error

Sampling error pertains to discrepancies between the sample and the population only due to the particular observations chosen for the sample.

Expanding the sample size will diminish this particular form of mistake.

Definition of Non-Sampling Error:

Non-sampling errors are more serious, and they occur as a result of data acquisition errors or incorrect sample observation selection.

Non-Sample Errors

Non-sample errors can be classified into:

- Non-response Error
- Response Error

Non-Response Error

A non-response error arises when the units chosen as part of the sampling process fail to provide a complete or partial answer.

Response Error

A response or data error refers to any consistent bias that arises throughout the process of data collection, analysis, or interpretation.

Response errors occur due to the following factors:

- Respondent Error (example: lying, forgetting, etc.,)
- Interviewer Errors
- Recording Errors
- Poorly designed questionnaires.
- Measurement errors.

Summary

- Information gathered through original or first-hand research is referred to as primary data. Surveys and focus group discussions, for example.
- Secondary data, on the other hand, is information that has already been gathered by someone else. For instance, internet research, newspaper articles, and company reports.
- Any study's goal determines whether primary or secondary data will be collected.
- For example, if a company wants to enter the women's apparel market in India and wants to know the size of the market, it can use secondary data such as industry reports and newspaper articles, whereas if it wants to learn about consumer preferences for a new type of fabric or style, it must conduct primary research.
- Designing a questionnaire involves multiple steps, each requiring meticulous attention to detail to ensure the collection of usable data.

- Researchers aim to create a survey that is easy to understand and complete, minimising the risk of respondents dropping out or providing inaccurate answers due to confusion.
- A sample, in research terms, is a group of people, objects, or items selected for measurement from a larger population.
- To ensure that the findings from the research sample can be applied to the entire population, the sample should be representative of the population.

Glossary

- **Primary Data:** Source data is what this word refers to. Referred to as “primary data,” this information is culled from unprocessed, first-hand sources such as surveys, experiments, and experimental results.
- **Secondary Data:** This means that the data has previously been gathered and evaluated by someone other than the user; in other words, the data is not new.
- **Direct Personal Investigation:** The interviewer collects information directly from the subjects involved.
- **Unstructured Interviews:** These interviews do not follow a predetermined set of questions and are characterised by a flexible approach to questioning.
- **Structured Interviews:** These interviews use predetermined questions and standardised recording techniques.
- **Focused Interview:** This type of interview aims to concentrate on the respondent's specific experiences and their impacts. The interviewer may ask questions in any order or manner to explore the respondent's reasons and motives.
- **Clinical Interviews:** These interviews delve into the underlying feelings, motives, and life experiences of individuals, using these aspects to gather information as directed by the interviewer.

Check Your Understanding

1. Which of the following is a primary source of data?
 - a) Internet research
 - b) Company reports
 - c) Surveys
 - d) Newspaper articles
2. What is a key characteristic of primary data?
 - a) It is always free to access
 - b) It is collected specifically for the current research purpose
 - c) It is usually less reliable than secondary data
 - d) It is always quantitative
3. Which step in constructing a questionnaire involves defining the purpose and topics?
 - a) Keep It Simple
 - b) Identify the Scope of Your Research

Notes

- c) Be Flexible with Your Options
- d) Know Your Audience
- 4. When designing a questionnaire, why is it important to avoid double-barrelled questions?
 - a) They require complex language
 - b) They are too short
 - c) They address multiple issues at once
 - d) They only offer open-ended responses
- 5. Why should sensitive or demographic questions be placed at the end of a questionnaire?
 - a. To ensure the survey is concise
 - b. To make respondents more comfortable completing the survey
 - c. To prioritize the most important questions
 - d. To allow for better statistical analysis

Exercise

1. Differentiate between primary and secondary data with examples.
2. Explain why primary data collection might be preferred over secondary data in certain research scenarios.
3. Why is it important to tailor a questionnaire to the target audience's language and comprehension level?
4. Describe the benefits of using a reliable survey platform, such as QuestionPro, when designing a questionnaire.
5. List and describe three key steps in constructing a questionnaire.

Learning Activity

1. Consider a research study aiming to understand consumer preferences for a new type of smartphone.
2. Why might researchers choose to use a sample instead of studying an entire population? Provide at least three reasons and explain each one briefly.

Check Your Understanding- Answers

1. c)
2. b)
3. b)
4. c)
5. b)

Further Readings and Bibliography

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Notes

Module - III: Data Collection and Experiments

Learning Objectives

At the end of this module, you will be able to:

- Discuss Design of Experiments
- Explain Data Collection
- State Basic Principles

Introduction

Experimental research is carried out using a systematic manner, where one set of variables is kept constant while the other set of variables is measured as the focus of the experiment. A highly effective quantitative research technique, experimental research is particularly useful when the researcher does not have enough information to arrive at meaningful conclusions and have to carry out experiments to prove facts.

3.1: Design of Experiments

Experimental research, thus, helps a researcher make important decisions. It is considered successful only when the researcher ensures that a change in the dependent variable is wholly due to a change in the independent variable

3.1.1: Design of Experiment: Objectives and Strategies

The results of experimental research should help a researcher establish a causal relationship between the dependent and independent variables in a given phenomenon. Effects are observed, during an experiment, due to the underlying cause. Experimental research is conducted when:

- There exists a constant and consistent relationship between the cause and the effect
- Time is a crucial factor for establishing a relationship between cause and effect
- The factors that drive the success of experimental research are:
 - ❖ A control group refers to a group of study participants who are similar to the experimental group but are not subject to the experimental research criteria.
 - ❖ The experimental group refers to the subset of study participants who are subject to the norms and protocols of experimental research.
- Manipulable variable controlled by the researcher
- Random allocation or distribution of individuals within a group

Strategies

The objective of experimental design is to get a maximum amount of knowledge in the least amount of time. Experiments are mostly designed by experts who possess the ability to:

- a. Define the problem and identify the specific set of desired characteristics.

For instance, there may be a need to manufacture glass bottles in a plant that adheres to stringent regulatory regulations.

- b. Elucidate all potential factors that may affect such characteristics, including manufacturing settings and the chemical makeup of the glass.

The production of high-quality glass bottles can be affected by various factors, including the chemical composition of the glass and its transition temperature, the suppliers of raw materials, the type and quantity of cullet used, the process parameters, the expertise, remuneration, and motivation of the production personnel, the day of production within a week (such as a weekend or workday), and numerous other influences.

- c. Identify the major factors and fundamental reasons without oversimplifying or making the situation unnecessarily complex.

A seasoned engineer may assess that the day of production, such as weekends, does not have a substantial impact on the quality of glass bottles, despite the variations in the working environment. Conversely, the engineer should acknowledge that the annealing schedule of the glass bottle and the transition temperature are crucial factors that must not be overlooked when considering quality. However, the transition temperature cannot be considered a significant reason as it mostly relies on the chemical makeup of the glass. Hence, the transition temperature, which serves as a potential variable, may be substituted with the glass composition.

- d. Quantify the effects in numerical variables.

Many factors may be quantified using meaningful numerical values, although aspects such as personnel experience should be simplified and classified into levels ranging from 1 (indicating little experience) to 3 (indicating significant experience). Likewise, the raw material provider might be classified.

Estimate the amount of anticipated impacts, D , of those factors on the desired attributes, if feasible, based on comparable research, and specify the necessary and accessible measurement accuracy for each technique employed.

It is crucial to estimate the size of the predicted effects in proportion to the measurement precision. This is because if the variables are tweaked too insignificantly, no observable impact will occur. For example, altering the pace at which the glass bottles are cooled during the annealing process by 1 degree Celsius per minute might lead to a significant variation in strength, ranging from 5% to 20%. This difference can be easily measured.

- f. Determine if the anticipated impacts of the variables are linear, regardless of the variables' values.

A change in cooling rate of 10C/min is anticipated to impact the strength of a glass bottle by 5-20%. However, a change in cooling rate of 20C/min is projected to have a greater impact, affecting the strength by 20-50%. This suggests that the relationship between cooling rate and bottle strength is likely non-linear.

- g. Determine potential interdependencies between variables (variable interaction).

For instance, previous experience suggests that a change in cooling rate of 10C/min may have a significant impact on the strength of glass bottles, either increasing or decreasing it by 5-20%, depending on modifications to the percentage of cullet used for glass melting or the introduction of alumina in the form of hydroxide instead of feldspar. The impact of cullet and alumina can be attributed to the glass composition as the main factor.

Notes

- h. Establish feasible upper and lower limits for the variables, as well as for the desired attributes. Additionally, set limitations for variable combinations if there are any variable interactions.

Variables are seldom possible to be updated in a totally arbitrary manner. During the annealing process of glass bottles, it is crucial to ensure that the cooling rate does not exceed the machine's safe operating limitations. Additionally, the chemical composition of the glass must adhere to financial and environmental constraints. It may not be justifiable to examine the durability of glass bottles below a specific threshold when even cautious handling results in breaking.

If there is a variable interaction between the cooling rate of glass bottles and the percentage of cullet used for glass melting (i.e., the glass composition), it is probable that it is not necessary to investigate the entire range of all possible combinations of cooling rate and glass composition. This is because it may lead to undesirably low strength data or there may be technical or financial limitations.

- i. Apply the appropriate experimental design matrix types.

Collecting more information in advance reduces the amount of effort needed for the final experimental design.

The average of several measurements is more dependable than a single or a few measurements. Put simply, the mean error diminishes as the number of measurements increases, as illustrated. Similarly, the most effective way to determine the impact of a variable is by comparing the average values of different groups. When examining experimental results, there are two potential types of errors: type 1 error, which occurs when an effect is mistakenly identified even if it does not actually exist, and type 2 error, which occurs when a genuine effect is not recognised. In order to effectively protect against both sorts of mistakes, it is necessary to conduct $n = 16 \cdot s^2/D^2$ trials at each variable level (i.e., variable value) and employ a statistical significance test to compare the mean of the outcomes.

To determine if a change in the glass composition reduces the annealing point by 10°C ($D = 10$) with a measurement precision of 5°C ($s = 5$), one would need to conduct 4 experiments for each of the two glass compositions. The mean of the results would then be compared for statistical significance. When several factors need to be taken into account, the number of experiments needed for a comprehensive inquiry might grow very enormous. An appropriately designed experiment can greatly minimise the amount of effort required.

3.1.2: Pre-Experimental Research Design

- The most basic sort of experimental study design
- Following the establishment of a cause and effect link, one or more groups are subjected to continuous monitoring.
- Typically performed to determine if more inquiry is necessary for the specific group(s) under consideration
- Cost-effective

Ex: An example of this would be a teacher using a new instructional method for their class.

3.1.3: Quasi-Experimental Research Design

- Resembles true experimental research but is not the same
- The control group is missing
- The participants of a group are not assigned in a random manner.
- This method is employed when random assignment is either not relevant or unnecessary.

Ex: We choose two homogeneous cohorts of children who are enrolled in separate educational institutions, with one cohort being exposed to the new programme while the other cohort serves as a control group.

3.1.4 True Experimental Research Design

- The most precise kind of experimental research design
- It relies on statistical analysis to validate or invalidate a hypothesis.
- It is the only type of experimental research design capable of establishing a cause-effect link within a group or a few groups.
- Commonly used in Physical Sciences

3.1.5 Factorial Experimental Design

Factorial design allows the researcher to simultaneously examine several factors. It aids in determining the impact of individual factors and measuring the combined effect of several variables.

A departmental store seeks to analyse the effects of price decrease on a certain product. In addition, there is also point-of-purchase (POP) promotion being used in the stores.

- a) near the entrance
- b) at the usual place, at the same time.

Let's consider a scenario where there are two different pricing levels: the usual price, denoted as P1, and the lowered price, denoted as P2. There are three sorts of POP, specifically R1, R2, and R3. There are a total of 6 potential combinations, which may be calculated by multiplying 3 by 2. The available combinations are R1P1, R1P2, R2P1, R2P2, R3P1, and R3P2. The researcher is interested in determining which of these combinations is most suitable. Assume that there are 120 departmental shops in the chain, organised into groups of 20 stores each. Now, allocate the aforementioned combination randomly to each of these 20 stores in the following manner:

Combinations	Sales
R1P1	S1
R1P2	S2
R2P1	S3
R2P2	S4
R3P1	S5
R3P2	S6

Notes

S1 to S6 reflect the sales generated by each variable. The collected data will offer specific information on product sales based on two separate characteristics. The following two inquiries will be addressed:

1. Does the discounted price yield greater effectiveness compared to the standard price?
2. Does the exhibit at the entrance have a greater impact than the display in the customary location?

Additionally, the research will provide insights into the interaction impact of the two factors.

The results of the sales trial are as follows:

1. Decrease in price accompanied by a prominently placed display at the entry.
2. Price decrease indicated at the regular location.
3. There is no display and the standard price is applied.
4. Exhibit at the door with the standard fee in effect

3.1.6: Designing Engineering Experiments

Experiments are an inherent component of the decision-making process in engineering. For instance, a civil engineer is examining the impact of various curing techniques on the average compressive strength of concrete. The experiment involves conducting tests on many concrete specimens using each recommended technique of curing. The compressive strength of each specimen is then measured. The data obtained from this experiment is utilised to ascertain the optimal curing process that yields the highest average compressive strength.

If there are just two curing procedures of interest, this experiment may be designed and analysed using statistical hypothesis methods for two samples. The researcher is focused on a certain variable, which is the procedures used for curing. There are only two options for this variable. If the researcher wants to establish which curing process yields the highest compressive strength, they may select the number of specimens to test using the operating characteristic curves. The t-test can then be used to evaluate if the means of the two methods vary.

Several single-factor studies need the consideration of more than two levels of the factor.

For instance, the civil engineer may choose to examine five distinct curing techniques.

Statistically based experimental design techniques are useful in the engineering world for increasing the performance level of a manufacturing process and also used to develop new processes. Many processes may be characterised by a number of manageable factors, such as temperature, pressure, and input rate. Engineers can utilize structured experiments to identify the specific subset of process variables that has the most significant impact on process performance. The outcomes of such an experiment might result in:

- Enhanced procedure.
- Minimised fluctuations in the process and adherence to specified nominal or goal criteria
- Decreased design and development duration

- Lowered operational expenses

These techniques are also valuable in engineering design endeavours for the creation of new goods and enhancement of current ones. Statistically designed experiments in engineering design have applicability in several areas:

- Assessment and analysis of fundamental design structures
- Assessment of various materials
- Selection of design parameters to ensure optimal performance of the product in diverse field situations.
- Identification of critical product design characteristics that significantly impact product performance

Utilising experimental design in the engineering design process can provide products that are more manufacturable, possess superior field performance and dependability compared to their rivals, and can be planned, developed, and manufactured more efficiently. Sequentially, designed experiments are typically utilized.

- First, experiment with many controllable variables, a screening experiment designed to determine which variables are most important.
- Subsequent experiments are applied on these critical variables to refine the collected information and determine which adjustments are required to improve the process.
- Finally, the experimenter optimized which levels of the critical variables result in the best process performance.
- Every experiment involves a sequence of activities like as:
 1. Conjecture refers to the initial hypothesis that serves as the basis for conducting the experiment.
 2. Experiment refers to the test used to examine the conjecture.
 3. Analysis refers to the process of statistically examining the data obtained from the experiment.
 4. Conclusion refers to findings on the first hypothesis based on the experiment.

Frequently, the experiment will result in an amended hypothesis, prompting a further experiment, and so on. Statistical approaches are crucial for conducting effective experiments. All trials are inherently designed experiments; nevertheless, regrettably, certain experiments are inadequately designed, leading to inefficient utilisation of important resources. Statistically planned experiments allow for efficiency and cost-effectiveness in the experimental procedure. Additionally, the application of statistical methods in analysing the data ensures scientific impartiality when making conclusions.

3.1.7: Causal Attributions

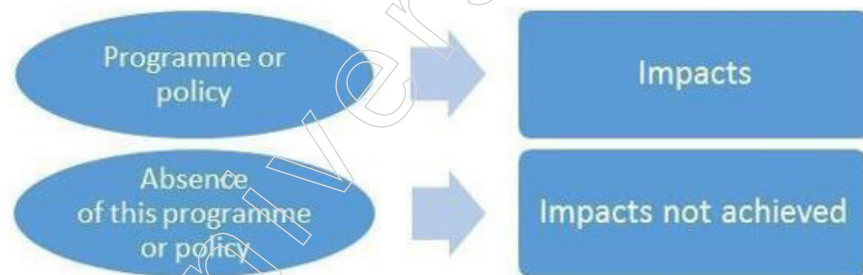
An crucial component of an impact evaluation is not just to quantify or characterise the changes that have happened, but also to ascertain the involvement of certain interventions (such as programmes or policies) in causing these changes. That process is named causal attribution, causal contribution or causal inference. In this article, we explore several methods of examining causal attribution using research design, data collecting, and analytic procedures. The definition of impact provided by the OECD-DAC states that an impact evaluation must determine the causal factors behind

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observed changes. It encompasses both positive and negative, main and secondary long-term consequences that result from a development intervention, whether they are planned or unexpected and whether they occur directly or indirectly.

1. Causal attribution, as defined by the OECD-DAC, refers to the process of determining the cause or causes of a certain event or outcome: "Ascription of a causal link between observed (or expected to be observed) changes and a specific intervention."
2. This criterion does not need that alterations are exclusively or entirely caused by the programme or policy being examined.
3. Put simply, it acknowledges the possibility that other variables, such as additional initiatives or policies in the relevant field or specific contextual elements (often known as 'external factors'), may have had a role. Evaluations provide more robust and valuable results when they not only examine the connections between activities and impacts, but also explore the connections across the causal chain encompassing activities, outputs, intermediate outcomes, and impacts.

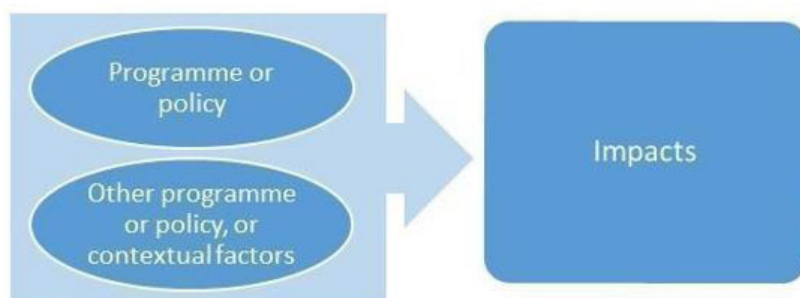
Sole causal attribution: In this sense, the programme or policy is seen to be both required and sufficient to generate the desired effects, regardless of external variables or other actions.



From a practical perspective, science programmes or policies alone are insufficient to accomplish the desired outcomes. Therefore, different alternative methods are employed to get the intended results. This model is said to be a seldom effective framework for evaluating the effects of a cause on an outcome.

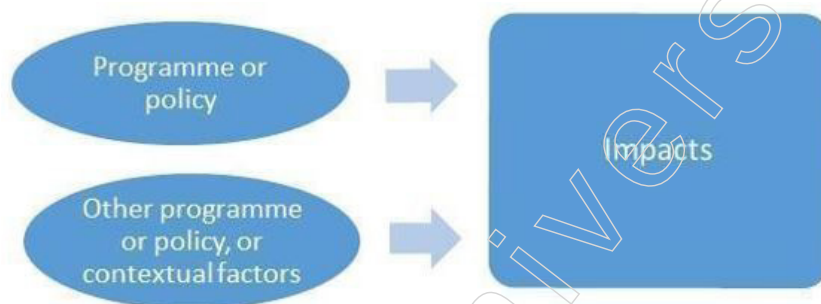
Joint causal attribution refers to the situation when a programme or policy has created impacts in concert with other programmes/policies or specific contextual conditions. It may also encompass variables related to the implementation environment, such as the capabilities and infrastructure of implementing partners or the characteristics of the participants. For illustrative purposes, in relation to their degrees of motivation and prior knowledge. The consequences will not be realised or will be significantly diminished if these contributing elements are either lacking or unfavourable.

This is a typical scenario in which a software can only function optimally when favourable conditions are present and unfavourable conditions are eliminated. For instance, if a project is determined to be effective only in a context where there is openness and clarity regarding government funding allocations, it should only be put into practice in settings where this transparency already exists or in combination with endeavours to attain transparency. To discover these additional contributing elements, it is crucial to construct a comprehensive theory of change that incorporates them, drawing on prior research and current knowledge. Additionally, it is essential to collect data specifically related to these aspects.



Alternative (or multiple) causal paths: Here, the plan or policy is only one of numerous potential methods to accomplish the desired effects. A specific programme or policy can generate the effects, but they can also arise from alternative actions and/or external variables.

This situation might occur when participants have the ability to utilise services from a different supplier, or when many initiatives from different providers are designed to achieve the same outcome. For instance, several projects may have the objective of diminishing infant mortality. Some programmes focus on enhancing nutrition, some on mitigating waterborne infections, and some on immunisation.



When employing a counterfactual design, it is crucial to examine the services that non-participants are utilising in order to make a comparison with the participants. When a programme aims to achieve the same results as previous programmes, it is crucial to identify intermediate outcomes in the theory of change and gather data on them. There are three primary approaches for determining the cause of an effect in impact evaluations:

- Estimating the counterfactual involves predicting what would have happened if the intervention had not taken place, then comparing it to the actual observed circumstance.
- Examining the coherence of data supporting the cause-and-effect links explicitly stated in the theory, while excluding other possible explanations using a logical and evidence-driven approach.
- Using a combination of these strategies can usually help to increase the strength of the conclusions.

Causal attribution should be routinely undertaken in all impact evaluations. However, it is important to acknowledge the practical limitations of doing a single review, particularly when there are time constraints that prevent repeated data collecting and analysis. While a single impact evaluation can contribute to the existing body of evidence, it is important to note that even if there is a reasonable level of certainty that a programme or policy caused the observed effects, this does not imply that the evaluation has thoroughly examined all aspects of the intervention or its

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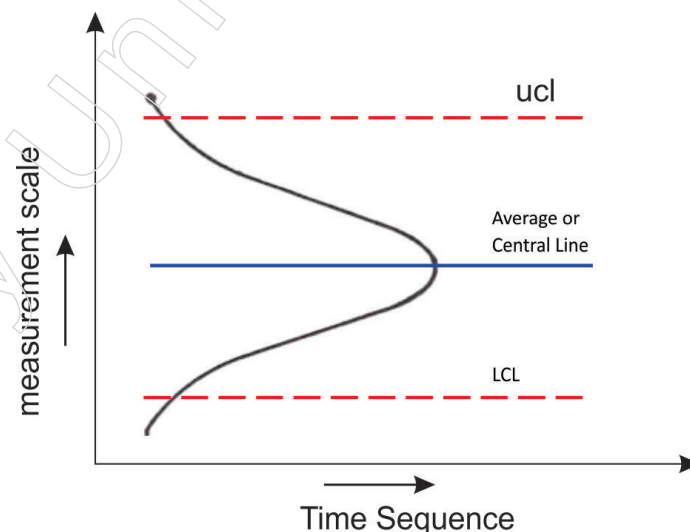
applicability in different contexts or time periods. When many impact assessments are combined, the synthesis of data offers more robust evidence than a single evaluation, regarding the generalisability of the observed effects.

Key highlights

- Causal attribution examines the causal connections between a programme or intervention and the observed alterations.
- L attribution is a crucial component of effect evaluation.
- Various methodologies may be employed to analyse causal attribution, all of which are enhanced by being grounded in a robust theory of causality.
- The most suitable approach for determining causal attribution relies on both the assessment environment and the specific subject being assessed.

3.1.8: Statistical Control

Statistical control mostly occurs through the use of several control charts, which give essential information that acts as a foundation for taking appropriate actions. Control charts are only useful if the individuals in charge of decision-making take action based on the information shown by the chart. Control charts are versatile instruments that can be effectively utilised, misapplied, or completely disregarded. Their efficacy is closely correlated with the comprehension and correct implementation by individuals. The control chart is a visual depiction of the process of monitoring. Control charts are essentially similar to the conceptual chart, with a vertical axis. A scale is developed to align with the collected data. The horizontal axis represents the succession of data or the interval of time. Once data is acquired, it may be graphed in relation to these two axes.



Control charts are divided into two broad categories based on the two types of data they handle: variable data and attribute data.

When a quality attribute is measured and recorded, such as a dimension represented in thousandths of an inch, it is referred to as a variable. The variables are:

Variables Charts

- $\bar{X} - R$ Average and Range
- $\tilde{X} - R$ Median and Range

$\bar{X}$ - S Average and Standard Deviation

$\bar{X}$ - Rm Individual and Moving Range

Notes

Attributes Charts

- p Fraction Nonconforming
- np Number of nonconforming Units
- c Number of Nonconformities
- u Number of Nonconformities per unit

Prerequisites for Utilising Control Charts

Prior to using control chart techniques on any process or action, several fundamental prerequisites must be met. Initially, individuals in authoritative roles must comprehend and endorse endeavours to consistently enhance procedures and quality by employing statistical techniques like control charts. In order for a corporation to truly experience process improvement utilising control charts, it is necessary to meet the following requirements once the initial goal has been accomplished.

- To ensure effective decision-making for chart selection, sample size, sampling frequency, and characteristics to be monitored, it's crucial to thoroughly understand the process under study. This understanding helps identify actions that can reduce variability.
- Identify which characteristics need monitoring. These are typically the ones critical for the process or product's optimal performance, those with stringent specifications, those frequently found nonconforming, or those costly to repair. Pareto analysis is a useful technique for selecting a primary characteristic to chart.
- It's essential to define and comprehend the measurement system. The quality of data on the chart directly depends on the measuring process or device. Establish the necessary accuracy to select appropriate measuring equipment. Regular reviews and calibrations are needed to ensure the accuracy and repeatability of the measuring devices.
- Choose a chart type that best meets the purpose. This choice often depends on the measurement costs and the potential loss if significant changes go undetected. The type of information required may also influence the chart selection.
- Generally, attribute charts are suitable if:
 - ❖ Visual inspection makes measurements impossible.
 - ❖ Measuring equipment is too costly or testing time is excessive.
 - ❖ There are many characteristics to evaluate in the part.
 - ❖ 100% inspection is planned.

Utilise a Variables Chart When

- Dealing with a critical or variable characteristic.
- Choose this type of chart when more precise control is needed than attribute charts can provide.

Also, determine the sample size and frequency, and in variables charts, the common sample size is five. The subgroup of a sample must be chosen in a manner

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that minimises the potential for variance within the group. Nevertheless, as the sample size increases, the distribution of subgroup averages tends to become more normal, even if the population itself is not normally distributed. Economics significantly influences the determination of sample sizes.

Attribute charts, in contrast, are designed to analyse data from far higher sample sizes, ranging from 30 to several hundred or more. Given that the attribute data may be accessible from reliable preceding reports or research, it implies that the effort and cost have already been expended.

Control Chart Selection

In manufacturing plants, when some type of process trouble arises, control charts may serve as useful tools for troubleshooting, benefiting both engineers and operators. Control charts primarily give information in four key areas:

- Primary variability of the characteristic
- Consistency of performance
- The average level of the characteristic
- Type of variability present

The subsequent instructions serve as a comprehensive flow chart selection guide for several categories of fundamental control charts.

- Process control charts are tools that help us analyse data, mainly distinguishing between variability as a result of common causes or special causes.
- If only common causes represent, then Processes are in control.
- If special causes are present, then Processes are out of control.
- There are two main types of control charts: Variables Control Charts and Attribute Control Charts.
- Tables of constants have been developed to assist in using control charts, utilizing the connection between the sample and the population.
- When the population is used to set control limits for an average and range chart, misleading signals of the state of control are present because the control limits are to be based on the sampling distribution of averages, not the population distribution of individual values. This holds true for other variable charts as well.

Steps for Developing and Plotting a Variables Control Chart

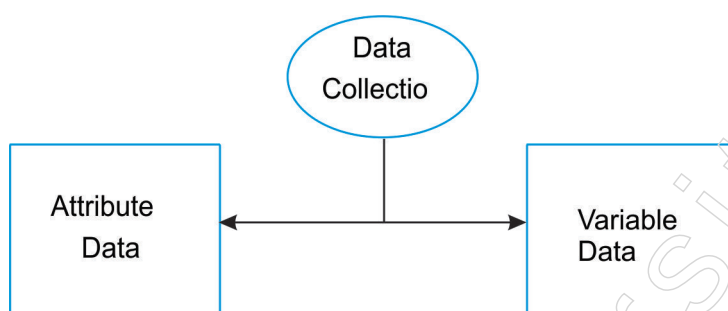
- Include heading information such as part name, part number, department, machine number, plant identification, and operation being performed.
- Record sample information, including date and time of sample collection, initials of the checker, and shift during which the samples were taken.
- Enter measured data into specified locations.
- Calculate statistical measures such as average, median, range, standard deviation, or other relevant statistics.
- Establish scales for the graphs; scales should facilitate plotting. Typically, the range of values for central tendency measures should either match the product specification or be twice the difference in the highest and lowest subgroup statistics.
- Plot the data.

Notes

- Calculate and plot the central lines for both portions of the chart - central tendency and variability. These should be drawn in as solid lines and labelled.
- Calculate and plot the control limits. These are calculated using the specific formulas for the statistics being monitored. The Control limits should be drawn in as dashed lines on the chart and labelled.
- Interpret the chart.

Control Limit Calculations:

As noted in the “Introduction to Control Charts” section, the upper and lower control limits represent the $\pm 3\sigma$ limits for the sample statistic. Let's examine the formulas developed for X and R charts.



SampleSize(n)	Chart Type
>20, notconstant	p chart
>20, constant	np chart
>1; <20, not constant	u chart
1	c chart

3.2 Data Collection

Data collection is the act of obtaining and examining precise information from a variety of sources in order to assess potential outcomes, trends, and probability, among other research concerns. Data is information that has been digitally transformed, at least according to the IT definition of data. Knowledge truly is power. Data is therefore power. However, you must first collect the data in order to use it to inform a profitable plan for your company or organization.

3.2.1: Collection of Data via Observation and Interview

Observation Method :Observation becomes a scientific tool and method of data collection when it serves a formulated research purpose, is systematically planned and recorded, and is subjected to checks and controls for validity and reliability.

1. Participant Observation: This involves the observer becoming a part of the group being studied. It is characterised by:
 - ❖ Careful definition of the units to be observed
 - ❖ Standardised conditions of observation
 - ❖ Systematic recording of observed information
 - ❖ Selection of pertinent data for observation
 - ❖ Importance of the respondent's role

Notes

2. **Non-Participant Observation:** This method occurs without the observer becoming involved in the group activities and lacks the systematic characteristics of participant observation.

Interviewing Method

The interviewing method involves presenting oral-verbal stimuli and receiving oral-verbal responses. It typically requires face-to-face contact between the interviewer and the respondent.

1. **Structured Interview:** This type uses a set of predetermined questions and highly standardised techniques of recording responses. It ensures consistency and comparability of data.
2. **Unstructured Interview:** This type does not follow a system of predetermined questions and standardized recording techniques. It is based on flexibility and the needs of the conversation.
3. **Clinical Interview:** Focuses on understanding broad underlying feelings or motivations, or the course of an individual's life experience. It is often used in psychological and therapeutic settings.
4. **Non-Directive Interview:** The interviewer encourages the respondent to talk about the topic with minimal direct questioning. The interviewer's role is to facilitate conversation rather than direct it.
5. **Focused Interview:** Confining the respondent to discuss a given experience and its effects. It aims to gather detailed insights about a specific event or experience from the respondent's perspective.

3.2.2: Collection of Data through Cases

A researcher doing case studies should use a rigorous approach to guarantee neutrality and clarity.

1. **Identify and Define the Research Questions**
 - ❖ **Establish Focus:** Begin by identifying the specific research object and the problem or issue surrounding it.
 - ❖ **Determine Research Object:** This could be a person, program, event, or entity.
2. **Select the Cases**
 - ❖ **Decide on Number of Cases:** Choose whether to study a single case or multiple cases.
 - ❖ **Type of Cases:** Determine whether to select unique or typical cases.
 - ❖ **Design Phase:** Plan how to collect, store, and analyse the data.
3. **Collect the Data**
 - ❖ **Gather Evidence:** Collect data from multiple sources related to the problem under study.
 - ❖ **Systematic Storage:** Store the evidence in a comprehensive and organised manner to facilitate easy referencing and sorting.
 - ❖ **Identify Patterns:** Aim to uncover converging lines of inquiry and patterns.
4. **Evaluate and Analyse the Data**

- ❖ Analyse Qualitative and Quantitative Data: Utilise varied methods to categorise, tabulate, and cross-check data.
- ❖ Address Initial Propositions: Ensure the analysis addresses the initial propositions or purpose of the study.
- ❖ Use Graphic Techniques: Employ arrays, matrices of categories, flow charts, etc., to explore data from different perspectives.
- ❖ Involve Multiple Investigators: Utilise multiple investigators to develop a wide variety of insights from the data.

5. Presentation of Results

- ❖ Corroborate Findings: Present results with sufficient evidence to show that all aspects of the problem have been explored.
- ❖ Highlight New Insights and Conflicts: Clearly present any new insights gained and conflicting propositions that have emerged.
- ❖ Enable Evaluation: Structure the results in a manner that allows the reader to evaluate the findings based on the evidence presented.

3.2.3: Collection of Data through Questionnaires

Questionnaires are a fundamental instrument for gathering information in survey research. They consist of standardised questions, often called items, designed to collect individual data on specific topics. There are two primary methods for collecting data through questionnaires:

1. Mailing Method

In the Mailing Method, the investigator prepares a questionnaire related to the research topic and mails it to various individuals with a request to fill it out and return it.

This method typically includes:

- ❖ Blank Space for Answers: Respondents are provided with space to write their answers.
- ❖ Self-addressed Stamped Envelope: For returning the completed questionnaire by post.
- ❖ Confidentiality Assurance: Respondents are assured of the secrecy of the information they provide.
- ❖ This method is commonly used by research workers and official and non-official agencies.

Suitability of Mailing Method

- Large Field of Investigation: Suitable when the investigation covers a large area.
- Educated Respondents: Ideal for respondents who are educated and likely to cooperate with the investigation.

2. Enumerator Method

The Enumerator Method involves a trained enumerator who personally visits the respondents, asks questions, and notes down their replies.

- ❖ Provide Assistance: Help respondents understand the questions to avoid ambiguous replies.

Notes

- ❖ Fill the Schedule: Record responses themselves to ensure clarity and accuracy.
- ❖ Characteristics of Enumerators: Should be well-trained, tactful, hard-working, and unbiased to gather reliable information.
- ❖ This method is typically employed by semi-government organisations, government bodies, research institutions, etc. The questionnaire filled out by the enumerator is known as a Schedule.

Suitability of Enumerator Method

- Availability of Resources: Suitable when adequate finance and trained enumerators are available.
- Illiterate Respondents: Ideal when respondents are not literate and need assistance in answering the questions.

3.2.4: Guidelines for Preparing Questionnaire

1. Identify Your Research Aims and the Goal of Your Questionnaire
 - ❖ Clearly define what information you want to gather and your main objective.
 - ❖ Check if there are existing validated questionnaires you can use or draw inspiration from, especially in psychological research.
 - ❖ If existing questionnaires are not available, draw inspiration from related research while paying close attention to the following steps.
2. Define Your Target Respondents
 - ❖ Identify the specific groups you need to target for your questionnaire to maximise the robustness of your study.
 - ❖ Choose between a longitudinal design (repeated measures over time) and a cross-sectional design (one-time measure across different groups).
 - ❖ Analyse how responses change over time.
 - ❖ Compare differences among groups at a single point in time.
3. Develop Questions
 - ❖ Ensure questions are clear and free from misunderstandings or ambiguities.
 - ❖ Tailor questions to the cognitive, attentional, and sensory competencies of your target group (e.g., children, adults, elderly).
 - ❖ Use simple language that can be easily understood by all participants.
4. Choose Your Question Type
 - ❖ Use for explorative questionnaires to gain in-depth understanding of topics (e.g., "What did you like about the webinar?").
 - ❖ Use for quantitative questionnaires with predefined answers (e.g., multiple choice or Likert scale).
 - ❖ Consider the benefits and drawbacks of each question type to design a balanced questionnaire.
5. Design Question Sequence and Overall Layout
 - ❖ Ensure a logical transition from one question to the next.
 - ❖ Position follow-up questions correctly.

- ❖ Allow participants to skip questions that do not apply to them, if necessary.
- ❖ Make sure the questionnaire is easy to navigate and visually appealing.

6. Run a Pilot

- ❖ Pilot the questionnaire with a sample from your target audience.
- ❖ Look for issues in readability, understanding, phrasing, and overall arrangement.
- ❖ Get feedback from pilot participants to understand their experience.
- ❖ Analyse pilot data statistically to ensure the questionnaire produces the desired insights and that analytic procedures can be applied effectively.

3.2.5: Data Editing

Data editing is a critical step in the research process that involves checking and correcting errors, inconsistencies, and omissions in collected data. This process is essential for ensuring the quality, accuracy, and reliability of research findings.

Techniques for Editing Data

1. Visual Inspection

- ❖ Description: This is the primary technique used for editing data. Researchers manually examine the data to identify and correct any errors, inconsistencies, or omissions.
- ❖ Advantages: Allows for the detection of errors that may not be easily identified by automated processes.
- ❖ Disadvantages: Time-consuming and prone to human error, especially with large datasets.

2. Computer Verification

- ❖ Description: This technique involves using software or computer programs to check the data for errors and inconsistencies.
- ❖ Advantages: Faster and more accurate than visual inspection. Can handle large volumes of data efficiently.
- ❖ Disadvantages: Requires programming skills and may not catch errors that require contextual understanding.

3. Double Data Entry

- ❖ Description: Involves entering the data twice by two different individuals to ensure accuracy and consistency. The two sets of data are then compared to identify and correct any discrepancies.
- ❖ Advantages: Highly effective in ensuring data accuracy and consistency.
- ❖ Disadvantages: Labor-intensive and can be costly due to the need for additional personnel.

Importance of Editing Data

1. Ensures Data Accuracy

Description: Editing data helps to ensure that the collected data is accurate. By correcting errors and inconsistencies, researchers can ensure that the data accurately reflects the research variables and parameters.

Notes

Impact: Accurate data is crucial for the validity of the research findings.

2. Enhances Data Quality

Description: Editing enhances the quality of the collected data by ensuring that it is complete and consistent.

Impact: High-quality data leads to more reliable and valid research findings, which can be confidently used for analysis.

3. Increases Credibility of Research

Description: Editing data is essential for increasing the credibility of research by ensuring that the findings are based on accurate and reliable data.

Impact: Credibility is important in research to gain the trust of the audience and stakeholders. Well-edited data helps in building confidence in the research outcomes.

3.2.6: Data Coding Rules

Data coding is a critical step in the research process, involving the transformation of raw data into a structured format. This enables researchers to efficiently organise, analyse, and interpret data, facilitating the identification of patterns, themes, and relationships within the dataset.

Importance of Data Coding

1. **Data Reduction:** Coding helps condense and summarise large volumes of data, making it manageable and analyzable. This reduction is crucial for handling extensive datasets efficiently.
2. **Data Organization :** Coding provides a systematic way to categorise and group similar pieces of information. It simplifies the management and analysis of data, making the research process more streamlined.
3. **Pattern Recognition :** Coding enables researchers to identify patterns, trends, and relationships within the data. These patterns might not be immediately apparent in raw data, and coding helps uncover them.
4. **Interpretation and Analysis :** Coded data serves as the foundation for statistical analysis and hypothesis testing. Researchers can run statistical tests on coded data to draw meaningful conclusions.
5. **Comparative Analysis :** Consistent coding allows for the comparison and contrast of information across different cases or groups. This aids in generating insights and developing theories.

Examples of Data Coding in Research

- **Qualitative Research:** A study on customer feedback about a new product.
 - ❖ **Coding Process:** Customer comments are categorised into themes such as “product quality,” “customer service,” “pricing,” and “delivery.”
 - ❖ **Outcome:** Each comment is assigned one or more codes based on its content, facilitating thematic analysis.
- **Survey Research :** A survey on job satisfaction.
 - ❖ **Coding Process:** Responses on a Likert scale are coded numerically (e.g., “strongly agree” = 5, “agree” = 4, “neutral” = 3, “disagree” = 2, “strongly disagree” = 1).

- ❖ Outcome: These numerical codes enable quantitative analysis of survey data.
- Content Analysis: Analysis of news articles about climate change.
 - ❖ Coding Process: Articles are coded as “supportive of climate action,” “neutral,” or “sceptical of climate change.”
 - ❖ Outcome: This coding allows researchers to assess the prevalence of different perspectives in the media.

Medical Research: Categorising patient data.

- Coding Process: Patient data is categorised into diagnostic groups based on symptoms, test results, or medical histories.
- Outcome: Researchers can analyse the effectiveness of different treatments for specific conditions.

Historical Research: Coding historical documents.

- Coding Process: Documents are coded based on themes, time periods, or key events.
- Outcome: This enables historians to identify patterns and trends across historical records and gain new insights into the past.

3.2.7: Don't Know" Responses

The intention behind incorporating “don't know” options in surveys is to provide respondents with an opportunity to abstain from questions they're uncertain about. However, this approach may backfire as respondents might opt for these choices to avoid engaging with the question or topic entirely. Consequently, the reliability of survey responses could be compromised. To mitigate this issue, it's crucial to explore alternative strategies, such as removing the option entirely or introducing follow-up questions to glean more insightful data.

Benefits of Offering an “Other” Option

Providing an “other” response alternative empowers respondents to furnish additional information or elucidate their perspectives. This can be especially beneficial in soliciting feedback on products or services, as it permits respondents to articulate negative opinions or express concerns. For instance, individuals may refrain from endorsing a product due to perceived deficiencies in quality or usability. Allowing them to elaborate on their rationale enables organisations to glean valuable insights into consumer perceptions and refine their offerings accordingly.

Appropriate Use of “Don't Know” Responses

While “don't know” responses may be suitable in surveys tackling sensitive topics, their deployment necessitates discretion. In surveys centred on brands or products, it's imperative to ensure respondents possess adequate familiarity with the subject matter before soliciting their opinions. Failure to do so may compel respondents to furnish ratings for items they have no experience with, potentially distorting the data.

Balancing Response Options for Optimal Survey Design: While “don't know” responses can fulfil a purpose in specific survey configurations, it's imperative to evaluate their impact on data quality and respondent engagement. Incorporating alternative response avenues, such as the “other” option, holds the potential to yield valuable insights and elevate the effectiveness of surveys.

Notes

3.2.8: Missing Data

In the realm of data analysis, the presence of missing data poses significant challenges, impacting the accuracy and reliability of research findings. Let's explore the various types of missing data and their implications, along with strategies to address them effectively.

Categories of Missing Data

- **Missing Completely at Random (MCAR):** In MCAR scenarios, missing data occur randomly across variables, unrelated to any other factors. For example, incomplete survey responses due to random dropouts exemplify MCAR data.
- **Missing at Random (MAR):** Despite its name, MAR data display a systematic pattern, albeit explainable by observed variables. For instance, in a holiday spending survey, missing values may be more common among certain age groups due to privacy concerns rather than spending amounts themselves.
- **Missing Not at Random (MNAR):** MNAR data arise due to reasons inherent to the values themselves. For instance, individuals with lower incomes may intentionally omit reporting holiday spending amounts, thereby skewing the dataset.

Implications of Missing Data

- The presence of missing data introduces biases that can compromise the representativeness and generalizability of study findings. While MCAR and MAR data may be considered ignorable, MNAR data introduce non-ignorable biases that can potentially distort conclusions.

Strategies to Handle Missing Data

- **Preventive Measures:** Minimise missing data by designing user-friendly data collection forms, employing data validation techniques, and providing incentives to participants. Additionally, limiting the number of follow-ups and the volume of collected data can help mitigate risks associated with missing data.
- **Secure Storage:** Store collected data securely with multiple backups to mitigate the risk of data loss and ensure data integrity.

By comprehensively understanding the nuances of missing data and implementing proactive strategies, researchers can effectively address missing data challenges, thereby enhancing the reliability and robustness of their analyses

3.3: Basic Principles

Numerous studies have demonstrated the significance of assessing the validity and reliability—also known as consistency and accuracy—of research instruments, particularly questionnaires. However, few health and social science researchers in developing nations routinely measure these factors. This has been connected to the lack of awareness of these assessments. This review article thoroughly examines and explains the reliability and validity of a research tool (specifically, a questionnaire). It then goes on to cover numerous validity and reliability test types with succinct examples, and at the end, it describes different approaches to test analysis that are guided by scientific ideas.

3.3.1: Validity and Reliability

Validity pertains to the degree of accuracy with which a technique assesses the desired measurement. High validity in research indicates that the results align with actual traits, characteristics, and fluctuations in the physical or social environment. High reliability indicates that measurement is valid, but if a method is not reliable, then it's probably not valid. For example, if the thermometer shows different temperatures each time, even though you have carefully controlled conditions to ensure the sample's temperature stays the same, the thermometer is probably malfunctioning, and therefore its measurements are not valid.

If a symptom of questionnaire results in a reliable diagnosis when answered at different times and with different doctors, this indicates that it has high validity as a measurement of the medical condition

However, reliability on its own is not enough to ensure validity. Even if a test is reliable, it may not accurately reflect the real situation.

Construct	The attachment of a measure to existing theory and knowledge of the concept being measured.	A self-esteem questionnaire could be assessed by measuring other attributes known or assumed to be related to the concept of self-esteem, like social skills and optimism). There has a correlation between the scores for self-esteem, and associated attributes would indicate high construct validity.
Content	The measurement covers all aspects of the concept being measured.	A test to measure a class of students' level of Bengali contains reading, writing and speaking components, but no listening component. Experts agree that listening comprehension is an essential aspect of language ability, so the test lacks content validity for measuring the overall level of ability in Bengali.
Concept.		A survey is conducted to measure the exit poll of voters in a region. If the results accurately predict the later outcome of an election in that region, this indicates that the survey has high criterion validity

Discriminant Validity

Discriminant validity, also known as divergent validity, is a type of construct validity that tests whether concepts or measurements that are supposed to be unrelated are, in fact, unrelated. This helps to ensure that the construct being measured is truly distinct from other constructs.

Key Points:

- Purpose: To confirm that constructs that are theoretically supposed to be unrelated are indeed not correlated.
- Assessment: Often assessed using techniques such as factor analysis, where low correlations between the measure of interest and measures of different constructs are expected.

Notes

- Example: If you are developing a test to measure job satisfaction, discriminant validity would be established if the test shows low correlations with a test measuring unrelated constructs such as physical health or political orientation.

Methods to Test Discriminant Validity:

- Correlation Analysis: Ensuring low correlations between measures that should not be related.
- Factor Analysis: Ensuring that items load onto their respective factors without cross-loading onto unrelated factors.

Convergent Validity

Convergent validity is another aspect of construct validity that tests whether concepts or measurements that are supposed to be related are actually related. It ensures that the construct is accurately being measured by demonstrating that it correlates well with other measures of the same construct or similar constructs.

Key Points:

- Purpose: To confirm that constructs that are theoretically supposed to be related are indeed correlated.
- Assessment: Often assessed using techniques such as correlation coefficients or factor analysis, where high correlations between the measure of interest and other measures of the same or similar constructs are expected.
- Example: If you are developing a test to measure job satisfaction, convergent validity would be established if the test shows high correlations with other established measures of job satisfaction, such as employee engagement surveys or job performance ratings.

Reliability

Reliability is a measure of the stability or consistency of every test. We can also think of it as the ability for a test or research findings to be repeatable. For example, a medical thermometer is a reliable tool that would measure the correct temperature each time it is used. In the same way, a reliable math test will accurately measure mathematical knowledge for every student who takes it, and reliable research findings can be replicated over and over. There are basically two types of reliability: Internal and External reliability.

Internal reliability, or internal consistency, is a measure by which our test is actually measuring what we want it to measure.

External reliability means that our test or measure can be generalized beyond what we're using it for.

For example, a claim that individual tutoring improves test scores should apply to more than one subject (e.g. to English as well as math). A test for depression should be able to detect depression in different age groups, for people in different socio-economic statuses, or introverts.

Coefficient Reliability

A reliability coefficient is a measure of how well a test measures achievement. It is

the proportion of variance in observed scores attributable to true scores (the theoretical “real” score that a person would get if a perfect test existed).

The several different coefficients are calculating through the several methods of test; these are test-retest, parallel forms and alternate-form:

- Cronbach’s alpha — the most widely used internal-consistency coefficient.

Range: The value of Cronbach’s alpha ranges from 0 to 1.

Interpretation:

- ❖ $\alpha \geq 0.9$: Excellent
- ❖ $0.7 \leq \alpha < 0.9$: Good
- ❖ $0.6 \leq \alpha < 0.7$: Acceptable
- ❖ $0.5 \leq \alpha < 0.6$: Poor
- ❖ $\alpha < 0.5$: Unacceptable
- A simple correlation between two scores from the same person is one of the simplest ways to estimate a reliability coefficient. If the scores are taken at different times, then this is one way to estimate test-retest reliability; Different forms of the test given on the same day can estimate parallel forms reliability.
- Pearson’s correlation can be used to estimate the theoretical reliability coefficient between parallel tests.
- The Spearman-Brown formula is a measure of reliability for split-half tests.
- Cohen’s Kappa measures integrator reliability.

The range of the reliability coefficient is from 0 to 1. Rule of thumb for preferred levels of the coefficient:

- ❖ For high stakes tests (e.g. college admissions), > 0.85 . Some authors suggest this figure should be above .90.
- ❖ For low stakes tests (e.g. classroom assessment), > 0.70 . Some authors suggest this figure should be above 0.80

3.3.2: Replication, Randomisation and Blocking

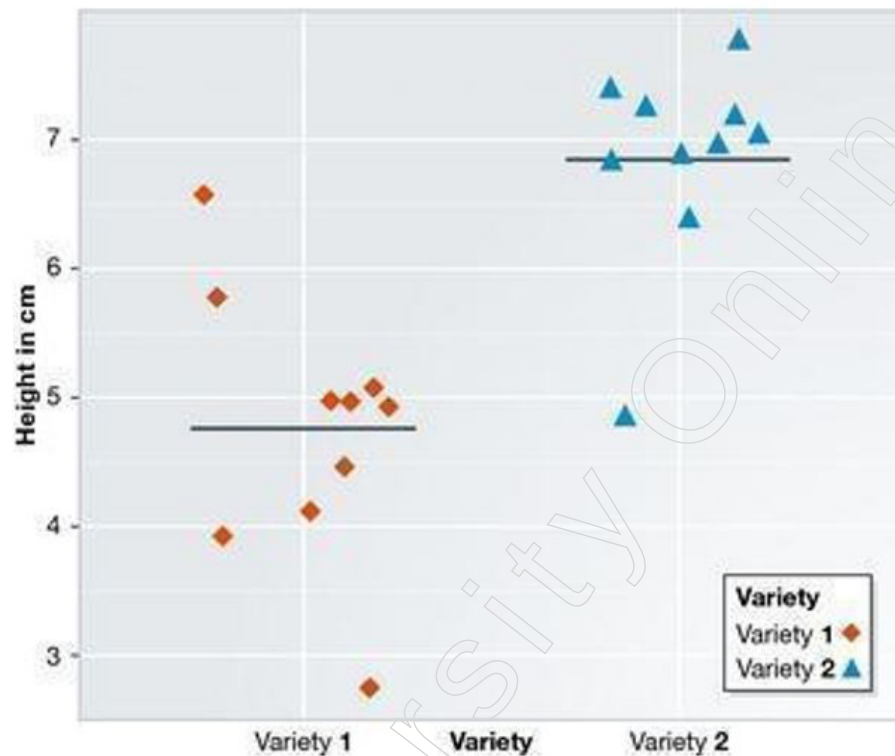
Replication

Without replication, it is impossible to judge whether there is an actual difference when comparing situations, it is important to determine if any observed difference is a result of chance or not.

For instance, suppose we aim to compare the vertical dimensions of two plant species by only doing a single measurement of plant height and noting a discrepancy of 20 cm. However, if numerous plants of each variety are measured and consistently show height variations of around 10 cm, it is less probable that the observed difference is a result of random chance.

The difference is strong relative to the variability between the measurements.

Notes



A comparison between two groups

Comparative analysis of two cohorts. The disparity is significant compared to the fluctuation among the measurements within each category.

Randomisation

Once the blocks have been meticulously identified, it is important to note that there may still be other variables that might impact the experimental objective, such as variations in mouse age, sex, and genetic backgrounds. Attempt to achieve equilibrium between these elements, and we can employ randomisation approaches. Randomisation mitigates confounding effects, which refer to the distortion of the relationship between an exposure and an outcome due to unaccounted factors that influence the experimental units. It achieves this by equalising the influence of these variables across the experimental units, therefore minimising their impact on the experimental design. This necessitates the random assignment of the experimental units to the experimental conditions. The assignment of units to conditions should be non-deterministic.

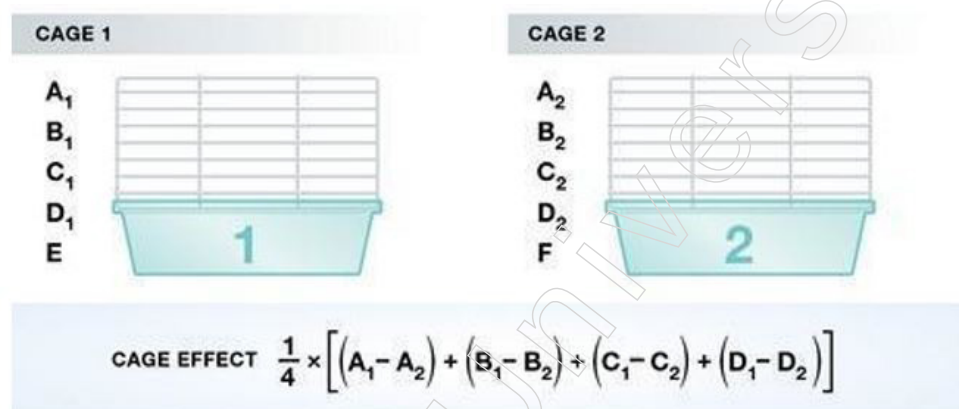
For instance, in an experiment that examines the impact of a genetic alteration on potato development, several intricate variables other than the genetic change itself may affect the growth: For instance, the growing chamber might have a minor temperature gradient, with one end being warmer than the other. Additionally, the quality of the compost may vary, and different watering systems may be employed. It will be imperative to randomise the placement of the plants in this scenario.

Blocking

We want to do experiments within a homogeneous group of experimental units. Those particular homogeneous groups called blocks try to reduce the variability between the units and increase the meaning of differences between conditions.

For instance, it is advantageous to conduct measurements across various experimental conditions simultaneously. If measurements are spread over an extended period, it becomes necessary to estimate and mitigate day-to-day variability between the measurements. When all control conditions are measured on one day and all treatment conditions on another, it becomes impossible to separate the day effect from the treatment effect, potentially rendering the data inconclusive. As a general guideline, it is crucial to include some “common conditions” to assess potential block effects.

Consider a scenario where we have six treatment conditions to apply to mice (the experimental units), but each cage (block) can only accommodate five mice. In such cases, not all treatments can be applied simultaneously within each cage/block. One approach is to apply four identical treatments to each cage and alternate the fifth condition each time. This allows estimation of the “cage effect” by calculating the mean differences between the four identical treatments, as described by the formula. Initially, conditions E and F are not directly comparable as they were measured on mice from different cages. However, by replicating treatments, we can compute a “cage effect” that represents the average difference between identical conditions measured in the two cages. Subsequently, the difference between E and F can be computed as $E - F - \text{“cage effect”}$.

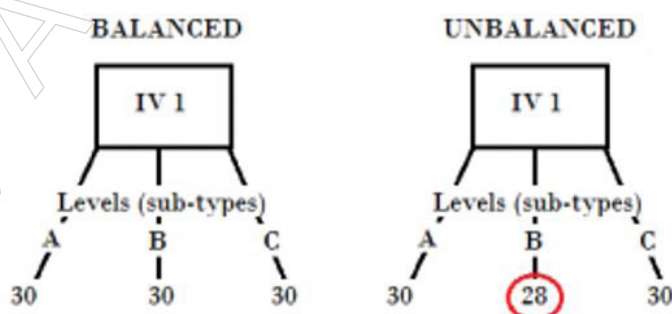


Here's an example demonstrating how to correct for batch effects:

Let's consider an experiment where all treatments except two have been applied to mice housed in two different cages (batches). The batch or cage effect can be calculated using the treatments that are administered to mice in both cages.

3.3.3: Orthogonality and False Positives

Orthogonality means “uncorrelated.” That means in an orthogonal model; all the independent variables are uncorrelated. However, if there is a correlation between one or more independent variables, the model is said to be non-orthogonal.



Notes

The design on the left is balanced because it has even levels.

The term “orthogonal” mostly pertains to traditional ANOVA. An orthogonal ANOVA is a statistical analysis that involves categorical independent variables. In this analysis, each cell in a two-way table contains an equal number of observations, meaning that the data is balanced. General linear models are not orthogonal because they always involve at least one independent variable that is not categorical, but rather a continuous variable.

In the field of calculus-based statistics, one may encounter orthogonal functions, which are defined as two functions with an inner product of zero. These functions are particularly valuable for solving partial differential equations such as Schrodinger's equation and Maxwell's equations.

Running Tests with Orthogonality

Orthogonality also impacts the execution of statistical tests. In orthogonal models, there is a singular method for estimating model parameters and conducting statistical tests. However, non-orthogonal models offer several approaches to do this, resulting in potentially more intricate interpretations of the outcomes. Therefore, a higher correlation among independent variables necessitates a careful interpretation of the results.

Determining whether a model is orthogonal or non-orthogonal might be subjective at times. For instance, consider a scenario where there are four cells in an ANOVA: three of these cells consist of 10 individuals each, whereas the fourth cell consists of nine subjects. This design is technically unbalanced, and as a result, it is also non-orthogonal. Nevertheless, the absence of a participant in one cell will have minimal influence on the outcomes. Put simply, you can consider this semi-unbalanced design to be orthogonal.

Roots in Matrix Algebra

The term “orthogonality” originates from the field of matrix algebra. Two vectors are considered orthogonal if the total of the products of their corresponding elements is equal to zero. For instance, the resultant of multiplying these two vectors is zero:

$$A = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 4 \\ -4 \\ 1 \\ 2 \end{pmatrix}$$

$$2(4) + 3(-4) + 4(1) + 0(2) = 0.$$

The identical matrix multiplication may be used for cells within a two-dimensional table. Orthogonality is a characteristic of a model where the total of the effects of any element is zero when considering the impacts of other factors in the table.

False Positives

A false positive occurs when a test yields a positive result instead of the expected negative result. Occasionally, this situation is referred to as a “false alarm” or a “false positive error.” Typically employed in the medical domain, it can also be applicable to other domains like software testing. Here are a few instances of false positives:

- The COVID-19 test shows a positive result, but the patient is not infected with COVID-19.

- A cancer screening test returns a positive result, but the patient does not have cancer.
- A prenatal test indicates Down's syndrome, but the patient does not have the disorder.
- Virus software on our computer wrongly identifies a harmless program as malicious.

False positives can be concerning, particularly in medical tests. Researchers are constantly investigating the causes of false positives to improve test sensitivity.

A related concept is a false negative, where a negative result is received when it should have been positive. For instance, a pregnancy test may show negative even though the patient is pregnant.

The False Positive Paradox

If a disease test is 99% accurate and we receive a positive result, what are the chances that the patient actually has the disease?

If we think it's 99%, we might be surprised to find out we're mistaken. When the disease is very common, our chances might approach 99%. However, differences

Guidelines for Designing Experiments

The suggested rules prioritise both statistical and non-statistical factors that are crucial for the outcome of experimental research. The rules are categorised into three distinct phases: planning, conducting, and analysis. These phases are designed to create a loop that allows management activities to be focused on continuously improving the efficiency and effectiveness of the experimental investigation. The stages are interconnected, and it may be essential to redefine what was previously specified at any given time.

The consistent infusion of subject matter expertise has the potential to alter the trajectory of an investigation, the conceptual framework being considered, the variables being employed, the responses being assessed, and the anticipation of potential enhancements (Box, 1996), illustrating the sequential and iterative nature of experimental research, as highlighted by Box et al. (1978). Strategizing Peace (1993) suggests that experimental investigations might fail not simply owing to a lack of technical understanding or incorrect use of statistical tools, but also because of inadequate preparation. In order to bridge this divide, it is crucial to focus on the technical and organisational management aspects that impact the successful implementation of experimental designs. This can be achieved by drawing from the experiences of Hahn (1984) in planning experimental programmes and considering the factors that either hinder or facilitate the process, as identified by Owen et al. (2003).

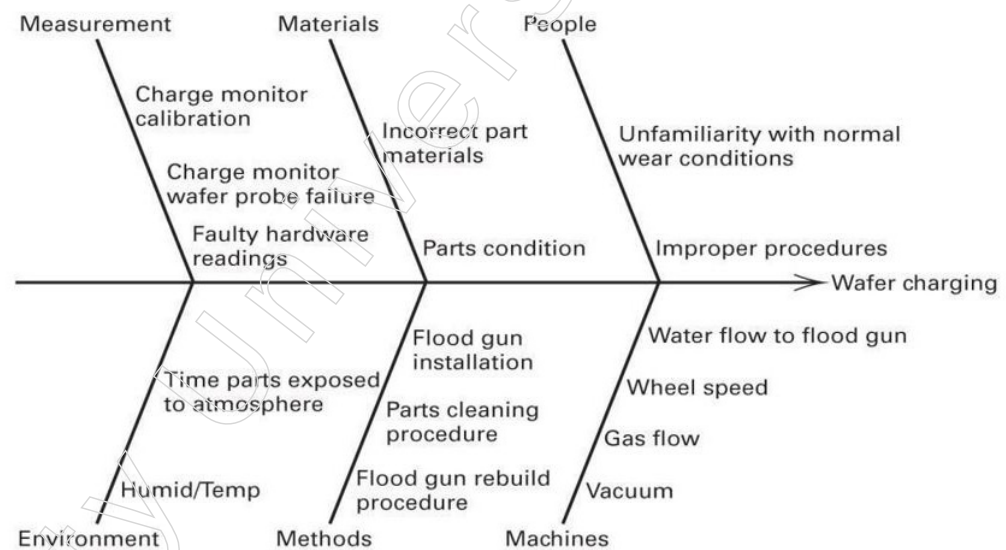
Organisational Management Aspects:

Responsibility of the staff. The duty for creating an atmosphere that fosters a culture of ongoing growth and change lies with senior management, who must demonstrate the right attitudes and provide effective leadership. The effectiveness of implementing statistical approaches in product and process development relies heavily on the motivation and dedication of the staff. In order to ensure the long-term sustainability of improvement efforts, it is essential to carefully allocate material and human resources based on cost-benefit analysis. Additionally, constant training of human resources in statistical techniques is vital to enhance product and process improvement. A well-designed training programme should assist workers in identifying

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novel methods to enhance the organisation and assume the responsibility for implementing these changes (Adebanjo and Kehoe, 1999).

Collaborative approach. Contemporary management approaches commonly employ interdisciplinary teams to gather and evaluate certain topics. In the biomedical and pharmaceutical business, interdisciplinary research teams usually collaborate closely (Bisgaard, 2005). Even when statistical approaches are used, the complexity of the process and product necessitates the formal establishment of interdisciplinary teams to exchange experiences and specialised technical expertise in order to achieve improvements. Statistical procedures, such as experimental design techniques, are effective only when used in combination with scientific knowledge, rather than as a replacement for it. The interdisciplinary team must consist of a formal supervisor who is appointed by senior management and has clearly defined tasks and authority. The supervisor should possess great communication skills, strong interpersonal abilities, and a good understanding of group dynamics. The team requires one individual with expertise in statistical approaches, as well as individuals from various hierarchical levels and activities, depending on the specific problem to be examined. Sharing experiences and individual expertise is essential to ensure a comprehensive comprehension of the process and the product, enabling more effective methods for designing experiments and achieving the research target.



When conducting experiments, it is important for the experimenter to remember the following key elements in order to effectively utilise statistical techniques:

- Apply your expertise in the issue area, utilising information specific to their respective disciplines.
- Strive to maintain a design and analysis that is as uncomplicated as feasible.
- Distinguish between practical and statistical importance.
- Experiments often include a process of repetition and refinement.

Summary

- Experimental research is carried out using a systematic manner, where one set of variables is kept constant while the other set of variables is measured as the focus of the experiment.
- A highly effective quantitative research technique, experimental research is

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particularly useful when the researcher does not have enough information to arrive at meaningful conclusions and have to carry out experiments to prove facts.

- Experimental research, thus, helps a researcher make important decisions.
- It is considered successful only when the researcher ensures that a change in the dependent variable is wholly due to a change in the independent variable.
- Validity pertains to the degree of accuracy with which a technique assesses the desired measurement.
- High validity in research indicates that the results align with actual traits, characteristics, and fluctuations in the physical or social environment.
- High reliability indicates that measurement is valid, but if a method is not reliable, then it's probably not valid.
- For example, if the thermometer shows different temperatures each time, even though you have carefully controlled conditions to ensure the sample's temperature stays the same, the thermometer is probably malfunctioning, and therefore its measurements are not valid.
- Data collection is the process of gathering information from various sources using different research methods and consolidating it into a single database or repository. This information is then used for further analysis to solve problems, track progress, and make informed decisions by individuals, businesses, and organisations.
- The primary goal of data collection is to provide accurate and relevant information that can aid in decision-making, problem-solving, and progress tracking.

Glossary

- Factorial Experimental Design: Factorial design allows the researcher to simultaneously examine several factors. It aids in determining the impact of individual factors and measuring the combined effect of several variables.
- Causal Attributions: A crucial component of an impact evaluation is not just to quantify or characterise the changes that have happened, but also to ascertain the involvement of certain interventions (such as programmes or policies) in causing these changes.
- Sole causal attribution: In this sense, the programme or policy is seen to be both required and sufficient to generate the desired effects, regardless of external variables or other actions.
- Alternative (or multiple) causal paths: Here, the plan or policy is only one of numerous potential methods to accomplish the desired effects. A specific programme or policy can generate the effects, but they can also arise from alternative actions and/or external variables.
- Statistical Control: Statistical control mostly occurs through the use of several control charts, which give essential information that acts as a foundation for taking appropriate actions.
- Validity: Validity pertains to the degree of accuracy with which a technique assesses the desired measurement.
- Reliability: Reliability is a measure of the stability or consistency of every test. We can also think of it as the ability for a test or research findings to be repeatable.

Notes

Check Your Understanding

1. What is a key characteristic of experimental research?
 - a) Keeping all variables constant
 - b) Measuring all variables simultaneously
 - c) Keeping one set of variables constant while measuring the other
 - d) Measuring variables randomly
2. Which of the following is NOT a factor that drives the success of experimental research?
 - a) A control group
 - b) Random allocation of individuals
 - c) Consistent manipulation of variables
 - d) Lack of specific objectives
3. In experimental research, what is the primary role of a control group?
 - a) To receive the experimental treatment
 - b) To be similar to the experimental group but not subject to the experimental criteria
 - c) To manipulate the variables
 - d) To ensure random allocation
4. Which type of experimental design is typically used when random assignment is not feasible?
 - a) Pre-experimental
 - b) True experimental
 - c) Quasi-experimental
 - d) Controlled experimental
5. What does the “manipulable variable” refer to in an experimental design?
 - a) The variable that remains unchanged throughout the experiment
 - b) The variable that is measured by the researcher
 - c) The variable controlled and changed by the researcher
 - d) The variable that is irrelevant to the experiment

Exercise

1. Explain the difference between the control group and the experimental group in an experimental research study. Why is it important to have both groups?
2. Describe the role of random allocation in experimental research. How does it contribute to the validity of the research findings?
3. List and explain three key factors that should be considered when designing an experiment. Provide an example of how each factor can influence the outcome of the experiment.
4. Discuss the significance of determining potential interdependencies between variables (variable interaction) in an experimental design. How can ignoring these interdependencies affect the results of an experiment?
5. Define pre-experimental research design and quasi-experimental research design.

Provide an example of a scenario where each design might be appropriately used and explain why.

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Learning Activity

1. Discuss with a peer or write down your thoughts on why experimental research is considered a highly effective quantitative research technique.
2. Discuss or write about the advantages and disadvantages of using quasi-experimental designs in educational settings compared to true experimental designs.

Check Your Understanding - Answers

1. c)
2. d)
3. b)
4. c)
5. c)

Further Readings and Bibliography

1. "Experimental and Quasi-Experimental Designs for Generalized Causal Inference" by William R. Shadish, Thomas D. Cook, and Donald T. Campbell
2. "Design and Analysis of Experiments" by Douglas C. Montgomery
3. "Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" by John W. Creswell and J. David Creswell
4. "Experimental Design: Procedures for the Behavioural Sciences" by Roger E. Kirk
5. "Introduction to Experimental Design" by John Lawson - An introductory textbook that covers experimental design principles and methods.
6. Campbell, D. T., & Stanley, J. C. (1963). "Experimental and Quasi-Experimental Designs for Research." Houghton Mifflin.
7. Cook, T. D., & Campbell, D. T. (1979). "Quasi-Experimentation: Design & Analysis Issues for Field Settings." Rand McNally College Publishing Company.
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9. Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). "Experimental and Quasi-Experimental Designs for Generalized Causal Inference." Houghton Mifflin.
10. Trochim, W. M. K., & Donnelly, J. P. (2008). "Research Methods: The Essential Knowledge Base." Cengage Learning.

Module - IV: Models and Hypothesis

Learning Objectives

At the end of this module, you will be able to:

- Discuss Hypothesis Testing
- Explain the concept of Parametric Test

Introduction

Hypothesis testing is a systematic approach to evaluating claims or ideas about a group or population. Also known as significance testing, it involves assessing a hypothesis or claim about a parameter within a population using data collected from a sample. This method helps determine the probability of selecting a sample statistic if the hypothesis regarding the population parameter were true.

4.1: Hypothesis Testing

Essentially, hypothesis testing is a statistical technique used to make decisions based on experimental data. It revolves around an assumption made about the population parameter, which is then tested against observed sample data.

During hypothesis testing, two mutually exclusive statements about a population are compared to ascertain which statement is better supported by the sample data. When a finding is deemed statistically significant, it indicates strong evidence supporting one of the hypotheses tested.

4.1.1 Introduction to Hypothesis

What is a Hypothesis

A hypothesis is a statement about the research problem and may or may not be true. It is a tentative and testable relationship between at least two variables that require research methodology for testing its authenticity. You are required to test the hypothesis and ascertain the truth in the way of research. Theory and hypothesis are completely different. Unlike a hypothesis, a theory is a tested, well-substantiated, complete explanation for a set of proven factors.

The term “hypo” means “subject to verification”, and “thesis” means “statement about the solution of a problem”. Thus, “hypothesis” is “a tentative statement, which is subject to verification and states the solution of a problem”. Hypothesis offers a solution to the research problem that must be verified empirically

Study the below illustration to grasp the concept better.

Illustration:

A renowned asset management company is in the process of launching a new Mutual Fund. The fund is aimed at catering to the wealth-building needs of the Indian middle-class population. The company wants to conduct detailed market research on the popular choices of investment options in India. It recruits a research team, which comes up with the research problem:

“Is Mutual Fund a popular investment option in India?”

The team constructs a trio of hypotheses, namely

H0: The Indian middle-class population does not invest in Mutual Funds

H1: At least 20% of the Indian middle-class population invests in Mutual Funds

H2: Less than 20% of the Indian middle-class population invests in Mutual Funds

Note that all three hypotheses have been phrased in a manner so that anyone is correct. However, note all three cannot be correct or incorrect simultaneously. Also, note that in order to make an effective decision, you need to select the first act if H1 is true and the second act if H2 is true. The decision of launching or not launching the Mutual Fund depends on whether H1 or H2 is true. The hypothesis H0 suggests that the Indian middle-class population is completely indifferent towards Mutual Funds

Decisions taken based on the hypothesis selected

If H1 Selected, then a decision is taken (Launching the Mutual Fund).

But If H2 selected, then a decision is (Not to launch the Mutual Fund).

Thus, the asset management company chooses to launch the mutual fund if H1 is true, i.e. if it is established from the market research that at least 20% of the Indian middle-class population invests in mutual funds. If established otherwise, i.e. if H2 is true, the company decides not to launch the mutual fund.

Formulation of Hypothesis

To discuss the process of formulating a hypothesis, we have to discuss a few concepts, which are related to that.

Types of Hypothesis

There are several types of hypothesis likes:

- Simple Hypothesis
- Complex Hypothesis
- Empirical Hypothesis
- Null Hypothesis
- Alternative Hypothesis
- Logical Hypothesis
- Statistical Hypothesis

Simple Hypothesis

In a simple hypothesis, there exists a relationship between two variables; one is called an independent variable or cause, and the other is a dependent variable or effect. For example

1. Smoking leads to Cancer
2. A high rate of unemployment leads to crimes.

Complex Hypothesis

In a complex hypothesis, there exists a relationship among more variables (more than two dependent and independent). For example

1. Smoking and other drugs lead to cancer, chest infections etc.

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2. The high rate of unemployment, poverty, illiteracy leads to crimes like dacoit, Robbery, Rape, prostitution & killing etc.

Empirical / Working Hypothesis

Working hypothesis is mainly applied to a particular field. At the time of formulation, it is an assumption, but when it is ready for a test to become an empirical or working hypothesis.

Logical Hypothesis

Here, the hypothesis is rationally validated. J.S. Mill has proposed four principles of this theory, including agreement, disagreement, difference, and residue.

Statistical Hypothesis

A hypothesis may be logical or illogical but can be verified statistically, called a statistical hypothesis.

4.1.2 Developing Hypothesis : Null and Alternate

Null Hypothesis

It is contrary to the positive statement of the empirical hypothesis, which has no relationship between dependent and independent variables. It is denoted by 'h₀'.

Alternative Hypothesis

Typically, several theories are considered, but one must be chosen that is both feasible and highly effective. The hypothesis is subsequently introduced as a result of modifications to the previously defined hypothesis. The term is represented by the symbol "h₁".

Testing Hypothesis and Significance Levels

Testing Hypothesis

The test statistic is a mathematical function or estimator that is calculated using the sample data and serves as the basis for making statistical decisions. The rejection region (RR) defines the range of values of the test statistic that lead to the rejection of the null hypothesis in favour of the alternative hypothesis.

- If the calculated value of the test statistic for a certain sample falls into the rejection region (RR), we will reject the null hypothesis (H₀) and accept the alternative hypothesis (H₁).
- If the value of the test statistic does not fall inside the rejection (critical) zone, we will accept the null hypothesis (H₀). The acceptance region refers to the region that is not part of the rejection region.

Significance Level

The significance level is the threshold probability used to make a decision between the null and alternative hypotheses. The probability level is insufficiently low to justify supporting the null hypothesis.

The significance level is typically denoted as a percentage, such as 5% or 1%. The level of significance, often set at 5%, represents the chance of rejecting the null hypothesis when it is actually true.

When the hypothesis is accepted at the 5% significance level, there is a potential danger that the statistician may make an incorrect judgement around 5% of the time in the long run.

4.1.3: Type I & Type II Error

Errors in Selecting Hypotheses

It is crucial to execute each stage of research with utmost precision, as even a small mistake in one phase might compromise the entire process and result in erroneous conclusions. It is important to recognise that the selection of a hypothesis has a significant risk of severe mistakes. The typical mistakes associated with selecting a hypothesis include Type1 & Type 2, which are described below.

Error	Explanation
Type I	H0 wrongly rejected
	H1 wrongly accepted
Type II	H0 wrongly accepted
	H1 wrongly rejected

Illustration:

A prestigious children's channel undertakes market research to assess the demand for advertising. The study team formulates the following hypotheses:

H0: Adults make up a minimum of 20% of the viewership for children's TV programming.

H1: Adults make up less than 20% of the viewership for children's TV programming.

It should be noted that if the null hypothesis H0 is correct, the corporation should engage in advertising. Conversely, if the alternative hypothesis H1 is valid, advertising is unnecessary.

It is crucial to formulate a hypothesis with great precision. There is a significant likelihood of mistakes occurring at every stage of creating a hypothesis. It is important to understand that hypotheses are designed in such a way that they cannot be both true and false at the same time. In order to prevent incorrect decisions, we must choose the first action (first hypothesis) if it is true, and the subsequent action (second hypothesis) if it is true.

Decision-making based on hypothesis testing carries the risk of mistakes at each stage. In order to elucidate flaws in hypotheses, it is crucial for us to familiarise ourselves with the concept of "state of nature," which refers to the authentic situation in the universe. The initial opportunity for mistake arises at the first stage of formulating a hypothesis, which involves establishing the connection between the different states of nature and the suggested course of action. A second opportunity for mistake may arise if the study findings are incorrect in relation to the true condition of nature.

The following table provides insight into the correlation between the various stages of nature and the suggested course of action. This consists of two possible scenarios and two different options for action.

Notes

True State		Conclusions	
		H0 True	H1 True
H0	True	No error	Type I error
H1	True	Type II error	No error

Based on the information provided in the table, it is clear that if the link between the two states of nature and the two acts is valid, there is no mistake in concluding that H0 is true. Similarly, it would be accurate to deduce that H1 is true without any mistake.

What are the consequences of the team accepting an incorrect hypothesis? Is it possible for the firm to make an incorrect decision? Allow me to elucidate by utilising two potential scenarios that may occur.

Hypothesis chosen	Status of acceptance	Type of error	Decision taken
H0	H0 wrongly rejected and H1 wrongly accepted	Type I	No advertisements
H1	H1 wrongly rejected and H0 wrongly accepted	Type II	Unnecessary costs incurred due to advertisements

4.1.4: One-tailed and Two-tailed Test

In a one-tailed test, the alternative hypothesis allows for the parameter being tested to be either bigger or less than the parameter under the null hypothesis, but not both simultaneously. The null and alternative hypotheses for a one-tailed test are formulated as follows:

Null Hypothesis	$H_0 : \mu = \mu_0$
Alternative Hypothesis	$H_1 : \mu < \mu_0 \text{ or } \mu > \mu_0$

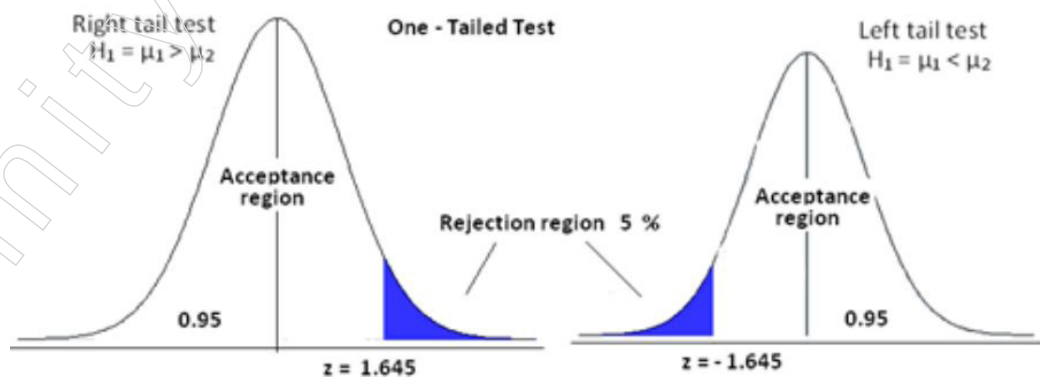


Figure: Image depicting one-tailed test

<https://prwatech.in/blog/wp-content/uploads/2019/07/onetailtest.png>

Think of a novel medicinal medication that they hope to demonstrate offers advantages over an earlier iteration. They complete their assigned task and carry out a flawless procedure. If the new medication is more successful than the previous one,

that is the only thing that matters right now. The same penalties apply to failing to reject the null hypothesis as they do to rejecting the null in the incorrect direction because if you don't reject it, the new drug's production and study won't proceed. In any case, the new medication won't be utilized, thus a one-tailed t-test will be set up to determine whether the medication is more effective at reducing headaches in particular.

In a two-tailed test, the alternative hypothesis permits the values of the parameter being researched to be either bigger or less than the values of the parameter under the null hypothesis.

The hypotheses for the two-tailed test are formulated as follows:

Null Hypothesis	$H_0 : \mu = \mu_0$
Alternative Hypothesis	$H_1: \mu \neq \mu_1$

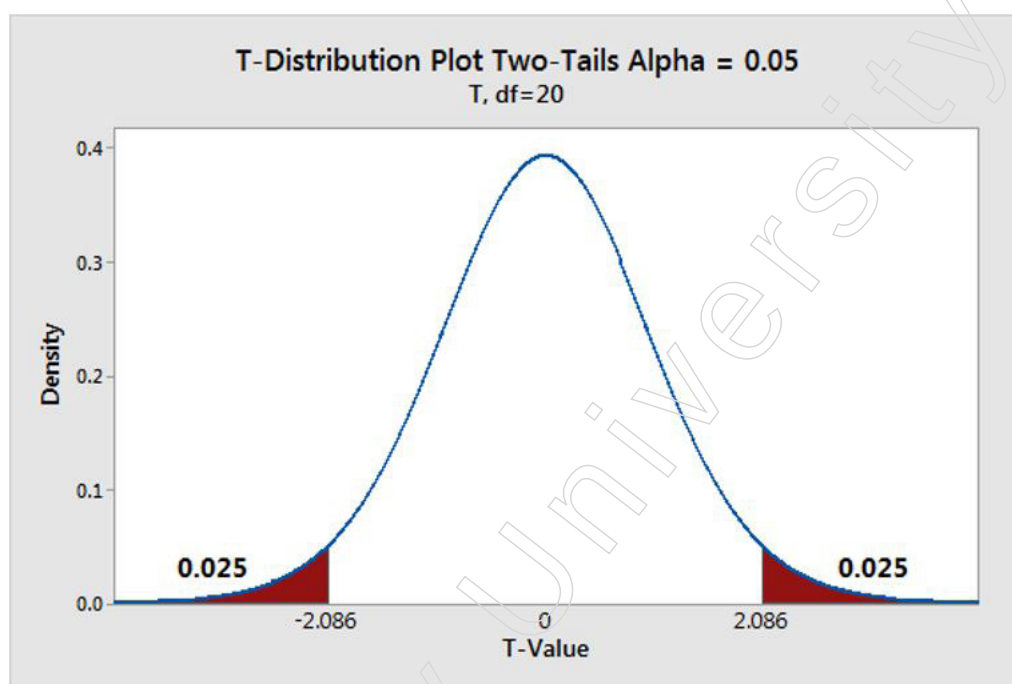


Figure: Image depicting two-tail test

Assume we compare the mean strength (4.1) of parts from a supplier to a target value (5) using a two-sided 1-sample t-test. Because it matters whether the mean is higher or lower than the desired value, we employ a two-tailed test.

Just compare the p-value to chosen significance threshold to understand the findings. You can determine which of the key zones the test statistic fell into if the p-value is smaller than the significance level. Take a look at the approximate impact. The test statistic fell in the key region in the left tail of the distribution, indicating the mean is smaller than the target value, as seen by the output below, where the t-value is negative.

4.1.5 Degree of Freedom

Definition

In a statistical calculation, the degrees of freedom represent the values involved in a calculation have the freedom to vary. Calculating the degrees of freedom is crucial

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for ensuring the statistical validity of chi-square tests, t-tests, and f-tests. These tests are frequently employed to compare observed data with data that would be anticipated based on a given hypothesis.

For example, let's suppose a vaccine trial is conducted on a group of patients, and it is hypothesized that the patients receiving the vaccine would show increased heart rates compared to those that did not receive the vaccine. The results of the test could then be analysed to determine whether the difference in heart rates is considered significant, and degrees of freedom are part of the calculations. Because degrees of freedom calculations identify how many values in the final calculation are allowed to vary, they can contribute to the validity of an outcome. Statistical calculations often rely on the sample size, or number of observations, and the parameters to be estimated.

Degrees of freedom = (The number of observations) – (The number of parameters).

The formula for Degrees of Freedom

Degrees of freedom equal the number of values in a data set minus 1 in a statistical formula like :

Degree of Freedom = $N-1$

Here N is the number of values in the data set (sample size). For example, we assume

If there is a data set of 4, that means $N=4$.

Let the data set is With the values: 15, 30, 25, 10

The mean value of the above data set is 20.

$$\begin{aligned}\text{Mean} &= \text{Sum of the values} / N \\ &= (15+30+25+10)/4 \\ &= (80)/4 \\ &= 20\end{aligned}$$

Using the formula, the degrees of freedom would be calculated as Degree of Freedom = $N-1$:

In this example, Degree of Freedom = $4-1 = 3$

This implies that within this dataset, three integers possess the capacity to fluctuate while ensuring that the average remains at 20.

4.1.6 Confidence Intervals

What exactly is a confidence interval?

A confidence interval is the mean of our estimate plus and minus the variation in that estimate. Exactly this is the range of values that we expect our estimate to fall between if we redo our test within a certain level of confidence.

Confidence, in statistics, is another way to describe probability. For example, if we construct a confidence interval with a 95% confidence level, then we are confident that 95 out of 100 times, the estimate will fall between the upper and lower values specified by the confidence interval.

In statistical test:

Confidence level = $1 - \alpha$

So if we use an alpha value of $p < 0.05$ for statistical significance, then our confidence level would be $1 - 0.05 = 0.95$, or 95%.

We can use confidence intervals for many kinds of statistical estimates, likes:

- ❖ Proportions
- ❖ Population means
- ❖ Differences between population means or proportions
- ❖ Estimates of variation among groups

All of these are single value estimates and do not provide any information regarding the range of values around the estimate. Confidence intervals serve the purpose of conveying the range of variance surrounding a certain point estimate.

For instance: To assess the level of deviation from an estimated value, we conducted a survey of 100 individuals from India and 100 individuals from America to inquire about their television-viewing behaviours. The results revealed that both groups have an average weekly television-watching duration of 35 hours.

Nevertheless, the Indian respondents exhibited significant disparity in the number of hours they watched, whereas the American participants all reported identical viewing durations.

Although the point estimate for both groups is the same (the average number of hours viewed), the Indian estimate will have a broader confidence interval than the American estimate due to more data volatility.

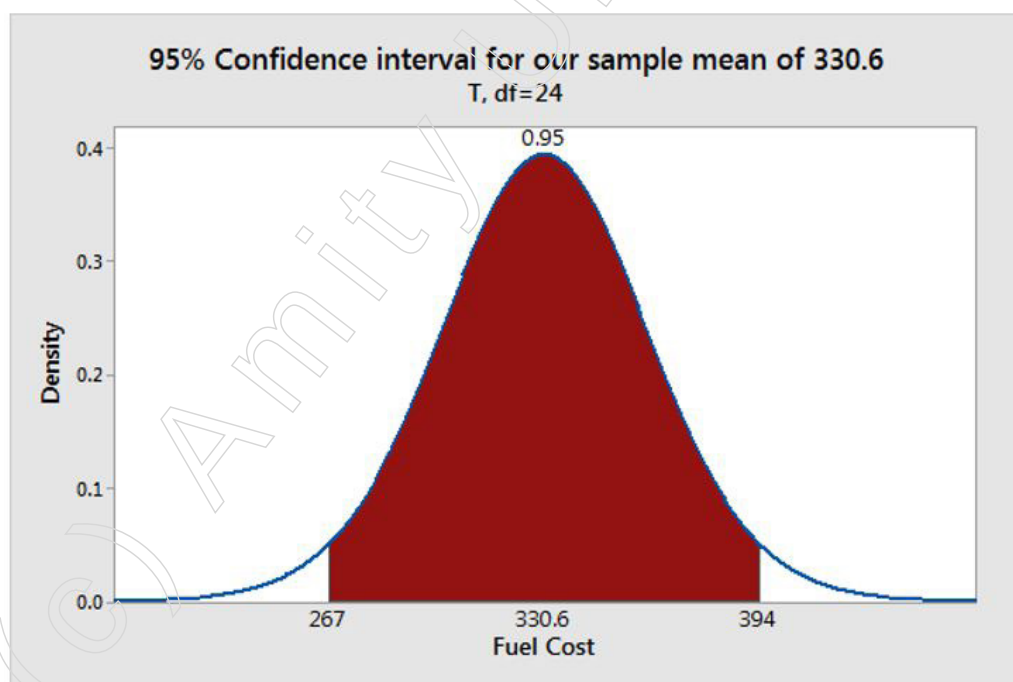
4.1.7: Level of Significance: The Confidence Coefficient

Figure: image depicting confidence interval

https://i0.wp.com/statisticsbyjim.com/wp-content/uploads/2017/04/CI_95.png?w=576&ssl=1

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On the other hand, we shade the center 95% of the distribution, and the sample mean is the center of the confidence interval graph.

The sampling distribution is covered by 95% of the sample means in the shaded range [267 394]. For our sample data, this range represents the 95% confidence interval. The population mean for fuel prices falls between 267 to 394, with a 95% confidence interval.

4.1.8: Rejection and Non-rejection Regions

The critical region, also known as the rejection region, encompasses all values of the test statistic that lead to the rejection of the null hypothesis.

A critical value is a threshold that distinguishes the rejection region from the non-rejection region.

The significance level (represented by α) denotes the probability of the test statistic falling within the critical region when the null hypothesis is true.

- I) For a left-tailed test, the alternative hypothesis (H_1) includes the less-than symbol ($<$), indicating that it's a test focused on detecting values significantly lower than the expected.
- II) In a right-tailed test, the alternative hypothesis (H_1) includes the greater-than symbol ($>$), indicating a focus on values significantly higher than the expected.
- III) A two-tailed test, indicated by the not-equal-to symbol ($\neq$) in the alternative hypothesis (H_1), involves both tails of the distribution and is used to detect deviations in either direction from the expected value. In a two-tailed test, each tail has an area of $\alpha/2$.

The decision rule based on the rejection (critical) region is as follows:

- ❖ Calculate the standardised test statistic (z or t).
- ❖ If the standardised test statistic:
- ❖ Falls within the rejection region, rejecting H_0 .
- ❖ Does not fall within the rejection region, fail to reject H_0 .

It's essential to understand that failing to reject the null hypothesis doesn't imply acceptance of it as true; rather, it indicates insufficient evidence to reject it.

A Type I error transpires when the null hypothesis is rejected despite being true. Denoted by α , it represents the probability of this error occurring—specifically, the likelihood of rejecting the null hypothesis when it's actually true.

On the other hand, a Type II error arises when the null hypothesis is not rejected even though it's false. This error is characterised by the probability of failing to reject the null hypothesis when it's actually false, typically denoted by β .

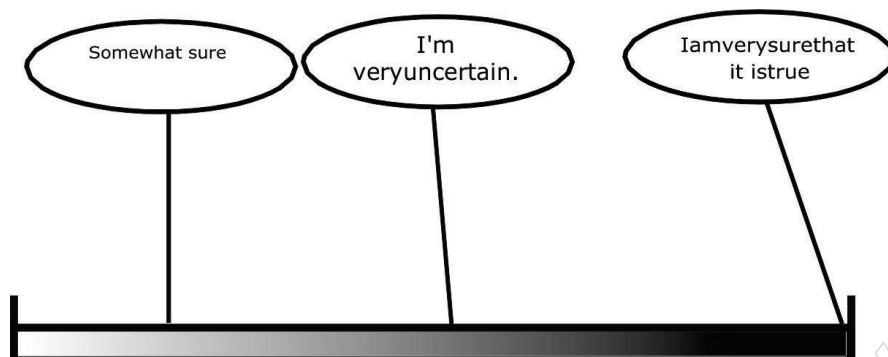
Bayesian Statistics

Bayesian statistics are a mathematical procedure that applies probabilities to statistical problems. In the field of Bayesian statistics, probability is defined as a measure of the degree of certainty regarding the truth of a statement or proposition.

- When the probability is 1, we can be certain that the assertion is true.
- When the probability is 0, we may confidently conclude that the claim is untrue.
- If the probability is 0.5, then our level of uncertainty is equivalent to that of a fair coin toss.

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If the probability is 0.95, then we have a high level of confidence that the statement is true, however we would not be overly shocked if we discovered that the statement was untrue.



The graphic above illustrates that probability may be employed to quantify levels of confidence or the plausibility of a given proposition. 0 and 1 represent the two ends of the spectrum and signify absolute confidence. Nevertheless, probabilities are not fixed quantities. As we acquire additional knowledge, our probabilities have the potential to alter.

Bayesian statistics is sometimes mistakenly seen as a simple process of pondering an issue and instinctively stating a probability that seems fitting. As an illustration, in a scenario where we are participating in the game show “Who Wants to Be a Millionaire?” and encounter a question for which we lack the solution, we may estimate that the likelihood of option A being correct is 25%. However, if we utilize the “phone a friend” option and our buddy unequivocally states, “It is undoubtedly A,” our level of certainty for option A would significantly increase. It is unlikely that our likelihood would reach 100%.

Now, we will examine a straightforward example to illustrate the fundamental principles of Bayesian statistics.

- The initial probability of the problem is referred to as prior probabilities.
- How precisely are these changed when we receive further information? The revised probabilities are referred to as posterior probabilities.
- In order to enhance clarity, we will employ a table known as a Bayes’ Box to facilitate the straightforward calculation of posterior probabilities.

Assume there are two balls in a bag, where:

- at least one of them is black,
- but we’re not sure whether they’re both black
- or whether one is black and one is white.

For the above, we consider two possibilities: either both are black in colour or one is white, and another is black so that we can label our two competing hypotheses BB and BW. So, at the beginning of the problem, we know that one and only one of the following statements/hypothesis is true:

BB: Both balls are black

BW: One ball is black and the other is white.

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Suppose an experiment is conducted to assist us in determining the veracity of either of these two theories. The researcher inserts their hand into the bag, extracts a single ball, and scrutinises its hue. The outcome of this experiment is

D: The ball that was removed from the bag was black.

The Bayes Box

In a Bayesian analysis, we begin by selecting prior probabilities for our two competing hypotheses, BB and BW. We also need to assign probability values that reflect our level of confidence in the truth of each hypothesis. Given that we are considering two hypotheses, there will therefore be two prior probabilities: one for the hypothesis of both parents having the same genotype (BB) and one for the hypothesis of the parents having different genotypes (BW). To simplify, we shall assume a lack of accurate knowledge and so rely on the following initial probabilities:

$$P(BB) = 0.5$$

$$P(BW) = 0.5.$$

The above two hypotheses are mutually exclusive (they can't both be true) and exhaustive (one of these is true, it can't be some undefined third option). The selection of 0.5 as the prior probability signifies our high level of uncertainty regarding the actual hypothesis before conducting the experiment. Now introduce a Bayesian Box, which displays a comprehensive list of all the possible hypotheses that may be true together with their corresponding prior probability. There are other columns that have not been addressed yet, and they are necessary to calculate the posterior probability in the last column. The initial column of a Bayes' Box comprises the roster of hypotheses under consideration.

Hypotheses	prior	likelihood	prior × likelihood	posterior
BB	0.5			
BW	0.5			
Totals:	1			

There are just two in this scenario. To create a Bayes' box for a new problem, just consider the potential solutions to the problem and record them in the first column. The second column displays the prior probability associated with each of the hypotheses. Prior to doing the experiment, we established a 50% likelihood for BB being true and a 50% likelihood for BW being true. This is why the column has values of 0.5. The preceding column must always have a total sum of 1. It is important to note that the prior probabilities just represent our initial level of uncertainty, prior to considering the evidence.

Likelihood

The third column is called likelihood by which we can calculate the posterior probabilities. It is equivalent to probability. Within the field of statistics, likelihood refers to a distinct type of probability. In order to complete the third column of the Bayes' Box, it is necessary to compute two distinct likelihoods, indicating that the likelihood varies for each hypothesis. However, could you perhaps provide a more precise definition?

Here is the Bayes' Box with the likelihood column completed.

Notes

Hypotheses	prior	likelihood	$h = \text{prior} \times \text{likelihood}$	posterior
BB	0.5	1		
BW	0.5	0.5		
Totals:	1			

Initially, compute the probability value for the BB hypothesis. It is important to note that the data being analysed pertains to the selection of a ball from the bag in a random manner, and the ball chosen happened to be black. The probability of the BB hypothesis is equivalent to the probability of obtaining a black ball if the BB hypothesis is correct.

Consider the scenario when BB is assumed to be true. Consequently, both balls are black. What is the likelihood that the experiment would yield a black ball? The answer is straightforward - it is 100%. We assigned the value of 1 to the Bayes' Box as the likelihood for the BB hypothesis.

Now envision a scenario when BW is indeed true. That would imply that one ball is black and the other ball is white. Assuming this hypothetical scenario and conducting the experiment, what would be the likelihood of obtaining the black ball? Given that one of the two balls is black, the probability of selecting the black ball is 50%. Thus, the probability of the BW hypothesis is 0.5, which is why we assign a value of 0.5 to the likelihood for BW in the Bayes' Box.

Hypotheses	Possible Data	Probability
BB	BlackBall	1
	WhiteBall	0
BW	Black Ball	0.5
	White Ball	0.5

Generally, the likelihood refers to the probability of obtaining the observed data, given that a specific hypothesis is correct. In this scenario, obtaining the likelihoods was rather straightforward by means of direct inquiry. "If this hypothesis is true, what is the probability of getting the black ball when we do the experiment?". Occasionally, this task may not be straightforward, and it might be advantageous to consider all conceivable experimental results or data that could have been seen - despite the fact that ultimately, we only need to choose the one that really happened.

This table illustrates a technique for computing the probability values by taking into account not only the observed data but also all possible data. In essence, the significance lies solely in the likelihood of the observed data, which is emphasised by being marked in the colour blue.

The Mechanical Part

The third column of the Bayes' Box is obtained by multiplying the prior probabilities and the likelihoods together. The outcome will be referred to as "prior times likelihood", with the occasional usage of the letter h to represent these values. This is the posterior that has not been normalised. The total of the values does not equal 1, which is the expected sum for posterior probabilities. However, the values are at least proportional to the actual posterior probabilities.

In order to calculate the posterior probabilities, we divide the values in the prior likelihood column by their sum, resulting in numbers that add up to 1. These are the

Notes

ultimate posterior probabilities, which were the intended objective from the beginning. Below is the Bayes' Box that has been finished.

Hypotheses	prior	likelihood	h = prior × likelihood	posterior
BB	0.5	1	0.5	0.667
BW	0.5	0.5	0.25	0.333
Totals:	1		0.75	1

It is evident that the posterior probability differs from the prior probabilities due to the acquisition of more information. The experimental outcome has increased the plausibility of BB to a certain extent. The likelihood has risen from 1/2 to 2/3.

Interpretation

The posterior probability of the hypotheses are directly proportional to the prior probabilities and the likelihoods. A hypothesis with a high prior probability is more likely to have a high posterior probability. To comprehend the implications for reasoning, it is necessary to examine the definitions of the prior and the likelihood. Two factors can contribute to the plausibility of a hypothesis:

Assume that the initial probability is elevated. Prior to obtaining the data, the hypothesis was already credible.

Assuming that the theory accurately anticipated the data. Essentially, the facts aligned with our expectations if the hypothesis had been accurate.

Bayes Rule:

Bayes' rule is an equation from probability theory and conditional probabilities.

For instance, the equation's left-hand side represents $P(A|B)$, which denotes the conditional probability of A given B. That is, it represents the likelihood of event A occurring after considering the information provided by event B. Put simply, $P(A|B)$ is a posterior probability, and Bayes' rule provides a method for computing it based on other probabilities. Bayes' theorem holds valid for any given propositions A and B.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

In Bayesian, Statistics A has been replaced by H, and B has been replaced by D. The purpose of these letters is to designate H as hypothesis and D as data. Bayes' rule may be understood as providing the likelihood of a hypothesis based on certain facts, in other words, a posterior probability.

$$P(H|D) = \frac{P(H) P(D|H)}{P(D)}$$

Bayesian statistics assigns specific names to the majority of the concepts in Bayes' rule. Some species even possess several designations, since various scientific societies choose distinct language. Below is a compilation of the different terms along with the corresponding names we will assign to them:

The posterior probability, denoted as $P(H|D)$, represents the likelihood of a hypothesis H given the observed data D. It quantifies the level of certainty or confidence in the truth of hypothesis H, based on the observed evidence D. The primary objective of Bayesian statistics is to compute posterior probabilities.

Notes

The prior probability $P(H)$ quantifies our initial level of confidence in the truth of hypothesis H prior to observing the evidence D .

The probability is denoted as $P(D | H)$. Given the assumption that H is true, the chance of observing data D is as follows.

The marginal probability is denoted as $P(D)$. This represents the likelihood of observing data D , regardless of whether H is true or not.

Phone Example

This example is derived from a question in David MacKay's remarkable book "Information Theory, Inference and Learning Algorithms" (accessible online as a complimentary PDF download). Please feel free to examine it, but it is a voluminous tome and only around 20% of its contents pertain to this particular course.

You relocate to a new residence that is equipped with a telephone. You are unable to recall the phone number, but you have a suspicion that it may be 555-3226 (some individuals may identify this as the phone number associated with Homer Simpson's "Mr Plough" enterprise). In order to verify this theory, you conduct an experiment by physically grabbing the telephone and inputting the numbers 555-3226 into the dial pad.

If your assumption regarding the phone number's accuracy is accurate, you will undoubtedly get a busy signal when you are attempting to contact your own number. If you make a mistake, the likelihood of hearing a busy signal is 1 in 100. Nevertheless, all of those assertions have validity just under the assumption that the phone is functioning, although there is a possibility that it is defective. When the phone is in a state of disrepair, it will consistently emit a busy signal.

During the experiment, the observed outcome, or data, confirms the presence of a busy signal. The prompt instructed us to evaluate four hypotheses and determine their posterior probability through calculation. It is important to understand that the four possibilities are completely separate and cannot occur at the same time.

Hypothesis	Description	Prior Probability
H1	The phone is working, and 555-3226 is correct	0.4
H2	The phone is working, and 555-3226 is incorrect	0.4
H3	The phone is broken, and 555-3226 is correct	0.1
H4	The phone is broken, and 555-3226 is incorrect	0.1

The table presented above contains four assumptions regarding the condition of the phone and its associated phone number. The initial probability is also provided.

Thorough. If you were to generate hypotheses independently, you may consider the possibilities of "the phone is functioning properly" and "the phone number 555-3226 is accurate." These two possibilities cannot be considered as mutually exclusive in a Bayes' Box. However, it is feasible to combine them using the logical operator "and" to create the four mutually exclusive options in the table.(DIY)

4.1.9 Hypothesis Testing Procedure

Hypothesis testing follows a structured procedure to evaluate the validity of a statement. Here's a stepwise guide:

Notes

- **Formulate Hypotheses:** Begin by stating the initial hypothesis, which is then reformulated into a null hypothesis (H_0) and an alternative hypothesis (H_a). The null hypothesis assumes no relationship between variables, while the alternative hypothesis predicts a specific relationship.

Example:

- ❖ H_0 : On average, there is no difference in height between men and women.
- ❖ H_a : On average, men are taller than women.
- **Collect Sample Data:** Obtain data through sampling, ensuring it represents the population under consideration.
- **Choose Hypothesis Testing Method:** Select an appropriate statistical method to test the hypotheses based on the nature of the data and the research question.
- **Calculate Test Statistic:** Compute a test statistic (e.g., z-score, t-statistic) based on the sample data and chosen method.
- **Determine P-value:** The p-value quantifies the probability of observing results as extreme as those obtained, assuming the null hypothesis is true. It's the probability of obtaining the observed result, or more extreme results, under the assumption that the null hypothesis is true.
- **Interpret P-value:** Compare the obtained p-value with a predetermined significance level (α), commonly set at 0.05. If the p-value is less than α , there's sufficient evidence to reject the null hypothesis, indicating a significant result. Conversely, if the p-value is greater than α , we fail to reject the null hypothesis due to insufficient evidence.
 - ❖ Low p-value ($< \alpha$): Reject the null hypothesis.
 - ❖ High p-value ($\geq \alpha$): Fail to reject the null hypothesis.
- A low p-value suggests that the observed difference is unlikely to have occurred by chance alone, supporting the alternative hypothesis. Conversely, a high p-value indicates that the observed difference is likely due to random variation, supporting the null hypothesis.
- **Draw Conclusion:** Based on the comparison between the p-value and the significance level, make a decision regarding the null hypothesis. Interpret the results in the context of the research question.

4.1.10 Types of Tests

Statistical tests serve as essential tools in data analysis, facilitating the examination of relationships between variables through hypothesis testing. They allow for drawing inferences about populations based on sample data. These tests are broadly classified into two categories: parametric and non-parametric.

- **Parametric Tests:** Parametric tests rely on specific assumptions about the underlying data distribution, such as normality and homogeneity of variance, to make accurate inferences. Examples of parametric tests include t-tests, ANOVA, and linear regression.
- **Non-Parametric Tests:** Non-parametric tests do not require any assumptions about the data distribution, making them more robust in certain situations, especially when data violates parametric assumptions. Examples of non-parametric tests include the Wilcoxon signed-rank test, Mann-Whitney U test, and Kruskal-Wallis test.

Categorisation by Research Focus: Quantitative and Qualitative Analysis

Statistical tests can also be categorised based on their application in quantitative or qualitative research. Quantitative research deals with numerical data, while qualitative research focuses on non-numerical data. Statistical tests used in qualitative research, particularly when analysing categorical data, play a crucial role in uncovering relationships and associations between different qualitative or categorical variables.

4.1.11 How to Select a Test

Selecting the Right Test: 7 Essential Steps

- **Define the Research Question:** Begin by identifying the specific research question you aim to answer. This question guides the entire process, from data collection to analysis.
- **Formulate Null Hypothesis:** Develop a null hypothesis that states there is no statistical significance in the expected observations. This serves as the baseline for comparison in hypothesis testing.
- **Determine Level of Significance:** Specify the level of significance in your study protocol, which dictates the threshold for statistical importance. This level determines whether to accept or reject the null hypothesis.
- **Choose Between One-Tailed and Two-Tailed Tests:** Decide whether your study warrants a one-tailed or two-tailed test. One-tailed tests are suitable when evidence points clearly in one direction, while two-tailed tests are appropriate when no specific direction is anticipated.
- **Consider Number of Variables:** Select statistical tests based on the number of variables to be analysed. Different tests are designed for single-variable or multivariable analyses.
- **Assess Data Type:** Identify whether your data is continuous, categorical, or binary. For continuous data, determine if it's normally distributed or skewed, as this influences the choice of statistical test.
- **Account for Study Design:** Distinguish between paired and unpaired (independent) study designs. Paired designs involve comparing dependent samples, while unpaired designs compare independent samples.

4.2 Parametric Test

Parametric tests are statistical tests that rely on assumptions about the parameters of the population distribution from which the sample data is drawn. These tests are based on a fixed set of parameters used to determine a probabilistic model, which can also be applied in machine learning.

Fixed Set of Parameters: Parametric tests operate under the assumption that specific parameters (such as mean and standard deviation) define the population distribution.

Population Distribution: Typically, parametric tests assume that the population follows a normal distribution. If the population distribution is unknown, the Central Limit Theorem can often be used to approximate it as normal.

Notes

Key Parameters for Normal Distribution:

- ❖ Mean (μ)
- ❖ Standard Deviation (σ)

Why Do We Need Parametric Tests?

The classification of a test as parametric depends on the assumptions made about the population. Parametric tests are useful for various purposes, including:

- Confidence Interval for Population Mean (Known Standard Deviation): Used to estimate the population mean when the standard deviation is known.
- Confidence Interval for Population Mean (Unknown Standard Deviation): Used to estimate the population mean when the standard deviation is unknown, often involving the t-distribution.
- Confidence Interval for Population Variance: Used to estimate the population variance.
- Confidence Interval for the Difference of Two Means (Unknown Standard Deviation): Used to estimate the difference between two population means when the standard deviation is unknown, often using the t-distribution.

4.2.1 Important Parametric Tests

Parametric tests are statistical tests that assume the underlying data follows a normal distribution. These tests are powerful tools in statistical analysis, relying on certain assumptions about the population parameters from which the samples are drawn.

Key Parametric Tests

1. t-test

The t-test is a statistical hypothesis test introduced by W.S. Gossett under the pen name "Student." It is commonly used to assess whether there is a significant difference between the means of two groups, particularly when dealing with small sample sizes (less than 30) and unknown population variance. The t-test statistic follows a t-distribution with

$n-1$ degrees of freedom.

Types of t-tests:

- ❖ One-sample t-test: Compares the mean of a single sample to a known population mean.
- ❖ Independent two-sample t-test: Compares the means of two independent samples.
- ❖ Paired t-test: Compares the means of two related samples, often used for "before and after" scenarios.

Example:

To determine if there is a significant difference in blood pressure before and after administering a new medication, a paired t-test is appropriate.

2. Z-test

The Z-test is used for comparing the mean of a sample to a hypothesised population

mean, especially when dealing with large sample sizes (greater than 30) or when the population variance is known. It is also used to compare the means of two independent samples in large samples.

Example:

To compare the average salaries of men and women in an organisation, a Z-test can be utilised.

3. F-test

The F-test is used to compare the variances of two independent samples. It helps determine if there is more variability in one sample compared to another. The F-test is also integral to ANOVA (Analysis of Variance), which compares the means of three or more groups simultaneously.

Example:

To compare the variability of bolt diameters produced by two different machines, an F-test can be applied.

4.2.2 Hypothesis Testing of Means: Single Population Mean Using Z-statistic

A Z-test is a statistical method used to determine whether the mean of a sample is significantly different from a known population mean when the population standard deviation is known. It is particularly useful for large sample sizes ($n > 30$). The Z-test can also be used to determine whether the distribution of test statistics can be approximated by a normal distribution.

Applications of Z-Test

- Comparing Sample Mean to Population Mean: Determines if the sample mean significantly differs from the population mean.
- Comparing Two Sample Means: Assesses if two sample means are approximately the same or different when their variances are known, and the sample sizes are large ($n \geq 30$).

A Z-test can be:

One Sample Z-test: To compare a sample mean with that of the population mean.

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$\bar{x}$ = sample mean
 μ = population mean
 σ = population standard deviation
 n = sample size

Two Sample Z-test: To compare the means of two different samples.

Notes

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where,

- ❖ $\bar{x}_1$ is the sample mean of 1st group
- ❖ $\bar{x}_2$ is the sample mean of 2nd group
- ❖ σ_1 is the population-1 standard deviation
- ❖ σ_2 is the population-2 standard deviation
- ❖ n is the sample size

4.2.3 Hypothesis Testing of Means: Single Population Mean Using T-statistic

The t-test is a parametric test used in hypothesis testing based on Student's t-distribution. It is primarily used to determine if there is a significant difference between the means of two groups, especially when the sample size is small (i.e., less than 30) and the population standard deviation is unknown.

Assumptions of the T-Test

- Normal Population Distribution: The population from which the sample is drawn is normally distributed.
- Random and Independent Samples: The samples are randomly selected and independent of each other.
- Small Sample Size: The sample size is small (typically less than 30).
- Unknown Population Standard Deviation: The population standard deviation is not known.

Types of T-Tests

- One-Sample T-Test: Compares the mean of a single sample to a known population mean.
- Independent Two-Sample T-Test: Compares the means of two independent samples.
- Paired T-Test: Compares the means of two related samples, often used for "before and after" measurements.

One Sample T-test: To compare a sample mean with that of the population mean.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where,

- ❖ $\bar{x}$ is the sample mean
- ❖ s is the sample standard deviation
- ❖ n is the sample size
- ❖ μ is the population mean

Two-Sample T-test: To compare the means of two different samples.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}}}$$

where,

- ❖ $\bar{x}_1$ is the sample mean of the first group
- ❖ $\bar{x}_2$ is the sample mean of the second group
- ❖ S_1 is the sample-1 standard deviation
- ❖ S_2 is the sample-2 standard deviation
- ❖ n is the sample size

4.2.4: Hypothesis Testing for Population Proportions

Hypothesis testing for population proportions involves determining whether a sample proportion significantly differs from a known or hypothesised population proportion. This is particularly useful when dealing with categorical data.

Steps in Hypothesis Testing for Population Proportions

1. State the Hypotheses:
 - ❖ Null Hypothesis (H_0): The population proportion is equal to a specified value.
 - ❖ Alternative Hypothesis (H_a): The population proportion is different from the specified value. This could be one-tailed or two-tailed depending on the context of the research question.
2. Set the Significance Level (α):
 - ❖ Common choices for α are 0.05, 0.01, or 0.10.
3. Calculate the Test Statistic:
 - ❖ For large sample sizes, the test statistic is a Z-score calculated using the formula:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where:

- $\hat{p}$ = Sample proportion
 - p_0 = Hypothesized population proportion
 - n = Sample size
4. Determine the Critical Value and Decision Rule:
 - ❖ Using the standard normal distribution, determine the critical value(s) for the given significance level. For a two-tailed test with $\alpha=0.05$, the critical values are ± 1.96 .
 5. Make a Decision:

Notes

Notes

- ❖ Compare the calculated Z-score to the critical value(s):
 - ◆ If the Z-score falls in the rejection region, reject the null hypothesis.
 - ◆ If the Z-score does not fall in the rejection region, fail to reject the null hypothesis.
- 6. Conclusion: Based on the decision, conclude whether there is enough evidence to support the alternative hypothesis.

4.2.5: Hypothesis Testing for Comparing a Variance to Hypothesised Population Variance

The chi-square test is a statistical method used to determine whether there is a significant difference between the observed variance of a sample and a hypothesised population variance. This test is particularly applicable to normally distributed populations. The hypotheses for this test can be structured in three ways:

1. Two-tailed test:
 - ❖ $H_0: \sigma^2 = \sigma_0^2$
 - ❖ $H_a: \sigma^2 \neq \sigma_0^2$
2. Right-tailed test:
 - ❖ $H_0: \sigma^2 \leq \sigma_0^2$
 - ❖ $H_a: \sigma^2 > \sigma_0^2$
3. Left-tailed test:
 - ❖ $H_0: \sigma^2 \geq \sigma_0^2$
 - ❖ $H_a: \sigma^2 < \sigma_0^2$

Chi-Square Distribution and Test Statistic

Chi-Square Test

Used for hypothesis tests concerning the variance of a normally distributed population

σ_0^2 : Hypothesised value of the variance

Null Hypothesis, H_0

Alternative Hypothesis, H_a

$$\sigma^2 = \sigma_0^2$$



$$\sigma^2 \neq \sigma_0^2$$

$$\sigma^2 \leq \sigma_0^2$$



$$\sigma^2 > \sigma_0^2$$

$$\sigma^2 \geq \sigma_0^2$$



$$\sigma^2 < \sigma_0^2$$

The chi-square test concerning variance uses the chi-square distribution characterised by $n-1$ degrees of freedom, where n is the sample size. Since variance cannot be negative, the chi-square distribution is asymmetrical and starts from zero. As the degrees of freedom increase, the shape of the chi-square distribution approaches that of the normal distribution.

The test statistic for the chi-square test is calculated as follows:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

where

- χ^2 = chi-square test statistic
- n = sample size
- s^2 = sample variance
- σ_0^2 = hypothesized population variance

Reading a Chi-Square Table

To illustrate how to read a chi-square table, consider a two-tailed test with a 10% significance level ($\alpha=0.10$) and 10 degrees of freedom. Here are the steps:

1. Determine the Critical Values:
 - ❖ For a two-tailed test, split the significance level into two tails, so each tail has $\alpha/2=0.05$.
 - ❖ Locate the column in the chi-square table where the probability to the right is 0.05. For 10 degrees of freedom, this critical value is approximately 18.307.
 - ❖ Locate the column where the probability to the right is 0.95 (which corresponds to 5% to the left). For 10 degrees of freedom, this critical value is approximately 3.94.
2. Decision Rule:
 - ❖ Reject the null hypothesis (H_0) if the calculated chi-square statistic is less than 3.94 or greater than 18.307.

4.2.6: Testing the Equality of Variances of Two Normal Populations

The F-test, developed by Snedecor and Cochran (1983), is used to determine if the variances of two populations are equal. This test can be conducted as either a two-tailed test or a one-tailed test, depending on the research question.

The F Test for Equality of Variances between two groups is a statistical test used to compare the variances of two samples to determine whether they are equal. It is based on the assumption that the samples are drawn from normally distributed populations.

Steps in the F Test for Equality of Two Variances:

1. Specify the null and alternative hypotheses. The null hypothesis is usually that the variances of the two samples are equal, while the alternative hypothesis is that the variances of the two samples are not equal.
2. Select two samples from the populations and calculate the sample variances and sizes.
3. Calculate the test statistic, which is the ratio of the larger sample variance to the smaller sample variance.
4. Determine the critical value of the test statistic based on the significance level (α) of the test and the degrees of freedom for the numerator and denominator.

Notes

The degrees of freedom for the numerator and denominator are calculated as the sample sizes minus 1.

5. Compare the calculated test statistic to the critical value to determine whether to reject or fail to reject the null hypothesis. If the calculated test statistic exceeds the critical value, the null hypothesis is rejected, and the alternative hypothesis is accepted.

Conditions for the F Test for Equality of Two Variances:

To conduct a valid F test for the equality of two variances, the following conditions must be met:

1. The samples must be drawn randomly from the populations.
2. Each observation in each sample must be independent of the others.
3. The population distributions must approximate a normal distribution.

Hypothesis Testing of Correlation Coefficients

When given a sample of bivariate data (data which include two variables), it is possible to calculate how linearly correlated the data are, using a correlation coefficient.

The correlation coefficient, r , tells us about the strength and direction of the linear relationship between X_1 and X_2 .

The sample data are used to compute r , the correlation coefficient for the sample. If we had data for the entire population, we could find the population correlation coefficient. But because we have only sample data, we cannot calculate the population correlation coefficient. The sample correlation coefficient, r , is our estimate of the unknown population correlation coefficient.

- ρ = population correlation coefficient (unknown)
- r = sample correlation coefficient (known; calculated from sample data)

The hypothesis test lets us decide whether the value of the population correlation coefficient ρ is “close to zero” or “significantly different from zero”. We decide this based on the sample correlation coefficient r and the sample size n .

If the test concludes that the correlation coefficient is significantly different from zero, we say that the correlation coefficient is “significant.”

- Conclusion: There is sufficient evidence to conclude that there is a significant linear relationship between X_1 and X_2 because the correlation coefficient is significantly different from zero.
- What the conclusion means: There is a significant linear relationship X_1 and X_2 . If the test concludes that the correlation coefficient is not significantly different from zero (it is close to zero), we say that correlation coefficient is “not significant”.

Performing the Hypothesis Test

- Null Hypothesis: $H_0: \rho = 0$
- Alternate Hypothesis: $H_a: \rho \neq 0$

Hypotheses Mean in Words

- Null Hypothesis H_0 : The population correlation coefficient IS NOT significantly different from zero. There IS NOT a significant linear relationship (correlation) between X_1 and X_2 in the population.

- Alternate Hypothesis H_a : The population correlation coefficient is significantly different from zero. There is a significant linear relationship (correlation) between X_1 and X_2 in the population.

Drawing a Conclusion

There are two methods of making the decision concerning the hypothesis. The test statistic to test this hypothesis is:

$$t_c = \frac{r}{\sqrt{(1-r^2)/(n-2)}}$$

Or

$$t_c = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Where the second formula is an equivalent form of the test statistic, n is the sample size and the degrees of freedom are $n-2$. This is a t -statistic and operates in the same way as other t tests. Calculate the t -value and compare that with the critical value from the t -table at the appropriate degrees of freedom and the level of confidence you wish to maintain. If the calculated value is in the tail then cannot accept the null hypothesis that there is no linear relationship between these two independent random variables. If the calculated t -value is NOT in the tail then cannot reject the null hypothesis that there is no linear relationship between the two variables.

4.2.7 ANOVA for Fixed Effect Model

Analysis of Variance

Introduction:

Definition: Variance analysis is that the study of deviations of actual behaviour versus forecasted or planned behaviour. This primarily focuses on how the disparity between actual and planned behaviours reflects the influence on company success. Variance analysis may be divided into two distinct stages:

1. Computing and documenting individual discrepancies
2. Comprehending the cause behind each deviation

We can use variance due to

1. Market conditions have changed, making it impractical to continue employing quality budgeting procedures. For example, there is a shortage of raw materials, leading suppliers to increase prices.
2. The budgeting rules followed may also be excessively utopian, for instance, the output of a machine might be incorrectly anticipated.
3. The quality of service delivery may not meet expectations, for example, the initial plan may have accounted for eight hours of work per day, but the real conditions on the ground may only allow for six hours of work per day.
4. In certain instances, planning may lack a foundation, such as when the results of creative endeavours cannot be accurately measured against a high standard.

Notes

Variances are also classified under the below-mentioned heads:

1. **Material Variances:** - These arise from the difference between actual costs of materials utilized in production and standard costs of materials specified for the product produced. This comes into play due to the difference in quantities consumed and the quantity initially allocated for production. This can also happen due to the difference in price paid and the price budgeted for materials used.
2. **Labour variances:-** This refers to the specific remuneration given to employees in comparison to the prevailing wage for the defined output. An adverse variance occurs when the actual labour expenses exceed the anticipated expenditures.
3. **Overhead variances:-** It may be described as the aggregate of indirect material, labour, and expenditure costs. Overhead variances can occur when there is a discrepancy between the planned standard overhead expenses and the actual overhead costs incurred.

ANOVA for Fixed Effect Model:

The concepts presented in the following portion of this module, which include the design and analysis of an actual experiment, may be easily comprehended with the use of concise explanations and practical illustrations. Curious and interested readers can consult the Oscar Kempthorne textbooks for more real sugar (Kempthorne 1952; Hinkelmann and Kempthorne 2005; Hinkelmann and Kempthorne 2008)!

While identical twins are selected as experimental units and subjected to similar experimental settings, it is important to acknowledge that natural variances exist in practically every aspect of the universe. Thus, the answer is determined by the factors or explanatory variables, together with some experimental error, as shown in Equation 1.

$$\text{Response} = f(\text{factor}) + \text{error}$$

Means Model

The research issue in the fuel economy study is if there is a disparity between fuel types in terms of the average miles per gallon (mpg). Hence, Equation 1 may be expressed to evaluate the disparity in average response (measured in mpg) across different treatments (fuel kinds, in this instance) as seen in Equation 2. The model is referred to as the means model since it is based on the mean treatment effect. The population means model is designed to generalise the results of a sample to the entire population. Additionally, it is worth mentioning that the model is linear due to the exponent of the mean term (μ) being one.

$$y_{ij} = \mu_i + \epsilon_{ij}$$

μ_i = mean response due to i th treatment.

The error term ϵ_{ij} is assumed to be identical, independent and normally distributed.

Effects Model

Let's assume that the three fuel kinds are created by blending three distinct chemicals that enhance fuel efficiency when combined with a standard base gasoline. The pure fuel, devoid of any additives, will effectively power the vehicle and enable it to go a certain distance. By isolating the impacts of the ingredients from the base fuel efficiency, the model Equation 2 may be expressed as Equation 3.

$$y_{ij} = \mu + \epsilon_{ij} + T_i$$

μ = mean response or overall mean

T_i = i^{th} treatment effect

The error term ϵ_{ij} is assumed to be identical, independent and normally distributed

Error

The error could occur from many different sources, including the following

Experimental Error

- Treatment error refers to the mistake that occurs when we are unable to reproduce a treatment from one instance to another.
- Error - An error caused by stochastic fluctuations in the physical condition of an EU selection.
- Issue - The problem is caused by the random selection of EUs for the trial.

Observational Error

- Measuring error refers to the inaccuracies that arise from the lack of precision in our measuring or scoring methods.
- Sampling error refers to the mistake that arises from the random selection of observational units (OUs) for a study or research.

4.2.8 Total, Treatment & Error of Squares

After determining the error sum of squares (SSE), we may proceed to determine the treatment sum of squares (SSTR) and the total sum of squares (SST). When calculating the sum of squares error (SSE), sum of squares treatment (SSTR), and total sum of squares (SST), the next step is to determine the mean square error (MSE) and mean square treatment (MSTR), which may be used to compute the test statistic.

How to compute the sum of squares for treatment:

Once the SSE has been determined, our subsequent task is to calculate the SSTR. This is a quantification of the extent of variability in the average lifespans of different battery types. Given a low SSTR (Systematic Storage Time Ratio), the average lifespans of the various battery types are comparable.

Firstly, it is necessary to compute the overall average for the sample, which is referred to as the overall mean or grand mean.

For instance, let's consider a scenario where a producer selects a random sample consisting of four Electrical batteries, four Ready forever batteries, and four Voltages now batteries, and subsequently evaluates the lifespans of these batteries. This table provides a comprehensive overview of the outcomes, which are measured in units of hundreds of hours.

Battery Lifetimes (in Hundreds of Hours)			
Sample	Electrical	Ready forever	Voltage now
Battery 1	2.4	1.9	2.0
Battery 2	1.7	2.1	2.3
Battery 3	3.2	1.8	2.1
Battery 4	1.9	1.6	2.2

Notes

To get the overall mean, we may add up the 12 sample values obtained from four batteries of each of the three battery types listed in the table, and then divide the sum by 12.

$$\frac{2.4+1.7+3.2+1.9+1.9+2.1+1.8+1.6+2.0+2.3+2.1+2.2}{12} = \frac{25.2}{12} = 2.1$$

we then compute the SSTR with the following steps for each column:

1. Calculate the square of the difference between the mean of a column and the mean of the entire dataset.
2. Compute the product by multiplying the outcome with the total count of items in the column.

So in this example, SSTR equals

$$SSTR = 4(2.3 - 2.1)^2 + 4(1.85 - 2.1)^2 + 4(2.15 - 2.1)^2 = 0.42$$

The calculations are based on the following results:

- ❖ There are four observations in each column.
- ❖ The overall mean is 2.1.
- ❖ The column means are 2.3 for column 1, 1.85 for column 2 and 2.15 for column 3.

After you compute SSE and SSTR, the sum of these terms is calculated, giving the SST.

How to compute the total sum of squares

The total sum of squares (SST) equals the sum of the SSTR and the SSE. So using the battery example, we get

$$\begin{aligned} SST &= SSTR + SSE \\ &= 0.42 + 1.52 \\ &= 1.94 \end{aligned}$$

4.2.8 Total, Treatment & Error of Squares

4.2.9: ANOVA for Random Effect Model

Two-way factorial analysis of variance (ANOVA) This is an expansion of the one-component scenario to incorporate a second element. The values of this second element are often established based on the categorisation of participants or units involved in the study. Therefore, it is sometimes referred to as a blocking factor as it categorises topics or units into similar groupings known as blocks. The design is referred to as a randomised block design.

A computer maker business seeks to evaluate the performance of four compilers developed by the company. The manufacturer can utilize a 50% proportion of experimental designs.

1. Utilise 20 identical programmes and distribute 5 programmes randomly to each compiler.
2. Utilise four duplicates of any five programmes, assigning one duplicate of each programme to each compiler.

Solution

In (1), the compilation timeframes and potential results may be influenced by any variations among the 20 programmes, notwithstanding their similarities. Therefore, in the worst-case situation, the 5 programmes assigned to the fastest compiler may actually have the longest compilation times, making the compiler appear to be the slowest. If utilised, the outcomes would necessitate a one-factor analysis of variance, with the component being the compiler at four distinct levels.

In (2), the findings should not be affected by any variations between the programmes, as all 5 programmes are executed on each compiler. Using 5 programmes with distinct characteristics will be beneficial for making more comprehensive comparisons of compilation times. This design consists of two factors: the compiler, which has four levels, and the programme, which has five levels. The primary element of interest is the compiler, but the opposing factor, the programme, can be seen as a hindering factor due to its creation of five blocks, each having four identical copies of the programme.

Therefore, inquiry (2) is the most well-designed option.

The table below displays the precise compilation times, measured in milliseconds, for this two-factor (randomised block) architecture.

1		Compiler			
		2	3	4	
Program	A	29.21	28.25	28.20	28.62
Program	B	26.18	26.02	26.22	25.56
Program	C	30.91	30.18	30.52	30.09
Program	D	25.14	25.26	25.20	25.02
Program	E	26.16	25.14	25.26	25.46

Assumptions and Interaction

The three assumptions for a two-way analysis of variance with just one observed measurement for each combination of values of the two components are as follows.

- The population at each combination of issue-level is typically dispersed in an approximate manner.
- The traditional populations have a characteristic variation, σ^2 .
- The impact of one issue is consistent across all degrees of the other issue.

Therefore, based on assumptions 1 and 2, when one component is at level i and the other at level j , the population follows a distribution that is

$$N(\mu_{ij}, \sigma^2).$$

Assumption 3 may be restated as the absence of any interaction between the two components. Interaction occurs when the impact of one element is influenced by the degree of another factor. For instance, let's examine the impact of two variables: the amount of sugar (with levels of none and 2 teaspoons) and the duration of stirring (with levels of none and 1 minute), on the perceived sweetness of a cup of tea.

Notes

Notes

Agitating has no impact on the level of sweetness in the absence of sugar, but it surely does have an impact when sugar is introduced. In a similar vein, the addition of sugar has minimal impact on the level of sweetness until the tea is agitated.

Therefore, it is believed that sugar and stirring have an interactive relationship.

Assessing interaction requires taking many measurements at each combination of component levels. Given the limitations of this book, it will always be assumed that there is no interaction between the two components in such cases.

Therefore, given that it is quite unlikely to discover a compiler that is specifically well-suited for a particular programme, it is acceptable to assume that there is no interaction between compilers and programmes.

Can we consider the premise of no interaction to be valid for the data acquired in Activities 3 and 4?

Notation and Computational Formula

As illustrated earlier, the data for a two-factor ANOVA can be displayed in a two-way table. It is thus convenient, in general, to label the factors as a row factor and a column factor.

Notation, similar to that for the one-factor case, is then as follows.

Number of levels of row factor $= r$

Number of levels of column factor $= c$

Total number of observations $= rc$

Observation in (ij) th cell of table $= x_{ij}$

(i) th level of row factor and $i = 1, 2, \dots, r$

(j) th level of column factor $j = 1, 2, \dots, c$

Sum of c observations in i th row $= T_{Ri} = \sum x_{ij}$

Sum of r observations in j th column $= T_{Cj} = \sum x_{ij}$

Sum of all rc observations $= T = \sum x_{ij} = \sum T_{Ri} = \sum T_{Cj}$

The degrees of freedom for SST (Total Sum of Squares), SSR (Regression Sum of Squares), and SSC (Column Sum of Squares) may be calculated when there are 20 observations in a table with 5 rows and 4 columns. What is the number of degrees of freedom for the sum of squared errors (SSE)?

Anova table and hypothesis tests

For a two-factor analysis of variance, the structure is as follows.

Source of variation	Sum of squares	Degrees of freedom	Mean square	F ratio
Between rows	SS _R	$r - 1$	MS _R	MS _R MS
				E
Between columns	SS _C	$c - 1$	MS _C	MS _C MS
				E
Error (residual)	SS _E	$(r - 1)(c - 1)$	MS _E	
Total	SST	$rc - 1$		

4.2.10: Single Factor Experiment

Frequently we wish to investigate the effect of a factor (independent variable) on a response (dependent variable). Then we carried out an experiment where the levels of the factor are varied. Such an experiment is known as a single-factor experiment. There are various types of designs available to carry out such experiments. The most popular ones are completely randomized design, randomized block design, Latin square design, and balanced incomplete block design.

Completely randomized design (CRD):

The CRD is the simplest design. Suppose there are v treatments to be compared. All experimental units are considered the same, and no division or grouping among them exist.

- In CRD, the v treatments are allocated randomly to the whole set of experimental units, without
- making any effort to group the experimental units in any way for more homogeneity. Design is entirely flexible in the sense that any number of treatments or replications may be used.
- The numbers of replications for different treatments need not be equal and may vary from treatment to
- treatment depends on the knowledge (if any) on the variability of the observations on individual treatments as well as on the accuracy required for the estimate of individual treatment effect.

Example: Suppose there are four treatments and 20 experimental units, then

- the treatment one is replicated, say three times and is given to 3 experimental units,
- the treatment two is replicated, say five times and is given to 5 experimental units,
- treatment three is replicated, say six times and is given to 6 experimental units and
- Finally, treatment four is replicated $[20 - (6 + 5 + 3) = 6]$ six times and is given to the remaining six experimental units.
 - ❖ All the variability among the experimental units goes into experimental error.
 - ❖ CRD is used when the experimental material is homogeneous.
 - ❖ CRD is often inefficient.
 - ❖ CRD is more useful when the experiments are conducted inside the lab.
 - ❖ CRD is well suited for the small number of treatments and for the homogeneous experimental material.

The layout of CRD Following steps are needed to design a CARD:

- Divide the entire experimental material or area into a number of experimental units, say n
- Fix the number of replications for different treatments in advance (forgiven total number of available experimental units).
- No local control measure is provided as such except that the error variance can be reduced by choosing a homogeneous set of experimental un

Procedure Let the v treatments are numbered from $1, 2, \dots, v$ and I_n be the number of replications required for i the treatment such that

Notes

Notes

v

$$\sum_{n_i=n}$$

$$n_{=i}$$

- Select n_1 units out of n units randomly and apply treatment 1 to these n_1
- Select n_2 units out of $(n - n_1)$ units randomly and apply treatment 2 to these n_2 units
- Continue with this procedure until all the treatments have been utilized.
- Generally, an equal number of treatments are allocated to all the experimental units unless no practical limitation dictates or some treatments are more variable or/and of more interest.

Analysis There is only one factor that is affecting the outcome – treatment effect. So the set-up of one-way analysis of variance is to be used. :

y_{ij} : Individual measurement of j^{th} experimental units for i^{th} treatment $i = 1, 2, \dots, v$, $j = 1, 2, \dots, n_i$.

y_i : Independently distributed following $N(a_i, \mu, \sigma^2)$ with $\sum_{n=i}^v$

$n=i$

μ : overall mean

α_i : i^{th} treatment effect

$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_v = 0$

H_1 : All ' α_i 's are not equal.

Data Layout for Single-Factor Experiments

Treatment observations totals averages

$$1. \quad y_{11} y_{12} \cdots y_{1n_1} y_{1\cdot} \bar{y}_{1\cdot}$$

$$2. \quad y_{21} y_{22} \cdots y_{2n_2} y_{2\cdot} \bar{y}_{2\cdot}$$

.....
.....

$$A. \quad y_{a1} y_{a2} \cdots y_{an_a} y_{a\cdot} \bar{y}_{a\cdot}$$

Analysis of Variance

- Statistical Model (Factor Effects Model): $y_{ij} = \mu + \tau_i + q_{ij}$ $i = 1, 2, \dots, a$
 $j = 1, 2, \dots, n_i$

μ - grand mean;

τ_i - i^{th} treatment effect;

iid

$q_{ij} \sim N(0, \sigma^2)$ - error;

Constraint: $\tau_a = 0$ Conceptual Approach; SAS: $\tau_a = 0$

Estimates for parameters:

$$\hat{\mu} = \bar{y}_{\cdot}$$

$$\hat{\tau}_i = (\bar{y}'_i - \bar{y}_{\cdot})$$

$$\hat{Q}_{ij} = y_{ij} - y'_{i.}$$

Table 3-3 The Analysis of Variance Table for the Single-Factor, Fixed Effects Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Between treatments	$SS_{\text{Treatments}} = n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2$	$a - 1$	$MS_{\text{Treatments}}$	$F_0 = \frac{MS_{\text{Treatments}}}{MS_E}$
Error (within treatments)	$SS_E = SS_T - SS_{\text{Treatments}}$	$N - a$	MS_E	
Total	$SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$	$N - 1$		

$$SS_t = SS_{\text{treatment}} + SS_e$$

$SS_{\text{Treatments}}$: sum of squares of differences between treatment averages and grand average.

- ❖ It is a measure of differences between treatment means.
- ❖ It has (a-1) DOF

SS_E : sum of squares of differences of observations within treatments from the treatment average.

- ❖ It is due to random error.
- ❖ It has (N-a) DOF.
- ❖ $E(MSE) = \sigma^2$

Example

Twelve lambs are randomly assigned to three different diets. The weight gain (in two weeks) is recorded. Is there a difference between the diets?

Diet 1 8 16 9

Diet 2 9 16 21 11 18

Diet 3 15 10 17 6

- ❖ $N = 12$, $\sum \sum y_{ij} = 156$ and $y'_{..} = 156/12 = 13$.
- ❖ $n_1 = 3$, $y_{1.} = 33$, $y'_{1.} = 11$; $n_2 = 5$, $y_{2.} = 75$, $y'_{2.} = 15$; $n_3 = 4$, $y_{3.} = 48$ and $y'_{3.} = 12$.
 $\hat{\tau}_1 = y'_{1.} - y'_{..} = 11 - 13 = -2$; Similarly, $\hat{\tau}_2 = 15 - 13 = 2$ and
 $\hat{\tau}_3 = 12 - 13 = -1$.
- ❖ $SS_T = \sum_i \sum_j (y_{ij} - y'_{..})^2 = 246$.
- ❖ $SS_{\text{Treatment}} = 3 * (-2)^2 + 5 * (2)^2 + 4 * (-1)^2 = 36$.
- ❖ $SS_E = 246 - 36 = 210$;

$$MS_E = \hat{\sigma}^2 = 210/(12 - 3) = 23.33$$

- ❖ $F_0 = (36/2)/(210/9) = 0.77$;
- P-value > 0.25;

Fail to reject H_0 : $\tau_1 = \tau_2 = \tau_3 = 0$.

Notes

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Estimation of Variance Components

Abstract

Estimating variance components is a commonly employed technique in the fields of population genetics and animal breeding. Even seasoned population geneticists may feel overwhelmed when faced with the extensive array of approaches available for variance component estimate. This is primarily because there is no universally superior strategy, making it challenging to determine which approach to use. Nevertheless, the one-way layout has little practical significance. However, it may be effectively utilised to elucidate the fundamental ideas, which are applicable to higher categories, to animal scientists. The symbols utilised are the conventional biometric symbols as outlined in Raschet et al. (1994). All the procedures provided by SPSS are highly recommended.

Introduction

We analyse techniques for estimating variance components in the simplest scenario, the one-way ANOVA, and illustrate most of these methods using a specific dataset. Assuming a vast universe of factor levels, we consider a sample of levels of a random factor A. To avoid excessive abstraction, let us suppose that the levels may be represented by sires. A random sample of daughters is taken from the i -th sire, and their milk output y_{ij} is recorded.

The scheme of the $N = \sum_{i=1}^a n_i$ observations are given in Table 1.

A case is considered balanced when an equal number of daughters has been picked for each sire. Balance in higher nested categorisation refers to the equal distribution of subclass numbers and the same number of layers of nested factors inside each level of factors of a higher order in the hierarchy. Some of the procedures discussed below only differ from others when dealing with imbalanced data, which implies that in the one-way ANOVA, the number of daughters for each sire is not the same. Consequently, we created a synthetic data set that was not evenly distributed.

Formulae are not provided for all procedures, and just the idea is discussed. The purpose of this article is to provide a fundamental grasp of various ways for non-mathematicians who frequently utilise many techniques in their application of specialised methods. The merits and drawbacks of the approaches are considered.

Observations in a one-way layout of ANOVA – unbalanced case

level A_1	level A_2	...	level A_a
y_{11}	y_{21}	...	y_{a1}
y_{12}	y_{22}	.	y_{a2}
.	.	.	.
.	.	.	.
.	.	.	.
y_{1n_1}	y_{2n_2}	...	y_{an_a}

Following Rasch et al (1999) the model equation is given in (1)

$$y_{Fehler! \text{ Textmarkenichtdefiniert}.ij} = E(y_{Fehler! \text{ Textmarkenichtdefiniert}.ij}) + e_{Fehler! \text{ Textmarkenichtdefiniert}.ij} \\ \text{Textmarkenichtdefiniert}.ij = \mu + \alpha_i + e_{Fehler! \text{ Textmarkenichtdefiniert}.ij} \quad i = 1, \dots, a; j = 1, \dots, n_i. \quad (1)$$

The α_i are the main effects of the levels A_i . They are random variables. The ε_{ij} are the errors and also random. The constant μ is the overall mean. Model (1) is completed by the assumptions ($E()$ = expectation of = mean of; $V()$ = variance of).

$$E(\alpha_i) = 0, \quad V(\alpha_i) = \sigma_a^2, \quad E(\varepsilon_{ij}) = 0, \quad V(\varepsilon_{ij}) = \sigma^2; \text{ all components on the right hand side of (1) are independent (2)}$$

σ_a^2 and σ^2 are called variance components. The total number of observations is always denoted by N , in the balanced case we have $N = an$. In the sequel we give the formula for the balanced case, generalisation can be found in the references.

Let us assume that all the random variables in (1) are normally distributed even if this is not needed for all the methods.

The normal distribution (Gauss-distribution) has the density (or likelihood) function

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2} \quad (\sigma > 0) \quad (3)$$

We say that $y \sim N(\mu; \sigma^2)$ – distributed. If we have a random sample $y^T = (y_1, \dots, y_n)$ its density is equal to the product of the densities of its components.

From (1), (2) and the normality assumption it follows that the α_i and the ε_{ij} are independently from each other $N(0; \sigma_a^2)$ – and $N(0; \sigma^2)$ – distributed respectively. The y_{ij} are not independent from each other $N(\mu; \sigma^2 + \sigma_a^2)$ – distributed.

The dependence exists between variables within the same factor level because $\text{cov}(y_j, y_l) = \text{cov}(\mu + \alpha_i + \varepsilon_j, \mu + \alpha_i + \varepsilon_l) = \text{cov}(\alpha_i, \alpha_i) = \text{var}(\alpha_i) = \sigma_a^2$, if $j \neq l$. A standardized measure of this dependence is the intra-class correlation coefficient

$$\rho_C = \frac{\sigma_a^2}{\sigma_a^2 + \sigma^2}$$

The Analysis of Variance (ANOVA) Table is that of Table 2

Table 2: ANOVA Table of the one-way ANOVA model II in the balanced case

Source of variation	SS	df	MS	E(MS)
Factor A	$S_A = \sum_{i=1}^a \sum_{j=1}^n (\bar{y}_{i.} - \bar{y})^2$	$a-1$	$M_A = \frac{S_A}{a-1}$	$s^2 + n s_a^2$ $\sigma^2 + \frac{n}{a-1} \sum a_i^2$
Residual	$S_R = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$	$a(n-1)$	$M_{Res} = \frac{S_{Res}}{a(n-1)}$	s^2
Total	$S_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y})^2$	$an-1$		

Notes

$$\bar{y}_{i.} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_j$$

Some of the methods are developed for unbalanced data. We assume therefore also a generalisation of (1) by a general linear model for one random factor in (5).

$$Y = X\alpha + U\hat{a} + e \quad (5)$$

with random vectors Y and e of length N , design matrices X ($N \times q$) and U ($N \times m$), a vector α of length q with fixed effects and a random vector β of length m . We complete the model (5) by the following assumptions:

$\text{rank}(X) = q$; $\text{rank}(U) = m$. e and β are independently normally distributed with expectation vector zero and covariance matrix $\sigma^2 I_N$ and $\sigma_a^2 I_m$ respectively.

Note: For more details see Sorensen, A. and Gianola, D. (2002).

Properties of Estimators

Estimation is a branch of statistics and signal processing that uses measured and observed data to determine the values of parameters. Estimation is performed to quantify and diagnose the accurate value of a function or a specific piece of information. It is conducted by making observations on samples, which are representative subsets of the target population or function. Various statistical measures are employed to carry out the process of estimation.

Two crucial aspects of estimators that are commonly utilised are the estimator itself and the estimate it produces. In order to have a comprehensive understanding of the idea of the estimator and estimate, we shall illustrate it using an example. Suppose we have a set of samples, denoted as a_1, a_2, a_3 , and so on, which are taken from a certain population with 'x' as its parameter. If $T = T(a)$ is a statistic, then the expected value of $T(a)$ is equal to x. These equations demonstrate that a statistical estimation has been performed, with the statistic T serving as the estimator and the parameter 'x' being estimated.

Estimation possesses several crucial attributes for the optimal estimator. These attributes encompass characteristics such as impartiality, effectiveness, uniformity, and adequacy.

Unbiased estimators are those that yield findings with zero bias for all parameter values during estimate. Unbiased estimators are characterised by having the mathematical expectation or mean equal to the parameter of the target inquiry. In the above example of estimating, T will be considered an unbiased estimator only if its estimate is equal to 'x'.

Consistent estimators are estimators that provide estimates that are reliable and accurate. As the number of random variables increases, the degree of concentration around the estimate should also grow in order to ensure that the estimator of estimation is a consistent estimator. A consistent estimator is one that provides an unbiased estimate and has a variance of zero. These two requirements must be satisfied only when the number of random variables approaches infinity.

estimating involves many estimators that frequently exhibit consistent behaviour. According to the principle of efficiency in estimating, these consistent estimators should follow a normal distribution. This feature was considered in the theory of estimating

due to instances where estimators produced consistent estimates but were not efficient estimators.

An adequate estimator is one that satisfies the requirement that the joint conditional distribution function of the sample/observation depends solely on the values of T_1, T_2, T_3, T_4 , and so on, which are the values obtained from the supplied estimator function 'T'. Therefore, the resulting joint conditional estimation must be completely independent of the parameter 'x'. During the process of estimating, it is important for a researcher to understand that the most optimal estimator is the minimum variance unbiased estimator (MVUE). This implies that the estimator exhibits the lowest level of variability in comparison to other estimators.

Methods of Variance Component Estimation

Before we discuss some of the existing methods of estimation let us make some general considerations.

Definition 1: An estimator is a mapping y on the parameter space of the parameter θ of the distribution of this sample. The realization of an estimator is called an estimate.

Definition 2: An estimator is unbiased if $E = \theta$. The difference $E - \theta = \text{vn}(\theta)$ is the bias of an estimator. The bias of an unbiased estimator is zero. The expression is

(MSE). The mean squared error of an unbiased estimator equals its variance. In Figure 1

Example 1 The estimator $\hat{\mu} = \frac{\sum_{i=1}^n y_i}{n} = \bar{y}$ of the mean μ of the components of a random sample $y^T = (y_1, \dots, y_n)$ is unbiased.

Definition 3 An estimator $\hat{\theta} = \hat{\theta}(y)$ of θ is called minimum variance unbiased linear (MVUL) or minimum variance unbiased quadratic (MVUQ) estimator, if its variance is minimum amongst all unbiased linear and quadratic estimators respectively.

Definition 4 : An estimator $\hat{\theta} = \hat{\theta}(y)$ of θ is called minimum mean squared error estimator (MMSE), if its mean squared error MSE is minimum.

The estimator $\hat{\mu} = \frac{\sum_{i=1}^n y_i}{n} = \bar{y}$ of the mean μ in example 1 is a minimum variance unbiased linear estimator.

Definition 5: Let $f(y, \theta)$ be the density function of θ , the parameter of the distribution of a random variable y . This is a function of two variables, θ and y . If we call it density function we consider it as a function of y for a fixed θ . But the same function $f(y, \theta)$ as a function of θ for fixed y is called likelihood function. If we estimate θ so that it maximizes the likelihood function in the parameter space of θ , we call the corresponding estimator a Maximum- Likelihood Estimator (MLE).

Notes

Definition 6 An estimator is called minimum norm quadratic unbiased invariant estimator - MINQUE, if it is a quadratic form of Y in (describe in 5), unbiased and invariant against translation of the fixed effect $X\alpha$ and minimizing some matrix norm.

Note :For more details see Rao (1971), the first paper on MINQUE.

remark. In previous topics we discussed the model equation of the random variables y . Mathematical operations as minimizing (least squares) or maximizing (likelihood) can be performed for the realisations only. The result of such an optimisation is the estimate, a function of the y . If this is an explicit formula, we obtain the estimator by replacing the y -s in that formula by the random variables y .

The various methods of variance component estimation are like as:

Analysis of Variance Method (ANOVA-Method)

The oldest and simplest method of estimating the two variance components is due to Fisher (1925). By using this method we can replace in the column $E(MS)$ of the ANOVA table the variance components any σ by its estimate s and put the resulting expressions = to the observed MS and finally we solve the resulting equations

$$s^2 + ns_a^2 = MS_A$$

$$s^2 = MS_{res}$$

Solving these equations gives:

$$s^2 = MS_{res}$$

$$s_a^2 = \frac{1}{n} (MS_A - MS_{res})$$

Properties of the estimators:

Minimum variance unbiased quadratic estimator for any continuous distribution with existing first two moments and Minimum variance quadratic estimator for normal distributions.

The probability of a negative estimator is given by

$$P(s_a^2 < 0) = P \left[F(a-1, a(n-1)) < \frac{1}{1 + n \frac{\sigma_a^2}{\sigma^2}} \right]$$

Where $F(a-1, a(n-1))$ is a random variable with an F -distribution with $\alpha-1$ and $\alpha(n-1)$ d.f. Tables for these probabilities are given by Verdooren (1982).

If we assume approximately a mean sample size of 26, then the above expression gives.

$$P(s_a^2 < 0) = P \left[F(9, 2497) < q = \frac{1}{1 + 8 \frac{1}{6}} = \frac{1}{5,3} = 0,1875 \right] = 0.$$

Notes

For some other values of σ_a^2 and σ^2 the corresponding probabilities can be found in below table

Table for Probabilities of negative estimates for $f_1=99$, $f_2=2497$, $n = 26$ and several values of σ_a^2

value of σ_a^2 / σ^2	h^2	q	probability
6	0,4	0,19	,000000000
20	0,17	0,43	,000000233
25	0,14	0,49	,000005901
30	0,12	0,54	,000052454
40	0,09	0,61	,000790015

In order to increase the likelihood of obtaining negative estimates, it is necessary to have a higher environmental variance, such as 30, rather than 6. Alternatively, the degrees of freedom should be reduced, indicating that negative estimates are expected only when the sample size is insufficient.

Quasi-Maximum-Likelihood Method (QML)

The maximum likelihood estimate is achieved by maximising the likelihood function of the sample, while ensuring that the answer falls inside the parameter space. This constraint is necessary to exclude the possibility of negative variance components. Since we lack an estimator that aligns with the specification provided in the preceding unit, we refer to the approach as the quasi-maximum-likelihood method. By substituting the realisations of the random variables with the random variables themselves, we obtain the quasi-maximum likelihood estimator for this minimum. The family of normal distributions is a two-parametric exponential family having a set of full minimal adequate statistics, as described in Rasch (1995).

$$\bar{y}_\cdot = \frac{1}{a} \sum_{i=1}^a \sum_{j=1}^n y_{ij}, \quad S_{res}, S_A,$$

the likelihood function can be written as

$$f(y_1, \dots, y_b) = \frac{e^{\left\{ -\frac{1}{2} \left[\frac{1}{\sigma^2} S_{res} + \frac{1}{\sigma^2 + n\sigma_a^2} S_A + \frac{a}{\sigma^2 + n\sigma_a^2} (\bar{y}_\cdot - \mu)^2 \right] \right\}}}{(2\pi)^{\frac{a}{2}} (\sigma^2)^{\frac{a(n-1)}{2}} (\sigma^2 + \sigma_a^2)^{\frac{a}{2}}}$$

With respect to μ deriving the logarithm of this likelihood function and the two variance components without side conditions to restrict the solutions to the parameter space leads to the following estimators:

$$\hat{\mu} = \bar{y}_\cdot$$

$$s^2 = MS_{res}$$

$$s_a^2 = \frac{1}{n} \left(\frac{a-1}{a} MS_A - MS_{res} \right)$$

Notes

The estimate of the population mean, denoted as μ , is required to substitute the μ in the equations derived only from the variance components.

Maximum-Likelihood method (ML)

Real maximum likelihood estimator (solutions restricted to the parameter space) derived by Herbach (1959) .

$$S^2 = \min \left[MS_{res}, \frac{S_{res} + S_A}{a} \right]$$

$$s^2_a = \max \left\{ \left[\frac{a-1}{a} M_A - M_{res} \right]; 0 \right\}$$

Both estimators are biased.

Restricted Maximum-Likelihood method (REML)

In 1952, Anderson and Bancroft proposed the constrained maximum likelihood (ML) technique. This technique employs a translation invariant limited likelihood function that depends solely on the estimation of variance components, rather than fixed effects such as μ . The limited likelihood function is defined as a function of the adequate statistics for the variance components. The latter is obtained by taking the derivative with regard to the variance components, while imposing the constraint that the solutions must be non-negative. The solutions are:

$$s^2 = \min \left[MS_{res}, \frac{S_{res} + S_A}{a-1} \right],$$

$$s^2_a = \frac{1}{n} \max \left\{ \left[M_A - M_{res} \right]; 0 \right\}$$

Modified Maximum-Likelihood method (MML)

ML-estimators which have uniformly smaller mean squared error compared with the ML estimators of Herbach , derived by Stein (1964) and Kotz et al. (1969) .

They are given by:

$$s^2 = \min \left[MS_{res}, \frac{SS_{res} + SS_A}{an+1}, \frac{SS_{res} + SS_A + an\bar{y}^2}{an+2} \right],$$

$$s^2_a = \frac{1}{n} \min \left\{ \max \left\{ \left[\frac{a-1}{a+1} MS_A - MS_{res} \right]; 0 \right\}; \max \left\{ \left[\frac{SS_A + an\bar{y}^2}{a+2} - MS_{res} \right]; 0 \right\} \right\}$$

Federer's Estimator:

Some of the truncated estimators above are inadmissible from a Bayesian point of view. The admissibility of the following estimators, which are both non-truncated and nonnegative, could not be shown. It is also not possible to establish that they are uniformly better than any other estimator. The advantage of the estimators proposed by Federer (1968) is that they and their distribution function can be expressed in an analytical form.

$$S^2 = MS_{res}$$

$$\frac{1}{s_{\alpha}^2 n} [MS_A - MS_{res} (1 - e^{-\delta MS_A})]$$

For δ Federer proposed to choose a value in the interval $\delta \in \left[0, \frac{1}{s_{res}}\right)$.

In our numerical example we will use $\delta = 0$ and $\delta = \frac{1}{s_{res}}$.

4.2.11 Model Adequacy Checking

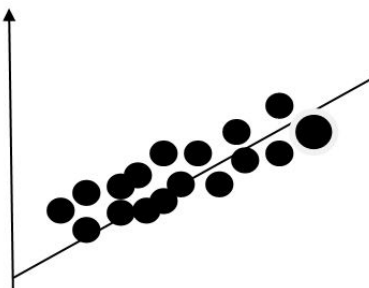
The estimation of parameters and testing of hypotheses about the qualities of the estimator are based on certain assumptions, such as:

1. The relationship between the research variable and explanatory factors is primarily linear, with a minimum value occurring around about.
2. The mistake has a mean value of zero.
3. The mistake exhibits a consistent variance.
4. The mistakes are not connected.
5. The mistakes follow a normal distribution.

The accuracy of such assumptions is necessary for the results to have significance. If these assumptions are not upheld, the outcome may be inaccurate. However, if these deviations are minor, the final result may remain unaffected. However, if the deviations are significant, the model derived may become unstable, meaning that a new sample may result in a completely different model with contradictory findings. It is important to note that these assumptions are based on the population and we are just working with a sample. Hence, the primary concern is to establish a selection of the population based on a sample of information. Various diagnostic approaches for assessing violations of regression assumptions involve analysing model residuals using a variety of graphical techniques.

Checking of the Linear Relationship between Study and Explanatory Variables

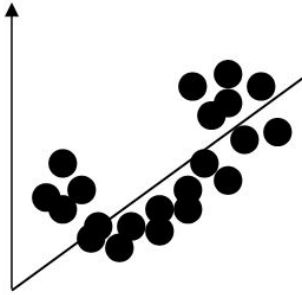
1. Case of one explanatory variable When there is just one explanatory variable in a model, it is simple to evaluate the presence of a linear connection between y and X by using a scatter plot using readily available data. If the scatter plot exhibits a linear trend, it signifies that there is a linear relationship between the variables y and X . Nonlinearity in the pattern indicates a nonlinear connection between y and X . For example, the following figure demonstrates a linear trend.



Notes

Notes

whereas the following graph suggests a nonlinear trend:

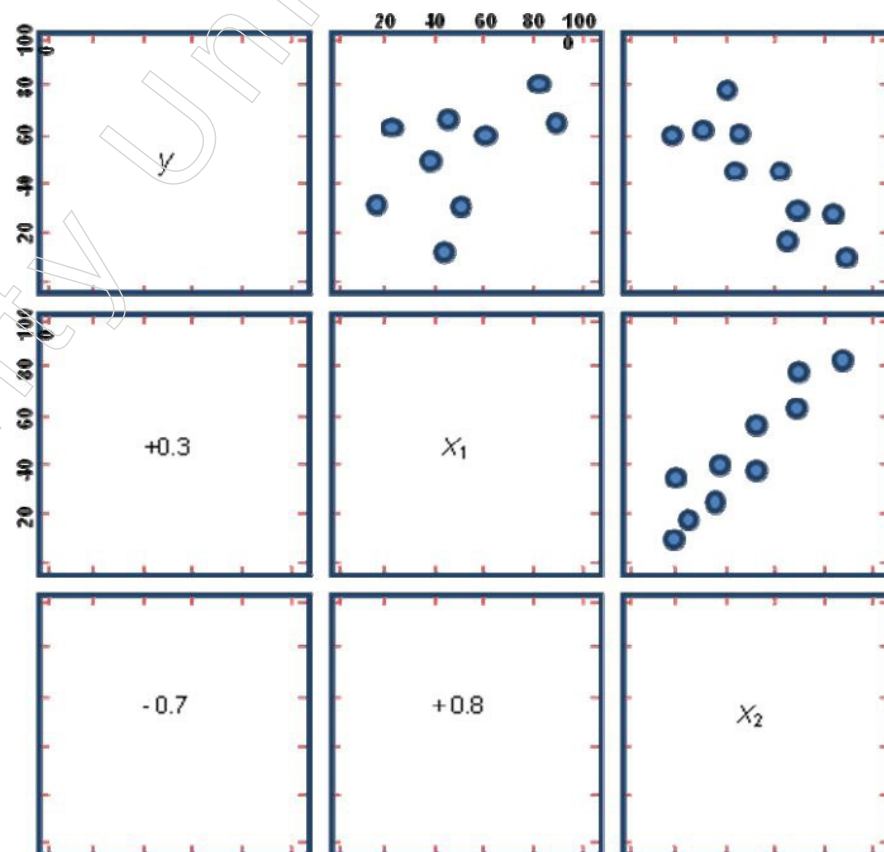


Case of more than one explanatory variables : The scatter plot matrix is utilised to examine the concept of linearity between the research variable and the explanatory factors. A scatter plot matrix is a two-dimensional array of scatter plots, where each cell includes a scatter diagram except for the diagonal. Every example demonstrated the connection between a set of variables. The measure of association provides additional information beyond the correlation coefficient by indicating the presence of linearity or nonlinearity in the connection between variables, as well as offering insights into the distribution of individual data points over the region. This is a scatter plot depicting (Y versus X_1), (Y versus X_2), ..., (Y versus X_3).

Another option to present the scatterplot is

- ❖ display the scatterplots within the upper triangular part of the plot matrix.
- ❖ Mention the corresponding correlation coefficients within the lower triangular part of the matrix.

Suppose there are only two explanatory variables and the model is $y = X_1\beta_1 + X_2\beta_2 + \epsilon$, the matrix looks as follows.



Residual Analysis

The residual is outlined because of the distinction between the discovered and fitted worth of study variable. The residual is outlined as

$$e_i = y_i - \hat{y}_i = y_i - \hat{y}_i, i=1, 2, \dots, n, \text{th}$$

where y_i is an observation and $\hat{y}_i$ is the corresponding fitted value

Residuals can be viewed as the deviation between the data and the fit. It is also a quantification of the extent to which the regression model fails to account for the variability in the response variable. Residuals might be conceptualised as the actual values of the mistakes in the model. If there are any deviations from the assumptions regarding random errors, it is believed that they will be revealed by the residual. The examination of residuals helps to identify model shortcomings.

Assuming that the OLS estimates the regression coefficients in the model $y = X\beta + \epsilon$, we find that:

- Residuals have zero means

$$\begin{aligned} E(e_i) &= E(y_i - \hat{y}_i) \\ &= E(X_i\beta + \epsilon_i - X_i\hat{\beta}) \\ &= X_i\beta + 0 - X_i\hat{\beta} \\ &= 0 \end{aligned}$$

Summary

- A hypothesis is a statement about the research problem and may or may not be true.
- It is a tentative and testable relationship between at least two variables that require research methodology for testing its authenticity. You are required to test the hypothesis and ascertain the truth in the way of research.
- Theory and hypothesis are completely different. Unlike a hypothesis, a theory is a tested, well-substantiated, complete explanation for a set of proven factors.
- The term “hypo” means “subject to verification”, and “thesis” means “statement about the solution of a problem”. Thus, “hypothesis” is “a tentative statement, which is subject to verification and states the solution of a problem”.
- Hypothesis offers a solution to the research problem that must be verified empirically.
- The posterior probability of the hypotheses are directly proportional to the prior probabilities and the likelihoods.
- A hypothesis with a high prior probability is more likely to have a high posterior probability.
- To comprehend the implications for reasoning, it is necessary to examine the definitions of the prior and the likelihood.
- After determining the error sum of squares (SSE), we may proceed to determine the treatment sum of squares (SSTR) and the total sum of squares (SST).
- When calculating the sum of squares error (SSE), sum of squares treatment (SSTR), and total sum of squares (SST), the next step is to determine the mean square error (MSE) and mean square treatment (MSTR), which may be used to compute the test statistic.

Notes

Notes

Glossary

- **Hypothesis:** A hypothesis is a statement about the research problem and may or may not be true. It is a tentative and testable relationship between at least two variables that require research methodology for testing its authenticity.
- **Simple Hypothesis:** In a simple hypothesis, there exists a relationship between two variables; one is called an independent variable or cause, and the other is a dependent variable or effect.
- **Complex Hypothesis:** In a complex hypothesis, there exists a relationship among more variables (more than two dependent and independent).
- **Empirical / Working Hypothesis:** Working hypothesis is mainly applied to a particular field. At the time of formulation, it is an assumption, but when it is ready for a test to become an empirical or working hypothesis.
- **Logical Hypothesis:** Here, the hypothesis is rationally validated. J.S. Mill has proposed four principles of this theory, including agreement, disagreement, difference, and residue.
- **Statistical Hypothesis:** A hypothesis may be logical or illogical but can be verified statistically, called a statistical hypothesis.

Check Your Understanding

1. What is a hypothesis?
 - a) A proven fact
 - b) A tentative statement subject to verification
 - c) An assumption without any basis
 - d) A well-substantiated explanation
2. Which type of hypothesis states that there is no relationship between the variables?
 - a) Alternative hypothesis
 - b) Simple hypothesis
 - c) Null hypothesis
 - d) Complex hypothesis
3. What does the significance level in hypothesis testing represent?
 - a) The probability of making a Type II error
 - b) The probability of making a Type I error
 - c) The probability of rejecting the null hypothesis when it is true
 - d) The probability of accepting the null hypothesis when it is true
4. What does the test statistic help determine in hypothesis testing?
 - a) The significance level
 - b) The p-value
 - c) Whether to accept or reject the null hypothesis
 - d) The type of error
5. Which error occurs when the null hypothesis is rejected, but it is actually true?
 - a) Type I error
 - b) Type II error
 - c) Sampling error

- d) Measurement error

Notes

Exercise

1. Explain the concept of degrees of freedom in statistical calculations. How is it calculated, and why is it important in statistical tests like chi-square, t-tests, and f-tests?
2. Describe what a confidence interval is in statistics. How is it constructed, and what does it represent?
3. Define Bayesian statistics. Explain the key components of Bayes' theorem and how it is used to update probabilities based on new evidence.
4. Differentiate between a hypothesis and a theory. Provide examples to illustrate each concept.
5. Discuss the significance level in hypothesis testing. What role does it play in accepting or rejecting the null hypothesis?

Learning Activity

1. Explain the concept of a confidence interval. Provide an example of how confidence intervals could be used in a real-world scenario.
2. Imagine you conducted a survey to measure customer satisfaction with a new product. You collected data from a sample of 200 customers and calculated the mean satisfaction score to be 4.2 on a scale of 1 to 5 (where 5 is very satisfied). Explain how you would use a confidence interval to interpret the results.

Check Your Understanding- Answers

1. b)
2. c)
3. c)
4. c)
5. a)

Further Readings and Bibliography

1. "Statistics for Business and Economics" by Paul Newbold, William L. Carlson, and Betty Thorne
2. "Hypothesis Testing: A Visual Introduction To Statistical Significance" by J.S. Kung
3. "Introduction to the Practice of Statistics" by David S. Moore, George P. McCabe, and Bruce A. Craig
4. "Statistical Methods for Psychology" by David C. Howell
5. "An Introduction to Statistical Learning: with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
6. Newbold, P., Carlson, W. L., & Thorne, B. (2020). Statistics for Business and Economics. Pearson.
7. Kung, J. S. (2017). Hypothesis Testing: A Visual Introduction To Statistical Significance. J.S. Kung.
8. Moore, D. S., McCabe, G. P., & Craig, B. A. (2018). Introduction to the Practice of Statistics. W. H. Freeman.
9. Howell, D. C. (2019). Statistical Methods for Psychology. Cengage Learning.
10. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2017). An Introduction to Statistical Learning: with Applications in R. Springer.

Module - V: Report Writing

Learning Objectives

At the end of this module, you will be able to:

- Discuss Research Report Writing
- Explain the concept of Presentation
- Infer Writing a Research Paper
- State Planning a Project Proposal

Introduction

A report is a very formal document that is composed for a variety of objectives, including sciences, social sciences, engineering, and business disciplines. Essentially, the process of looking for a certain assignment is documented in a report.

5.1 Research Report Writing

It is important to recognise that reports are regarded as legal papers in all workplaces.

5.1.1 Meaning of Research Report

There are three fundamental characteristics that define report writing at a basic level:

- a predefined structure,
- independent sections
- reaching unbiased conclusions.

Predefined structure: In a broader context, these headers may serve as indicators of different parts within a report, such as an introduction, discussion, and conclusion.

Independent sections: Each section in a report is separately written, because if the reader wants to selectively identify the report sections they are interested in, rather than reading the whole report in one go from top to bottom.

Unbiased conclusions: Another aspect of report writing is that it is a kind of writing that is impartial and objective.

5.1.2 Structure of Scientific Report

The preliminaries

- Title page.
- Preface or foreword, acknowledgements.
- Graphs or illustrations, tables, charts.
- Table of contents.

Contents of the Report:

1. Introduction
 - ❖ The research aims to achieve specific goals, provide a clear characterisation of the issue, propose hypotheses, and define key concepts.

- ❖ Literature review and analysis of research.
 - ❖ Please provide the specific details on the time, location, and resources required for the survey.
 - ❖ Scope refers to the extent or range of anything. Assumptions are beliefs or ideas that are taken for granted without proof. Limitations are restrictions or constraints that affect the boundaries or capabilities of something.
 - ❖ Methods of organisation and selection of samples.
 - ❖ Approaches, instruments, and methodologies employed for the acquisition of knowledge.
2. Analysis and presentation of results:
- ❖ Report containing factual information on the type, volume, and dimensions.
 - ❖ Quantitative analysis of knowledge.
 - ❖ Summary of the research results and suggested actions.
3. The reference materials:
- ❖ References.
 - ❖ Appendices consist of surveys, statistics tables, and other related materials.
 - ❖ Glossary: A comprehensive list of terms and their definitions.
 - ❖ Index: An alphabetical list of terms with corresponding page numbers for easy reference.

5.1.3 Components of a Scientific Report

A technical report is primarily intended for other academics and, as such, may be structured in a fundamentally different manner. In this type of report, the researcher is required to provide a comprehensive explanation of the technical components, including both the sample techniques and the topic matter. Colleagues in the same field are mostly focused on the techniques employed. The extent of the search is determined by the strategies employed. The design of the sample for the conceptual and analytical framework should be well elucidated. A technical report possesses the following characteristics:

- Main searching and contents: The primary search functionality and content will be limited to two or three pages.
- Nature of the research work: Which describes the general objectives of the study, formulation of the problem in operational items, the working hypothesis, the type of analysis, data required, etc.
- Research methodology: Provides an explanation of the different techniques employed and their inherent constraints. For example, factors such as sample size and sample selection.
- Data analysis: The report should thoroughly examine the data, including their sources, qualities, and limits. If secondary data are utilised, they must be appropriate for the given situation. When conducting a survey, it is essential to provide a comprehensive description of the data collection procedure.
- Findings presentation: The researcher delivers the primary results of the study together with corroborating evidence in the form of tables and figures. The primary content of the report often spans many chapters.
- Bibliography: This contains the main sources of secondary data.

Notes

- Technical appendices: These encompass all technical aspects related to surveys, mathematical derivations, detailed explanations of specific analytic methodologies, and similar topics.

5.1.4 Types of Reports

There are two types of reports (1) Oral report (2) Written report

1. Oral Report When the researchers are asked to make an oral presentation then they prepare this type of oral report. It is more difficult to prepare as compared to the written report. Because here the reporter interacts directly with the audience. Any delay during an oral presentation can leave a negative impression on the audience. This may also lower the self-confidence of the presenter. Communication plays a big role in an oral presentation. A lot of planning and thinking is required to decide 'What to say', 'How to say', 'How much to say'. The presenter should have to face a bunch of questions from the audience. A lot of preparation is required; the broad classification of an oral presentation is as follows. Nature of an Oral Presentation Opening: A descriptive statement may be made on the basis of the nature of discussion that will follow. The opening statement should provide a clear description of the project's attributes, its origins, and the objectives that were pursued.

Finding/Conclusion: Every conclusion may be articulated and supported by empirical evidence. Suggestion: It is essential for every advice to be backed with a conclusion. Following the lecture, there may be a question-and-answer session conducted by the audience. Presentation procedure: If it is necessary to display visuals, they can be utilised. The utilisation of tabular format for presenting statistical data is essential in facilitating comprehension for the audience.

- (a) A root question is a form of presentation. Is the content being recited from a written document, memorised, or presented spontaneously without prior preparation? It is not advisable to rely on memorisation, as there is a possibility of making a mistake during a presentation.
- (b) Secondly, it generates a speaker-centric approach. To effectively deliver an impromptu speech, it is advisable to create concise notes on the essential ideas, which may then be elaborated upon. It is important to adhere to logical sequencing.

Points to remember in oral presentation:

- ❖ The language employed should be uncomplicated and comprehensible.
- ❖ Time management should be followed strictly.
- ❖ The utilisation of visual aids such as charts and graphs will augment the comprehension of the audience.
- ❖ Vital material, such as numerical information, can be printed and distributed to the audience. This allows them to enhance their understanding by referring to it throughout the presentation.
- ❖ In order to deliver a customised presentation, it is crucial for the presenter to have a thorough understanding of the target audience beforehand. The researcher should possess an understanding of the intended objective of the report, such as whether it is intended for decision-making purposes or for the dissemination of information.

The following are the different categories of written reports:

A) Reports can be classified based on the time-interval such as:

- 1) Daily
- 2) Weekly
- 3) Monthly
- 4) Quarterly
- 5) Yearly

B) Type of reports:

- 1) Short report
- 2) Long report
- 3) Formal report
- 4) Informal report
- 5) Government report

A short report is composed when the issue is clearly identified and the extent of the investigation is restricted.

As an illustration, consider the monthly sales report. It will span around five pages. The report contains information on the progress accomplished in relation to a given product in a specified geographical area.

2. The long report is a fusion of both a technical and non-technical report. It can depict the findings of the investigation in a comprehensive manner.

a) Technical report: This report shall encompass the data sources, study methodology, sample design, tools utilised for data collection, information analysis technique, appendix, conclusion, and comprehensive suggestions based on particular results. When referring to any magazine, article, or periodical, it is important to include references for the benefit of the reader.

b) Non-technical report: This report is intended for those who do not possess technical qualifications.

For example, the head of the financial department. He may be solely focused on the financial ramifications, such as profit margins, sales volumes, and so on. He lacks interest in the process.

3. Formal report: The marketing head has prepared a report to be sent to the Vice-President of marketing about the quarterly performance, specifically focusing on test marketing.

4. Informal report: The supervisor's report, published by completing the shift log book, for the benefit of his colleagues.

5. Government report: These reports are often generated by either state governments or the federal government to address specific issues. The programme for the rural employment strategy has been announced as part of the five-year plan.

5.1.5: Steps in Report Writing

Once the type of report has been determined, the subsequent stage is to prepare the report. The subsequent text outlines the structure of a research report:

1. Title Page

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2. Page Contents
 3. Executive Summary
 4. Body
 5. Conclusions and Recommendations
 6. Bibliography
 7. Appendix
1. Title Page: The title page should clearly state the subject of the report. The report should contain the name of the individual or organisation responsible for its preparation.
 2. Table of Contents: The table of contents provides a comprehensive overview of the materials included in the report. The table of contents should clearly delineate the distinct components or divisions of the report. It might potentially signify the titles of the chapters combined with their corresponding page numbers.

Chapter no.	Title of the chapter	Page no.
1	Declaration	
2	Certificates	
3	Acknowledgement	
4	Executive summary	
5	Introduction to the project	
6	Research design and methodology	
	Theoretical perspective of the study	
	Company and industry profile	
	Data analysis and interpretation	
	Summary of findings, suggestions and conclusions	
	Bibliography	
	Appendix	

3. Executive Summary: If the report is excessively lengthy and verbose, the intended recipient of the report may not have the available time to thoroughly read and comprehend it. Introducing an executive summary will aid in emphasising key elements. This is a concise rendition of the whole report. The document should be limited to one or two pages in length. Given that top-level executives exclusively peruse the executive summary, it is imperative that it be both precise and eloquently composed. An executive summary should facilitate the process of decision-making.

An executive summary should have,

- ❖ Objectives
 - ❖ Brief methodology
 - ❖ Important findings
 - ❖ Key results
 - ❖ Conclusion
4. The Body: This section includes
 - ❖ Introduction
 - ❖ Methodology

Notes

- ❖ Limitations
 - ❖ Analysis and interpretations
 - ❖ Introduction: Provide a concise and explicit description of the decision issue and the research purpose. The background knowledge should be allocated to the product and services offered by the organisation being examined.
 - ❖ Approach: The process of data collection is detailed in this section.
 - ❖ For example, was original data gathered or was secondary data utilised? Was a survey instrument employed? Could you please provide information on the sample size, sampling plan, and method of analysis? Was the design process characterised by exploration or a definitive conclusion?
 - ❖ Constraints: Each report will inevitably be subject to some limitations, such as temporal constraints, geographical limitations, the chosen technique, accuracy of the replies, and so on.
 - ❖ Data analysis and interpretation: The acquired data will be organised into tables. If there are any statistical tools available, they will be used. Notes can be utilised for the purpose of analysis and decision-making.
5. Conclusion and Recommendation:
 - a) What were the outcomes drawn from the study?
 - b) Based on the study, what recommendation do you make?
 6. Bibliography: When the report relies on secondary data, provide a bibliography section to enumerate the publications or sources that you have referenced. The bibliography should include the book's title, the journal's name for articles, the volume number, page number, edition, and other relevant details.
 7. Appendix: An appendix is a section of a report that is used to provide non-essential content. The appendix contains duplicates of data collecting forms known as questionnaires, as well as information on the company's annual report, graphs/charts, images, CDs, and interviewers' instructions. The following items are to be included in this section.
 - ❖ Data collection forms
 - ❖ Project related paper cuttings
 - ❖ Pictures and diagrams related to project
 - ❖ Any other relevant things.

5.1.6 Format and Writing Style

A well-formed physical layout is most important, as it may help the report these are:

- Create a favourable first impression.
- Motivate the readers by providing a clear overview of the organisational structure of the information,
- Enabling them to promptly decide which section to read first.

Particular attention should be paid to make sure there is:

- An efficient design for the title page and a concise table of contents.
- Uniformity in margins and spacing.

Notes

- Ensure uniformity in headings and subheadings by using specific font sizes and styles. For instance, use font size 16 or 18 in bold for chapter headings, font size 14 in bold for main section headings, and type size 12 in bold for subsection headings.
- High-quality printing and photocopying services. Thoroughly revise documents by using spell check and engaging in critical reading to ensure clarity, with the assistance of other team members.
- Allocate a distinct numerical identifier to each figure and table, ensuring that tables have unambiguous names and columns and rows have clear headers.
- Precision and uniformity in citations and sources.

Style of Report Writing

Remember that the reader:

- Has short of time,
- Has many other urgent matters demanding his or her interest and attention,
- Is probably not knowledgeable concerning 'research jargon'.

Therefore, the rules are:

- Simplify. Keep to the essentials.
- Justify. Make no statement that is not based on facts and data.
- Quantify when you have the data to do so. Avoid large, small, instead, say 50%, one in three.
- Be precise and specific in your phrasing of findings.
- Inform, not impress. Avoid exaggeration.
- Use short sentences. Use adverbs and adjectives sparingly. Notes
- Avoid the passive voice, if possible, as it creates vagueness (e.g., 'patients were interviewed leaves uncertainty as to who interviewed them) and repeated use makes dull reading.
- Aim to be logical and systematic in your presentation.

5.1.7 Illustrations and Tables

Tables, figures, drawings, and similar things can be designated using the term "Table", "Figure", or another suitable descriptor, and should include a title and/or caption. The title or caption should be incorporated into the List of Tables, List of Figures, or any other relevant list.

Consistency in the format of titles and captions for tables, figures, drawings, and similar things is necessary for the entirety of the thesis.

- The text in tables and figures should have a minimum height of 2 mm to ensure legibility.
- Additionally, these elements, including tables, figures, and drawings, must be accompanied by titles or captions and be appropriately numbered.
- The placement of headings or captions for tables, figures, or illustrations might be either above or below them.

- Headings must be replicated on the second and subsequent pages of tables that are divided over two or more pages.
- Tables should be divided at a suitable location, such as right before a completely new subheading.
- The formatting for titles and captions of tables, figures, illustrations, and similar items must remain consistent throughout the thesis.

Numbering

Tables, figures, drawings, and other similar elements should be sequentially numbered based on their order of presentation in the thesis.

There are two approaches for assigning numbers to Tables, Figures, and other items:

- sequentially throughout the thesis, e.g. 1, 2, 3...
- chapter number first, then numbered sequentially within each chapter, e.g.:

Tables in Chapter 1: Table 1.1, 1.2, 1.3...

Figures in Chapter 3: Figure 3.1, 3.2, 3.3...

Regardless of the chosen technique, the numbering style must be consistent for all numbered items. For instance, it should be either Table 1.1 and Figure 1.3, or Table 1 and Figure 3, but not Table 1 and Figure 1.3.

Location

There are three acceptable locations for tables and figures:

- within the chapter immediately following first relevancy them
- grouped at the bottom of the relevant chapter
- grouped at the bottom of the thesis before the bibliography

Regardless of the strategy we use, we desire to maintain consistency.

If tables and figures are consolidated at the end of the thesis, it is necessary to provide an item in the table of contents that guides the reader to their specific place.

If the caption for a figure, table, or any other visual representation cannot be accommodated on the same page as the corresponding picture, it should be placed on a separate page.

Reproducing and Reducing

Copying or lowering the size of figures such as charts, drawings, graphs, pictures, maps, etc. may result in visuals that are difficult to read. Following the reduction, it is imperative that all lettering is sufficiently large to comply with the font size specifications, and must be easily legible and unambiguous.

Copyrighted Material

If we exclude copyrighted tables, figures, or illustrations from our thesis, we must enter the corresponding replacement at the same location where the table, figure, or image was originally located:

- An assertion that the material must be eliminated as a result of copyright limitations.

Notes

- The text provides a detailed account of the material and the corresponding information it encompasses, along with a hyperlink to an online source, if accessible.
- The complete reference to the original source of the material

For longer reports that contain dozens of figures and tables, produce separate lists of figures and tables. Place them along on a similar page if they work, as shown in the illustration below. We can combine the two lists under the heading, "List of Figures and Tables," and establish the things as figures or tables as is done in the illustration below.

LIST OF FIGURES

Figure 1. Natural Gas Use by SH and EEH 7

Figure 2. Annual Electricity Use by SH and EEH 8

LIST OF TABLE

Table 1. EEH and SH Systems.....

Table 2. Energy-10 Simulation.....

Table 3. Energy Efficient Strategies

Table 4. Cost Comparisons for SH and E.....

iv

Topic overview: Always provide a brief idea of the contents of the report in the introduction.

1.0 INTRODUCTION

1.1 Purpose of the Report

This report analyzes the results of using various energy-efficient strategies to determine if such practices actually make a difference in the amount of energy consumed by an average house. Additionally, it analyzes which home system improvements provide the greatest reductions in energy and whether such improvements are cost-efficient in the long run.

1.2 Background of the Report

Annually, 24% of the natural gas and 35% of the electricity in the US is consumed by the residential housing sector. Consequently, 1.3 metric tons of greenhouse gases are emitted annually [6,7]. Understanding energy consumption and taking measures to reduce it is essential if a systematic and comprehensive reduction of environmental impacts is desired. Reductions in home energy consumption will not only reduce utility costs but also reduce the impact on the environment.

1.3 Scope of the Report

This report provides technical background on the construction of the standard and the energy-efficient house, the energy-efficient strategies used in the latter, energy-consumption rates, construction costs, and other relevant details. Not included in this report are discussions of the receptiveness of the American home-building industry or American home buyers to energy-efficient housing design or of pending legislative to promote energy-efficient housing design.

Note: A basic understanding of terminology for housing constructing, HVAC, and cost analyses is assumed.

Audience: Introductions must alert readers about the technical background they must possess to understand the report.

Ref: <https://openoregon.pressbooks.pub/technicalwriting/chapter/10-5-list-of-figures-and-tables>

5.1.8 Citing and Referencing Sources

Footnotes in a research paper contain numerical references. For instance: 'Recent studies suggest that...' This number corresponds directly to the references.

The following list presents the recorded articles in consecutive sequence as they

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are referenced in our study report. This feature will be advantageous since it allows us to include comments, footnotes, and references.

- The arrangement of individual citations inside the bibliography follows the same format as the Harvard method.
- If we find that we are referring to the same thing many times, we shall utilise common bibliographic abbreviations to avoid duplicating the references in their entirety.
- The publications mentioned exclusively include those that we have referenced in our study. The section should be titled 'References' instead of 'Bibliography', as indicated below.

Abbreviation	Explanation
Op. cit. (opereciato)	Meaning, in the work cited. This refers to a work previously referenced and so you must give the author and date and if necessary the page number, like: Robson (1993) op. cit. pp. 23-4.
Loc. Cit. (loco ciato)	Meaning, in the place cited. This refers to the same page of a work previously referenced. So you must give the author and date, like: Robson (1993) loc. Cit.
Ibid. (ibidem)	Meaning, the same work given immediately before. This refers to the work referenced immediately before and replaces all details of the previous reference other than a page number if necessary.

5.1.9: Footnotes

Researchers must insert footnotes within the appropriate places. These fulfil two purposes:

- The correct identification of materials utilised in quotations in the report.
- The footnotes provide supplementary value to the main part(body) of the text. Based on the footnotes' description, one can easily refer to the cross references, citation of authorities and sources, acknowledgement and elucidation or explanation of a degree of view. The recent trend is to avoid footnotes. Some people feel that they enhance the display of the scholarship of the researchers. But it's neither an end nor a means of displaying scholarship.

When using footnotes, a variety of references are shown within the research report.

For example: 'Recent research indicates that...' This number refers directly to the references

5.2 Presentation

As reports are better understood when accompanied by a presentation, presentations have grown in importance as a communication tool in businesses. Readers can engage in a conversation with the report's author(s) to learn about the process of thought that went into writing it. Consequently, it is required of the manager to be equally proficient in presenting as well as drafting reports.

5.2.1 Oral Presentation

Oral Report

When the researchers are asked to make an oral presentation then they prepare this type of oral report. It's more difficult to prepare as compared to the written report.

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Because here the presenter directly interacts with the audience. Any delay during an oral presentation can leave a negative impression on the audience and also violates the self-confidence of the presenter. Communication plays a big role in an oral presentation. A lot of planning and thinking is required to decide 'What to say', 'How to say', 'How much to say'. Also, the presenter should have to face a bunch of questions from the audience. A lot of preparation is required; the broad classification of an oral presentation is as follows.

Nature of an Oral Presentation

Opening: A brief statement will be created on the character of dialog that may follow. The opening statement ought to make the nature of the project, however it took place and what was tried.

Finding/Conclusion: Every conclusion is also declared to protect by findings.

Recommendation: Every recommendation should have the support of conclusion. At the bottom of the presentation, a question-answer session should follow from the audience.

Method of presentation: Visuals, if they have to be compelled to exhibit, can be made use of. The use of tabular form for statistical information would help the audience.

- a) What form of presentation could be a root question? Is it browsed from a manuscript or memorized or delivered extempo. Memorisation is not recommended, since there could be a slip during the presentation.
- b) Secondly, it produces a speaker-centric approach. Even reading from the manuscript is not recommended, because it becomes monotonous, dull and lifeless. The best procedure to present an ex-tempo, is to make main points notes, so that the same can be expanded.

Logical sequences should be followed.

Points to remember in oral presentation:

- ❖ Language used must be simple and understandable.
- ❖ Time Management should be adhered to.
- ❖ Use of charts, graphs, etc., will enhance understanding by the audience.
- ❖ Vital data such as figures may be printed and circulated to the audience so that their ability to comprehend increases, since they can refer to it when the presentation is going on.
- ❖ The presenter should know his target audience well in advance to prepare tailor-made presentations.
- ❖ The presenter should know the purpose of the report such as "Is it for making a decision", "Is it for the sake of information", etc.

5.2.2 Making Presentation

Golden Rule

Human attention is extremely limited. Don't cram an excessive amount of information, either in each slide, or within the whole talk. Avoid details: they will not be remembered anyway.

Notes**Organisation**

- Have a really clear introduction, to motivate what we are doing and to present the matter we want to resolve. The introduction isn't technical in nature, but strategic.
- If we have got a companion paper, mention it during the talk and recommend it for more details. Don't put all the main points within the talk. Present only the important ones.
- Use only 1 idea per slide.
- Have a good conclusion slide: put there the most ideas, those we actually want people to recollect. Use only 1 conclusions/slide.
- The conclusion slide should be the last one. Don't put other slides after conclusions, as this may weaken their impact.
- Having periodic "talk outline" slides (to show where we are within the talk) helps, especially for extend talks. A minimum of one "talk outline" slide is extremely useful, usually after the introduction.
- Don't judge the audience to recollect any detail from one slide to another (like color-coding, applications you measure, etc.). If we want to remember, re-state the information a second time.
- Especially if we've got to present many various things then try to build a unifying thread. The talk should be sequential in nature (i.e. no big conceptual leaps from one slide to the next).
- Try to chop out the maximum amount as possible; less is healthier.
- Help the audience understand where we're going. Often it's best to allow them a high-level overview first, and then plunge into the details; then, while paying attention to the details they can relate to the high-level picture. This also helps them save important brain power for later parts of the talk which may be more important.

Mechanics

- Use a decent presentation-building tool, like MS PowerPoint. Avoid Latex, aside from slides with formulas.
- Humour is extremely useful; prepare one or two couple of puns and jokes beforehand. However, if we're not familiar with jokes then better ways to avoid them altogether.
- The more we rehearse the talk, the better it will be. A rehearsal is most helpful once carried out loud. 5 rehearsals is a minimum for an important talk.
- The more people criticize our talk (during practice), the better it will be; pay attention to criticism, not necessarily to all suggestions, but try to see what and why people misunderstood your ideas.
- Everything hasn't to be written down.
- Be enthusiastic.
- Present to explain, ask instant questions, act surprisingly, etc.
- Give individuals some time to think about the vital facts by slowing down, or by putting on a mute for a moment.
- Do not go overtime under any circumstance.
- Listen to the queries terribly carefully; several speakers answer totally different questions than the ones asked.

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- Don't treat the audience as mentally-impaired: don't explain the completely obvious things.

Text

- Slides ought to have short titles. An extended title shows one thing is wrong.
- Use uniform capitalisation rules.
- All the text on one slide ought to have the identical structure .
- Put little or no text on a slide. Don't put more than one plan in one slide. If we have lots of text, people will read it faster than we talk, and will not pay attention to what we say.
- Don't use small fonts.
- Use very few formulas (one per presentation). The identical goes for program code (maximum one code fragment for one presentation).
- Do not place useless graphics on every slide: logos, grids, affiliations, etc.
- Spell-check. A spelling mistake is an attention magnet.

Illustrations

- Use suggestive graphical illustrations to the maximum amount as attainable. Don't shun graphical metaphors. Prefer an image to text.
- Do not put in the figures details which will not be mentioned explicitly. The figures ought to be as schematic as attainable.
- Don't "waste" info by victimising spare colour. Each different colour should signify something different, and something important but don't use too many different colours. Have high-contrast colours.
- For some strange reason, rectangles with shadows appear to look far better than without .
- Sometimes a matte pastel background looks far better than a white one.
- Exploit animation with restraint. Don't use fancy animation effects unnecessarily.
- However, there are places where animation is much more effectible, e.g., to depict the evolution of a complex system, or to introduce related ideas one by one.
- Use strong colours for important stuff, pastel colours for the unimportant.
- Employ information cleverly: e.g. make arrow widths showing flows proportional to the flow capacity.
- Use thick lines in drawings (e.g. 1 1/2 points or more).

5.2.3 Use of Visual Aids

Visual aids can enhance our presentations - they can increase the audience's understanding about a topic, explain points, make an impact and create enthusiasm. It's become more useful to make data visual

What are visual aids?

Visual aids are peripherals of a visual manner, like as graphs, photographs, video clips etc are rather than spoken information. Visual aids shall be selected depending on their purpose, as an example:

Summarise Information

Reduce the amount of spoken words, for example, show a graph of results rather than reading them out.

Clarify and Show Examples

Create more of an impact, for example, if our presentation is on the health risks of smoking, then show images of the effects of smoking on the body rather than describing this.

- Emphasise what you're saying.
- Make a point memorable.
- Enhance credibility.
- Engage the audience and maintain their interest.
- Make something easier for the audience to understand
- Preparation and use of visual aids

If we decide that we want to use it as a visual aid, then we must make sure that the audience is able to easily understand the image (it must be clear). They should be employed throughout our speech but try to only use visual aids for essential points as it can be tiring for the audience to skip from one visual to another.

Preparation

Think about how a visual aid can support our message and what do you want the audience to do?

- Ensure that our visual aid follows what we're saying or this will confuse the audience.
- Avoid cluttering of the picture.
- Visual aids must be clear, concise and of a high quality.
- Keep the style consistent, such as, the same font, colours, positions etc
- Use graphs and charts to present data.
- The audience should not be trying to read and listen at the same time - use visual aids to highlight points.

One message per visual aid, for example, on a slide there should only be one key point.

- Employ visual aids in moderation to emphasise and support main points.
- Ensure that the presentation still works without visual aids in case of technical problems.

During the Presentation

- Make sure that the visual aids should be seen by everyone in the audience.
- Avoid reading from the visual aid.
- As soon as the visual aid is shown the audience's attention will be drawn to it so must immediately explain it.
- Make it clear to the audience why they are using it.
- When there is no need for visual aid, ensure that the audience can't see it.

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There are a variety of different types of visual aids like as:

PowerPoint

Microsoft PowerPoint is vastly employed for presentations because it's too easy to create. We can insert a range of visual items into the slides which will improve the audience's focus. Also, the audience can generally see slideshows better than other visual aids.

Tips:

1. Have a clear and simple background.
2. Avoid using too many different types of fonts or font sizes.
3. Only use animations for a purpose, such as, to reveal the stages of a process, otherwise this can be distracting and look amateurish.
4. Use a large font size - a minimum of 24pt.
5. Use bullet points to summarise key points.
6. Consider providing handouts of diagrams because the audience will find the diagrams easier to read.
7. Avoid putting too much text on a slide.
8. Avoid using red or green text as it's difficult to read.
9. There should only be one key point for each slide.

Whiteboards

Whiteboards are great for providing further explanations, such as, showing the order of a process, creating diagrams or explaining complex words or phrases. They're often used to display headings and write down audience suggestions. Whiteboards are also ideal for displaying vital data for the whole period of the presentation, such as, key definitions, as a result the audience will simply look at the whiteboard for a reminder.

Tips:

- ❖ Ensure that enough time has passed for the audience to take notes before rubbing something off of the whiteboard.
- ❖ Write concisely to avoid facing away from the audience for too long.
- ❖ Handwriting must be large and legible.

Handouts

Handouts are papers that contain key information from the presentation or they may provide further information that prevent us from overwhelming the audience as there'll be less data on the slides and so less data they have to put in while writing down.

Tips:

- ❖ If given at the beginning and middle of the presentation the audience may be reading rather than listening.
- ❖ If given at the end of the presentation the audience may be trying to take lots of notes which may reduce the amount of information they are actually understanding.

To manage this, provide the audience with partially completed handouts so they will have to listen to what we're saying to be able to fill in the gaps. Providing the audience with graphs and charts beforehand is also beneficial because the audience will find them easier.

Video Clips

Using videos are a good way to engage the audience and increase their attention. Use video to bring motion, pictures and audio into the presentation.

Tips:

- ❖ Ensure that any videos used are relevant to the presentation's content.
- ❖ Only show as much of the video as necessary.
- ❖ Never show a really long clip.
- ❖ Inform the audience how long the video will last.

Flip Chart

Flip charts offer a low cost and low tech solution to record and convey information. They're more beneficial for smaller audiences and they are favoured for brainstorming sessions for gathering ideas easily. Flip charts are also widely used for summarising information and, like with a whiteboard.

Posters

Poster boards can be created using a variety of visual devices, such as graphs and images. They're generally quite portable and we can make them as elaborate as we want. However, they can be expensive to produce if the poster is quite complex.

5.2.4: Importance of Effective Communication

Good communication plays an essential role to get success in the workplace and in personal life also. Leaders who know how to communicate effectively with those around them and will see better productivity and improved relationships in every aspect of their lives.

Effective communication involves knowing how to listen attentively. It's the ability to offer empathy, open-mindedness, and helpful feedback. Also, a friendly demeanour, confidence, and quality nonverbal communication will also help us, as a manager, develop good relationships with the members of our team.

Let's explore some benefits of effective communication you will see in- and outside your office when you take the time to nurture these skills.

1. Building trust

Effective communication fosters trust with others. Our ability to listen attentively and embrace different points of view helps others keep a trust that we are making optimal decisions for everyone in the group.

2. Preventing or resolving problems

The ability to communicate effectively plays a large role in resolving conflicts and preventing potential ones from arising. The key is to remain calm, make sure all parties are heard and find a solution that is ideal for everyone involved.

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3. Providing clarity and direction

With effective communication skills, we're able to deliver clear expectations and objectives for our team. This involves finding constructive ways to point out when something isn't working as well as providing helpful feedback to get people back on track. They will understand their specific tasks and responsibilities, as well as those of their teammates, which will help eliminate conflicts and confusion.

4. Creates better relationships

Good communication also improves relationships, both with employees and in our personal life with friends and family members.. This, in turn, nurtures mutual respect.

5. Increases engagement

With people feeling more confident in their work and in their understanding of what they need to do, they become more engaged with their work as a whole. According to a recent study exploring the psychology of employee engagement, only 15% of adult employees are engaged with their employers. By prioritizing effective communication, we can increase engagement, and thus boost satisfaction, among our team members.

6. Improves productivity

When team members understand their roles, the roles of others and our expectations, they can focus more on their work and less on workplace issues. With effective communication, conflicts are resolved quickly, employees can better manage their workload and distractions are minimized. These benefits contribute to greater productivity .

7. Promotes team building

With improved communication, team members will be better able to rely on each other. Build positive feelings and relationships between the team members, which leads to improved morale and work experiences.

Good communication skills can play an important role in nurturing positive work experiences for your entire team.

5.2.5 Convention and Strategies of Authentication

Endless description without interpretation is another pitfall. Tables need conclusions, not detailed presentation of all numbers or percentages within the cells which readers can see for themselves.

Neglect of qualitative data is additionally quite common. Still, quotes of informants as illustrations of our findings and conclusions make our report lively. They even have scientific value in allowing the reader to draw his/her own conclusions from the knowledge we established. (Assuming we aren't biased in our representation!)

Sometimes qualitative data (e.g., open opinion questions) are just coded and counted like quantitative data, without interpretation, whereas they'll be providing interesting illustrations of reasons for the behaviour of informants or of their attitudes. This can be a serious maltreatment of data that needs correction.

The following must be avoided while preparing a report:

- ❖ The inclusion of careless, inaccurate, or conflicting data.
- ❖ The inclusion of outdated or irrelevant data.

- ❖ Facts and opinions that aren't separated.
- ❖ Unsupported conclusions and proposals.
- ❖ Careless presentation and proofreading.
- ❖ An excessive amount of emphasis on appearance and not enough on content.

5.3 Writing a Research Paper

Writing a research paper is a fundamental component of academic and professional research. It involves presenting original findings or insights on a specific topic, grounded in systematic research and analysis. A well-written research paper contributes to the body of knowledge in a particular field and communicates your research findings clearly and effectively.

5.3.1 Preparing Research Papers for Journals

Here we described a series of steps to make a research paper for journals. Such as

1. Determine the authors. When designing a search project, we recommend preparing an initial list and order of authors. Such inventory authors should be supported by established guidelines and should make explicit the estimated contribution of every individual to the project. We recommend that every research group establish and make known to its members the criteria for authorship on papers resulting from the work to be conducted. A list of authors make sure that all individuals involved within the project understand at the outset whether or not they will expect to be an author and, if so, what their contribution is to be.
2. Start writing before the experiments are complete. We have to Start writing while we are still doing the experiments. Writing often evokes new ideas: we may realize that there are additional experiments to run or additional controls that we need to add. If we wait until completion in the lab, have dismantled the equipment, and possibly moved on to another position, then we will not have the opportunity to test these ideas.
3. Decide it is time to publish. It is time to publish when our findings represent a complete chapter, one that will make a significant contribution to the scientific literature.
4. Draft a title & abstract. Drafting a working title and an abstract helps define the contents of the paper, identifying which experiments we'll want to publish during this paper, and which studies we will save for inclusion in another paper.
5. (Re)examine the list of authors. When we are determined which experiments will be included in this paper then we must select the authors and the order in which they will appear. Re-evaluate it based on the contributions that were made to those experiments and the additional contributions that will be made through the preparation of the manuscript. If an inventory already exists, make adjustments to make sure about the compliance with guidelines.
6. Determine the basic format. There are three basic formats for peer-reviewed research articles:
 - ❖ Full-length research articles: These articles contain a comprehensive investigation of the subject matter and are viewed as the standard format. It uses the format: Introduction, Methods, Results and Discussion.

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- ❖ Short (or brief) communications: While not as comprehensive in scope as full-length research articles, these papers also make a big contribution to the literature. Their length is going to be set by the journal but is typically 3500 words or less and can contain up to 2 tables and figures.
 - ❖ Rapid communications: These articles quickly disseminate particularly “hot” findings, usually during a brief communication format. Articles that have immediate implications for public health would be appropriate for such a format, as it might be found during a highly competitive and quickly moving field.
7. Select the journal. There are several factors to think about when choosing a journal. It's unlikely that one journal will have all the features we are looking for, so we may have to compromise. However, there's one essential feature we should always not compromise on – manuscripts must be peer reviewed for publication if they're to be considered research articles.
- ❖ Language: If we are interested in communicating our results widely to the international scientific community, then it is essential to publish in English. If, on the other hand, we wish to communicate to a more localized community then we can choose a journal that permits another language.
 - ❖ Focus: What type of research does the journal publish? Is its focus broad or narrow? Which disciplines are represented? What is the journal's orientation – for instance, is it clinical or basic, theoretical or applied?
 - ❖ Indexing: Is the journal indexed in the major electronic databases .
 - ❖ Availability: Is the journal broadly available? Is there an online version of the journal? Are papers provided in PDF format?
 - ❖ Reputation: Ask colleagues which journals they respect. Look at recent articles and judge their importance. Check the members of the editorial board and determine if they are leaders in their fields. Determine the journal's impact factor. How selective is the journal in accepting papers for publication? Try to find out the acceptance rate of the journal.
 - ❖ Format: Do we like the appearance of published articles – the format, typeface, and style used in citing references? If relevant, does the journal publish short and/or rapid communications?
 - ❖ Figures: Do figures published within the journal have the resolution that we simply need? Time to Print: Using the “date submitted” and a “date accepted” that are published on the article, along with the date of the issue, we can estimate the length of the review process also because the time from acceptance to publication in print.
 - ❖ Charges: Some journals bill the author for page charges, a price per final printed page. Most journals have a separate charge for colour plates. This may be as much as \$1000 per colour plate. Many journals will waive page charges if this presents a financial hardship for the author; colour plate charges are less-readily waived and would at least require evidence that the colour is important to the presentation of the information.
8. Stock the sections of your paper. As we think about your paper, store relevant material in folders marked with Introduction, Methods, Results, and Discussion. This may save time and avoid frustration once the writing begins. Stored items might include figures, references, and ideas.

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9. Construct the tables, figures, and legends. Create figures and tables before the writing begins! The entire paper should be organized around the data needed to present. By preparing the tables and figures we will be certain of our results before we worry too much about their interpretation.

Note: except under unusual circumstances, we may not include any data that we have already published.

10. Outline the paper. An outline is like a road map which define how we will get from here to there, and helps ensure that we take the most direct and logical route. If they have co-authors, then we may wish to get feedback from them before proceeding to the actual writing phase.
11. Write the first draft. Write the primary draft of the complete manuscript. If we are writing with co-authors, then we may wish to assign different aspects of the manuscript to different authors. This will save time, enable a lot of people to feel that they are making substantive contributions to the writing method, and make sure that the best use of expertise. However, it also can result in a mix of design. Thus, if you take this approach, be certain that the final product is carefully edited to provide a single voice. "Components of a Research Article" discusses what goes into each section of the manuscript.
12. Revise the manuscript. This step involves three major tasks, each to be carried out in the order given:
 - ❖ Make major alterations: Fill in gaps, correct flaws in logic, restructure the document to present the material in the most logical order.
 - ❖ Polish the style: Refine the text, then correct grammar and spelling.
 - ❖ Format the document: Make manuscript attractive and easy to read. It is important to do the tasks in the stated order. Otherwise, we may be spending a lot of time revising material that we delete later.
13. Check the references. Make sure that the citations are correct and complete. Do one last literature search to make certain that we're up to date.
14. Write the final title and abstract. Several changes are created throughout the editing method. Make certain that our title and abstract match the final version.
15. Reread the journal's Instructions to Authors. Review the main points of however the manuscript is to be formatted and submitted.
16. Prepare the final illustrations. Ensure that tables, figures, and figure legends are complete, clear, self-contained, and in the format required by the journal.
17. Get feedback on manuscript and then revise manuscript again. Getting feedback is one of the most important things by which we can improve our article. First, be sure our co-authors have had a chance to read and comment on the draft. Then, once it's prepared, provide the manuscript to some colleagues. After you get their comments (on the science, logic, language, and/or style), revise our manuscript to address their concerns. Do not submit manuscript until it is ready for publication. Once it is accepted.
18. Submit the manuscript to the editor. Follow the Instructions to Authors to determine what items we need to submit, how to submit them, and to whom we should send them. Note that some journals permit (or even require) a "pre-review," i.e., a letter indicating the content of the article so that the editors can determine whether they will accept the manuscript for a full review. At this point we may wish to list possible

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reviewers (or individuals to be avoided). If required then contact the editor to make sure that the manuscript was received.

19. Deal with reviewers' comments. Most manuscripts are not accepted on the first submission and ask for submitting a revised manuscript. If we agree with it then we need to respond to the reviewer's comments. Answer every comment of the reviewers, and highlight where the corresponding changes were made in the manuscript if happened, indeed, made. We do not need to make all of the changes that the reviewer recommended, but we do need to provide a convincing rationale for any changes that we did not make. When we resubmit the manuscript, indicate in our cover letter that this is a revised version.
20. Check the proof. Once the manuscript is accepted and ready for print, the publisher may send the corresponding author page proofs of the article. It should be accompanied by a list of queries, such as missing information regarding a reference. The proofs either be sent through e-mail or as hard copy. If there is a chance that we will be away when the proofs arrive, have a plan for making certain that they are received and we are notified.
20. Celebrate! Robert Day says in *How to Write and Publish a Scientific Paper* (1998), "The goal of scientific research is publication.... A particular experiment, no matter how spectacular the results, is not complete until the results are published." Your experiment – at least one phase of it – is now complete.

5.3.2 Design of Paper using Template

Notice that no correct number of letters or numbers exists; the sole determining factor is the number of points we need to create for the specified length of our paper.

- I. Introduction
 - A. Background information connecting the reader to the topic
 - B. Thesis statement
- II. First Main Point
 - A. First sub point
 1. Corresponding example or detail—either our insight or a critic's comment paraphrased or directly quoted
Never start a sentence with a quote that we simply haven't introduced.
 - a) First comment for support
 - b) Second comment for support
 2. Supporting example or detail (our discuss on supporting detail from a critic)
 - a) First comment
 - b) Second comment
 - B. Second sub point
 1. Supporting example or detail
 2. Supporting example or detail
 - C. Third sub point (same as A, B above)
At this time continue with D, E... if needed
- III. Second Main Point

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- A. First sub point—Comment on sub point (only one inquiry into this detail)
- B. Second sub point
 - 1. Supporting example—only one comment on this example:
 - 2. Supporting example
 - a) Comment
 - b) Comment

Continue with the identical sequence alternating numerals and letters

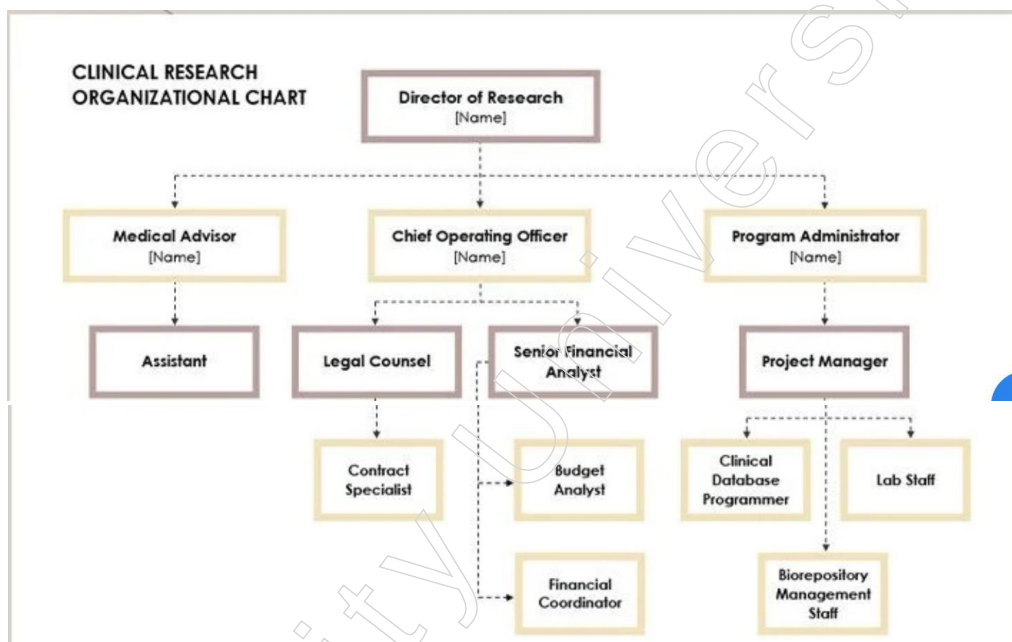
IV. Critical Thinking Section

- A. Incorporate reactions to the source
- B. Include insights about the research topic
- C. Synthesize critical thinking threads

V. Conclusion

Affirm that the thesis has been proven

Example:



5.3.3: Calculation of Impact Factor of a Journal

The impact factor (IF) may be a calculation of the frequency with that the typical article in a very journal has been cited in an explicit year. It's employed to measure the importance or rank of a journal by calculating the days its articles are cited.

The way of the Impact Factor is Calculation?

The calculation is predicted on a 2yr period and involves dividing the quantity of times articles were cited by the quantity of articles that are citable.

Calculation of 2010 IF of a journal:

A = the quantity of times articles published in 2008 and 2009 were cited by indexed journals throughout 2010.

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B = the overall range of “citable items” published in 2008 and 2009.

A/B = 2010 impact factor

What is the vary of IF:

The impact factor of ten or higher is taken into account as a wonderful score where 3 is flagged nearly as good and also the average score is less than 1. This is often a rule of thumb. However, the wild card to concentrate to is that impact factor and examination journals are most effective in the same category.

Example:

Cites in 2008 to items published in:		2007 =263	Number of items published in:	2007 =55
		2006 =374		2006 =49
		Sum: 637		Sum: 104
Calculation=		Cites to recent items	637	
		Number of recent items	104	= 6.125

5.3.4 Citation Index

A citation is the text reference and acknowledgement of a documented information. A scientifically written article comprises a reference section at the top where all the references mentioned within the document are cited serially, and every reference may be a citation. A citation count is the frequency of a writing article cited by other articles. A citation index could be a kind of bibliographic database, an index of citations between publications, allowing the user to simply establish which later documents cite which earlier documents

Major Citation Indexing Services

There are two major publishers of general-purpose academic citation indexes:

- Institute for Scientific Information (ISI): Citation indexing has long been dominated by the ISI, which is now a part of Thomson Reuters. It publishes the citation indexes in print and CD forms, which are generally accessed online under the name 'Web of Science' (WOS). WOS provides access to 7 databases: Science Citation Index (SCI), Social Sciences Citation Index (SSCI), Arts & Humanities Citation Index (A&HCI), Index Chemicals, Current Chemical Reactions, Conference Proceedings Citation Index: Science and Conference Proceedings Citation Index: Science and Humanities.
- Elsevier Which publishes Scopus, is offered online only. Scopus is one among the biggest abstract and citation databases of peer-reviewed literature and web sources with smart tools to trace.

There are also a variety of other readily available citation databases. Few of the notable ones among these are:

- CiteSeer is both a citation engine and a digital library. CiteSeer is predicted on the SmealSearch engine and provides citation, citation graph analysis, related document retrieval and other searching of scientific literature.
- Research Papers in Economics (RePec) provides databases in economics and other discipline specific indexes.

- Google Scholar (GS) provides citation and search of scholarly literature across many disciplines and sources. Although limited to the recent articles, it's a freely-accessible web program. Google Scholar is mostly praised for its speed.
- EBSCOhost is one of the world's foremost, most used, premium, reference database services.
- Compendex (Engineering Index, COMPuterizedENGINEERINGinDEX) is the computerized version of the Engineering Index and is one of the foremost comprehensive engineering literature databases.

Indian Citation Index Indian Citation Index (ICI) database was developed with specific motives to market knowledge contents, published in Indian journals and bridge the gap between the content sources and content users via the World Wide Web. ICI database covers about 1000 Indian scholarly journals, encompassing all disciplines of information including the sciences, social sciences, arts, and humanities'

The ICI database also produces other useful by-products like Indian Science Citation Index (ISCI), Indian Social Science and Humanities Citation Index (ISSHCI), Indian Journals Citation Reports (IJCR), Indian Science and Technology Abstracts (ISTA) and Directory of Indian Journals (DOIJ).

5.3.5 ISBN & ISSN

"ISBN" stands for "International Standard Book Number" whereas "ISSN" is "International Standard Serial Number." Both ISBN and ISSN are codes that are utilized by publishers for numbering or serialising their publications.

International Standard Book Number is assigned for monographs or books but the International Standard Serial Number is given to a series of monographs or books.

ISBN is assigned for one or separate book, and ISSN is assigned for a series of books.

ISBN identifies the particular volume or issue, the ISSN only identifies the series of the amount or issue.

ISSN is simply optional, which implies that the publisher isn't legally sure to use it. But The ISBN is obligatory if the book falls under the ISBN application.

Unlike the ISBN, the ISSN are identical altogether in all the volumes or the problems of a single series. But the ISBN is different for each volume and issue.

The ISBN could be a 13-digit standard code. It absolutely was Gordon Foster who created for the beginning a nine-digit ISBN code. Later the World Organisation for Standardisation developed the ten-digit standard number code. After 2007, the ISBN incorporated a 13-digit number standard code.

The International Standard Serial Number is an eight-digit standard number. It's the amount given to a single set of the series, and once the series changes, another ISSN code is allotted. It was in 1971 that the ISSN system was first drafted. The ISSN standard number codes are assigned by the ISSN National Centres and are coordinated by the ISSN International Centre that's based in Paris.

Summary:

1. "ISBN" is "International Standard Book Number" and "ISSN" is "International Standard Serial Number."

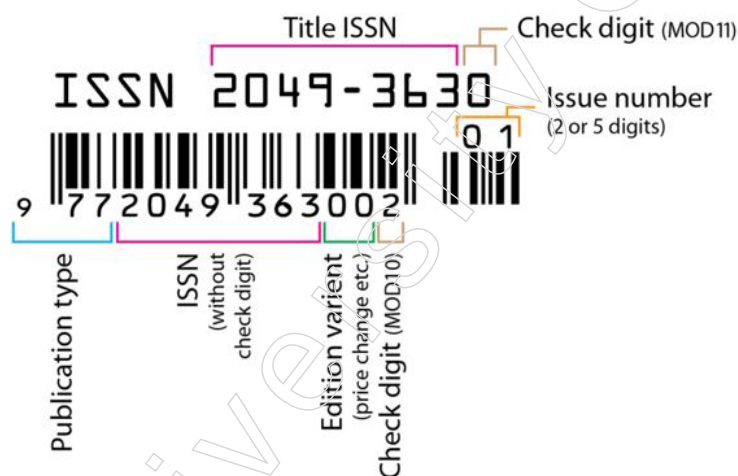
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2. One of the personal differences between the ISBN and ISSN is that the previous one identifies the publisher whereas the latter one doesn't identify the publisher.
3. International Standard Book Number is given for monographs or books whereas International Standard Serial number is given to a series of monographs or books.
4. In the case of the ISSN, it's only optional, which suggests that the publisher isn't legally bound to use it. But the ISBN is obligatory if the book falls under the ISBN application.

Below are the examples.

Sample ISSN numbers of the publications ISSN 0027-9633 (online)

ISSN 0027-9634 (print)



ISBN 978-951-45-9693-3 (hardback)

ISBN 978-951-45-9694-0 (paperback)

ISBN 978-951-45-9695-7 (PDF)

ISBN 978-951-45-9696-4 (EPUB without DRM)

ISBN 978-951-45-9999-5 (EPUB with ACS4 DRM)



Planning a Project Proposal

Discuss with an example

Colour Psychology Paper

Research Question:

How does colour affect one's mood?

Review of Literature:

Colours may just seem simple and unimportant, but they affect our daily lives more than we may know. If someone is feeling angry, it could just be because they're angry, or it could be perhaps that they are surrounded by or looking at the colour red. That's right! People's moods can change just because they are looking at different colours! There are many theories on how just a simple colour can change one's whole mood.

According to Johnson (2007), colour does affect mood by producing certain chemicals and stimulating different feelings such as hunger. For example, blue can make one feel calm because it releases calming chemicals, and red can make one hungry because it is an appetite stimulant. Yellow can make one feel irritated, and it is a fact that people lose their temper most in yellow rooms. However, pink is tranquilizing and can make one feel weak. In conclusion, Johnson says that depending on the colour, one's body can do things (like producing chemicals) that cause a certain emotional reaction (mad, sad, etc.).

Another idea, by Smith (2007), is that the effect colour produces is based on what one's body does in response. For example, yellow is mentally stimulating, and activates memory, whereas red increases confidence. Also, brown can make a person feel orderly and stable, while a dark blue can make one feel sad. Therefore, Smith says that different colours do in fact change one's mood and the consequences can be negative or positive.

A third writer, Wollard, (2000) seems to think that colour can affect one's mood, but the effect also can depend on one's culture and what one's personal reflection may be. For example, someone from Japan may not associate red with anger, as people from the U.S. tend to do. Also, a person who likes the colour brown may associate brown with happiness. However, Wollard does think that colours can make everyone feel the same, or close to the same, mood. According to Wollard, pink reduces aggression, which is why the walls of the jail cells in the Seattle prison are pink! Also, brown can make one feel comforted. Wollard feels that colours do affect one's mood, but there are other factors that can alter what one is supposed to feel.

Eric, John, and Parag's (2007) main point about colour psychology is that colour has both a physiological and psychological effect. For example, green makes people feel relaxed because it relaxes their muscles and makes them breathe deeper and more slowly. Furthermore, blue lowers blood pressure, which makes one feel calm. Eric, John, and Parag conclude that colour affects one's mood because of what it does to the body.

Yet another idea, by Airey (2006), is that colour is energy, and it can have a physical, mental, spiritual, and/or emotional effect on people. He states that black can make one feel sophisticated and secure, but it can also make one feel depressed.

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Also brown can make a person feel reliable and serious, while yellow lifts self-esteem. Therefore, Airey concludes that different colours can have different kinds of effects on people.

Analysis:

There are three ideas about colour psychology in these sources, and they all say that colour affects one's mood. They differ based on what factors influence the effects of colour, such as culture, opinion, and what goes on inside one's body. One of the three ideas is that colour affects mood based on one's personal opinions. For example, if a person dislikes the colour pink, he may associate pink with hate. Another idea states that colour affects mood based on one's culture. For example, someone from the U.S. may think of the colour green when referring to envy, while people in Japan think of yellow in connection with wanting what someone else has. However, the majority of the sources consulted say that colour affects mood by influencing what goes on inside of people. For example, seeing the colour blue releases calming chemicals, which in turn makes one calm. Also, because yellow is the hardest colour for the eye to focus on, people may become irritated when looking at yellow, and it is a proven fact that babies cry most in yellow nurseries. These theories do not seem to have much in common.

Methodology:

In order to test ideas about how colour affects one's mood (colour psychology) this researcher will test sixty middle school students (thirty girls and thirty boys) on how the colours green, blue, black, yellow, red, pink, and brown make them feel. The students will mark the feeling that each colour makes them feel, according to the choices shown on the example survey below.

Example Survey

(The only thing that changes in each different survey is the colour being tested and the gender of the student responding):

Choose as many answers as fit your feeling for the colour. Does the colour make you feel:

Mad/Angry?

Sad/Depressed/Cold?

Secure/Safe?

Irritated/Annoyed?

Calm?

Relaxed?

Fun/Excited?

Happy/Cheery?

Tired/Sleepy/Drowsy?

Bored?

Losing Energy/Weak?

Neutral/Balanced?

Love/Affection?

Dislike/Hate?

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Survey Results:

Most female middle school students associated these feelings with these colours.

Black	Green	Yellow	Pink	Brown	Red	Blue
Sad/Cold/ Depressing (43%)	Neutral/ Balanced 27%)	Happy/ Cheery (53%)	Happy/Cheery (27%) and Love/Affection (27%)	Bored (40%)	Mad/Angry (30%) and Love/ Affection (30%)	Calm (23%)

So, the greatest number of female middle school students thought that black made them feel sad, cold, and depressed; and green made most of them feel neutral and balanced, and so on. The numbers in parentheses by the feelings show the percentage of girls that felt that way. The reason most of these percentages are so low is that there were many different responses, and for most feeling categories there were only one or two girls who felt that way.

Most male middle school students associated these feelings with these colours.

Black	Green	Yellow	Pink	Brown	Red	Blue
Sad/Cold/ Depressed (37%)	Secure/Safe (27%)	Happy/Cheery (17%)	Love/ Affection (33%)	Bored (20%)	Mad/Angry (27%)	Relaxed (23%)

So, the greatest number of male middle school students thought that black made them feel sad, cold, and depressed; and green made most of them feel secure and safe, and so on. The numbers in parentheses by the feeling show the percentage of boys that felt that way. The reason these percentages are so low is that there were many different responses, and for most feeling categories there were only one or two boys who felt that way.

5.4 Planning a Project Proposal

Planning a project proposal is a crucial first step in the successful execution of any project. It involves outlining the goals, scope, and methods of a proposed project to secure approval, funding, or support from stakeholders. A well-crafted project proposal provides a detailed plan that guides the project's implementation and helps ensure its objectives are met efficiently.

Key Aspects of Project Proposal Planning:

1. Defining Objectives:
 - ❖ Clearly articulate the project's goals and objectives. Specify what the project aims to achieve and how these goals align with broader organizational or strategic objectives.
2. Identifying Scope:
 - ❖ Outline the project scope, including the boundaries and limitations of the project. This helps to manage expectations and prevent scope creep.
3. Conducting Feasibility Analysis:
 - ❖ Assess the feasibility of the project in terms of technical, financial, and

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- operational aspects. This includes evaluating the availability of resources, expertise, and potential risks.
4. **Developing a Work Plan:**
 - ❖ Create a detailed work plan that includes the project timeline, key milestones, and deliverables. This plan should also define roles and responsibilities for team members.
 5. **Budgeting and Resource Allocation:**
 - ❖ Prepare a comprehensive budget that covers all project expenses, including personnel, materials, and overhead costs. Ensure that resources are allocated efficiently to support project activities.
 6. **Risk Management:**
 - ❖ Identify potential risks and develop strategies to mitigate them. This includes analyzing possible challenges and developing contingency plans.
 7. **Stakeholder Engagement:**
 - ❖ Identify key stakeholders and outline strategies for engaging them throughout the project. Effective communication with stakeholders is essential for project support and success.
 8. **Evaluation and Metrics:**
 - ❖ Define criteria for evaluating the project's success. Establish metrics and performance indicators to measure progress and assess outcomes.
 9. **Proposal Documentation:**
 - ❖ Prepare the project proposal document, ensuring it is well-organized and includes all necessary sections, such as the executive summary, project description, objectives, methodology, budget, and appendices.
 10. **Review and Revision:**
 - ❖ Review the proposal thoroughly to ensure accuracy and completeness. Revise as needed based on feedback from stakeholders or reviewers.

5.4.1: Identifying and Defining Problem

In the research process, the first and foremost step happens to be that of selecting and properly defining a research problem. A researcher must find the problem and formulate it so that it becomes susceptible to research. Like a medical doctor, a researcher must examine all the symptoms (presented to him or observed by him) concerning a problem before he can diagnose correctly. To define a problem correctly, a researcher must know: what a problem is?

Identifying a research problem, in general, refers to some difficulty in the context of either a theoretical or practical situation and wants to obtain a solution for the same.

Hence, identifying a research issue or problem is a phenomenon as it might be called in many forms of qualitative research, if the topic is to be addressed, investigated, or studied, whether descriptive or experimentally. It is the focus or reason for engaging in the research. It is typically a topic, phenomenon, or challenge that is required to be addressed.

Defining issues or Problems

Defining a research problem is the fuel that drives the scientific process and is the foundation of any research method and experimental design.

Research problem: A situation or circumstances that require a solution to be described, explained or predicted.

The components of a research problem can be:

- There must be an individual or group which has some difficulty or a problem.
- There must be some objective(s) to be attained at.
- There must be an alternative means for obtaining the objectives one wishes to attain.
- There must remain some doubt in the mind of a researcher with regard to the selection of alternatives. This means that research must answer the question concerning the relative efficiency of the possible alternatives.
- There must be some environment to which the difficulty pertains.
- The research problem undertaken for study must be carefully selected.
- If there is a knowledge gap in an area that needs to be investigated, the research problem identifies this gap.
- This difficulty in selecting a problem or the gap arises not due to the limited range of reachable problems but the inability to locate the socially relevant topic without duplicating earlier studies.

5.4.2 Research Problem aspects and considerations

Problem Aspects

Let us start with the question: What does one mean when he/she wants to define a research problem? The answer may be that one wants to state the problem along with the bounds within which it is to be studied. In other words, defining a problem involves the task of laying down boundaries within which a researcher shall study the problem with a predetermined objective in view.

How to define a research problem is undoubtedly a herculean task. However, it is a task that must be tackled intelligently to avoid the perplexity encountered in a research operation. The usual approach is that the researcher should himself pose a question (or in case someone else wants the researcher to carry on research, the concerned individual, organisation or an authority should pose the question to the researcher) and set-up techniques and procedures for throwing light on the question concerned for formulating or defining the research problem. But such an approach generally does not produce definitive results because the question phrased in such a fashion is usually in broad general terms and as such may not be in a form suitable for testing.

Defining a research problem properly and clearly is a crucial part of a research study and must in no case be accomplished hurriedly. However, in practice this is frequently overlooked which causes a lot of problems later on. Hence, the research problem should be defined in a systematic manner, giving due weightage to all related points. There are a lot of aspects which need to be kept in mind before we deal with it. The aspects are:

- (i) **Statement Of the problem:** Problems should be stated in either practical or scientific or some intellectual interest. It can be stated through a preliminary field survey or observation. This helps in stating the problem clearly; it also helps in narrowing the concern in a phrase the problem in operational terms. In case the problem is stated in a broad and general way it will involve a lot of ambiguities which can be resolved

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- by looking at the detailed observations and surveys available. Thus the feasibility of a particular solution can be considered.
- (ii) Understanding the nature of the problem: Another aspect to be kept in mind is to understand the origin of the problem. The best way to understand is to find out the source of occurrence or need for study.
 - (iii) Surveying the available literature: All available literature concerning the problem at hand must necessarily be surveyed and examined before a definition of the research problem is given. This means that the researcher must be well-versed with relevant theories in the field, reports and records as also all other relevant literature. Sufficient time must be devoted to reviewing the problem. This is done to find out what materials are available at the moment and what other materials are required. "Knowing what data are available often serves to narrow the problem itself as well as the technique that might be used. "This would also help a researcher to know if there are certain gaps in the theories, or whether the existing theories applicable to the problem under study are inconsistent with each other, or whether the findings of the different studies do not follow a pattern consistent with the theoretical expectations and so on. All this will enable a researcher to take new strides in the field for furtherance of knowledge i.e., moving up starting from the existing premise. Studies on related problems are useful for indicating the type of difficulties that may be encountered in the present study as also the possible analytical shortcomings. At times such studies may also suggest useful and even new lines of approach to the present problem.
 - (iv) Deduction from the theory: This is related to deductions made from the available source. Once the irrelevant details of the problems are kept aside and the core of the problem is identified then we can carry out a systematic investigation of the research to assess the data.
 - (v) Rephrasing the research problem into a working proposition: Rephrasing is important because the nature of the problem has been clearly understood, defined. Through rephrasing, the researcher puts the research problem in as specific terms as possible so that it may become operationally viable and may help in the development of the research.

Considerations:

The research problem undertaken for study must be carefully selected. The task is a difficult one, although it may not appear to be so. Help may be taken from a research guide in this connection. A research guide can thus help in considering a problem. Considering a problem can be defined in three categories:

1. Identify concepts and terms that make up the topic statement: The research problem can have many aspects which needs to be kept in mind but identifying the concept and terms which comprises the topic makes it easier for the consideration and thus the research.
2. Review related literature to help refine how you will approach examining the topic and finding a way to analyse it: it can be done by reading through background information from other books and materials related to it. A preliminary review of the research literature can be done using multidisciplinary databases.
3. Additional sources: since the research work can be extensive it can be helped with additional data which can broaden the knowledge and strengthen the initial thoughts and arguments related to the research.

5.4.3 Research Plan

Planning a Project Proposal

Locating and Defining Issues or Problems

This step focuses on determining the nature and parameters of a situation or question involving marketing strategy or implementation. The researcher should consider the study's purpose, relevant background information, what information is needed, and how it will be used in decision-making when defining the issues or problems.

Designing the Research Project

This step focuses on developing a research strategy or overall approach to resolving the identified issue or problem. A research plan or approach is a blueprint or framework for carrying out a marketing research project. Its purpose is to design a study that will test the hypotheses of interest, determine possible answers to the research questions, and provide the information required for decision making.

The steps in the research design are as follows:

- ❖ Analysing secondary data
- ❖ Qualitative analysis
- ❖ Quantitative data collection methods (survey, observation, and experimentation)
- ❖ Definition of the data required
- ❖ Procedures for measurement and scaling
- ❖ Designing a questionnaire
- ❖ Process of sampling and sample size
- ❖ Data analysis strategy

Collecting Data

This step was all about gathering the data you'll need to solve the issue or problem you've identified. A field force or staff collects data in the field (in-home, mall intercept, or computer-assisted personal interviewing), over the phone (telephone or computer-assisted telephone interviewing), or through the mail (traditional mail and mail panel surveys with recruited households).

The following are some examples of data collection techniques:

- ❖ Interviews entail questioning people about their prior knowledge.
- ❖ Observations are the act of gathering information without asking questions.
- ❖ Questionnaires: Ask a group of people a series of questions.
- ❖ Interviewing and observing a group of people is what a focus group is.
- ❖ Old-fashioned research with documents and records

Interpreting Research Data

Interpreting research data: The goal of this step is to examine the data and come up with a solution to the problem.

Begin by putting your findings and the information you gathered in Step 3 into order. After that, write a rough draught of your conclusion, recommendations, and

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findings. The rough draft will assist you in organising your thoughts. The last step is to refine the draft into your final research conclusion. Before the final product is ready for Step 5, you will most likely revise the draft as many times as necessary.

Report Research Findings

The final step is to present the findings of the research to those who require the information to make decisions. The results should be presented in an easily understandable format so that they can be used in the decision-making process quickly. In addition, to improve clarity and impact, an oral presentation to management should be made using tables, figures, and graphs.

Formats for Research Reporting:

- Formal Paper
- Published Article
- PowerPoint Presentation
- Audio or Video
- Spreadsheet

Summary

- A report is a very formal document that is composed for a variety of objectives, including sciences, social sciences, engineering, and business disciplines. Essentially, the process of looking for a certain assignment is documented in a report.
- It is important to recognise that reports are regarded as legal papers in all workplaces.
- When the researchers are asked to make an oral presentation then they prepare this type of oral report.
- It's more difficult to prepare as compared to the written report.
- Because here the presenter directly interacts with the audience.
- Any delay during an oral presentation can leave a negative impression on the audience and also violates the self-confidence of the presenter.
- Communication plays a big role in an oral presentation. A lot of planning and thinking is required to decide 'What to say', 'How to say', 'How much to say'. Also, the presenter should have to face a bunch of questions from the audience.
- The International Standard Serial Number is an eight-digit standard number.
- It's the amount given to a single set of the series, and once the series changes, another ISSN code is allotted.
- It was in 1971 that the ISSN system was first drafted. The ISSN standard number codes are assigned by the ISSN National Centres and are coordinated by the ISSN International Centre that's based in Paris.

Glossary

- Predefined structure: In a broader context, these headers may serve as indicators of different parts within a report, such as an introduction, discussion, and conclusion.
- Independent sections: Each section in a report is separately written, because if the reader wants to selectively identify the report sections they are interested in, rather than reading the whole report in one go from top to bottom.

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- Unbiased conclusions: Another aspect of report writing is that it is a kind of writing that is impartial and objective.
- Bibliography: This contains the main sources of secondary data.
- Technical appendices: These encompass all technical aspects related to surveys, mathematical derivations, detailed explanations of specific analytic methodologies, and similar topics.

Check Your Understanding

1. What is one of the fundamental characteristics of report writing?
 - a) Personal opinion
 - b) Informal language
 - c) Predefined structure
 - d) Subjective conclusions
2. Which of the following is included in the preliminaries of a scientific report?
 - a) Analysis and presentation of results
 - b) Methods of organization
 - c) Title page
 - d) Bibliography
3. In the structure of a scientific report, where would you find the research aims and hypotheses?
 - a) Introduction
 - b) Analysis and presentation of results
 - c) Appendices
 - d) Glossary
4. Which type of report is prepared for direct interaction with the audience and requires a lot of preparation and communication skills?
 - a) Written report
 - b) Oral report
 - c) Informal report
 - d) Government report
5. Which sampling method ensures every item in the population has an equal chance of being selected?
 - a) Non-Probability Sampling
 - b) Stratified Sampling
 - c) Probability Sampling
 - d) Convenience Sampling

Exercise

1. Define a report and explain its significance in different disciplines.
2. What are the three fundamental characteristics of report writing? Explain each briefly.
3. List and describe the components included in the preliminaries of a scientific report.

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4. What is the importance of the 'Introduction' section in a scientific report? List the key elements it should cover.
5. Differentiate between a 'technical report' and a 'non-technical report'. Provide examples of when each might be used.

Learning Activity

1. You are tasked with preparing an oral report on the impact of social media on academic performance. Outline the key points you would include in your presentation and explain how you would engage your audience using visual aids and addressing potential questions.
2. You have been given a scientific report. Identify the components within the report and provide a brief explanation of their purpose.

Check Your Understanding- Answers

1. c)
2. c)
3. a)
4. b)
5. c)

Further Readings and Bibliography

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